



AOS-CX 10.18.xxxx IP Services Guide (4100i, 6000, 6100 Switch series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

Latest version available online

Command syntax notation conventions

About the examples

Identifying switch ports and interfaces

Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A, S4R20A, S4R21A, S4R22A, S4R23A, S4R24A, S4R25A, S4R26A, S4R27A, S4R28, S4R29A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands , code examples, filenames, pathnames, and output displayed in a command window. Items that appear li

Convention	Usage
	ke the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch (config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch (config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch (config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 4100i Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6000 and 6100 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

IRDP

ICMP Router Discovery Protocol (IRDP), an extension of the ICMP, is independent of any routing protocol. It allows hosts to discover the IP addresses of neighboring routers that can act as default gateways to reach devices on other IP networks.



NOTE

On the switches covered by this guide, IRDP is configured on a VLAN interface.

IRDP operation

IRDP uses the following types of ICMP messages:

- Router advertisement (RA): Sent by a router to advertise IP addresses (including the primary and secondary IP addresses) and preference.
- Router solicitation (RS): Sent by a host to request the IP addresses of routers on the subnet.

An interface with IRDP enabled periodically broadcasts or multicasts an RA message to advertise its IP addresses. A receiving host adds the IP addresses to its routing table, and selects the IP address with the highest preference as the default gateway.

When a host attached to the subnet starts up, the host multicasts an RS message to request immediate advertisements. If the host does not receive any advertisements, it retransmits the RS several times. If the host does not discover the IP addresses of neighboring routers because of network problems, the host can still discover them from periodic RAs.

IRDP allows hosts to discover neighboring routers, but it does not suggest the best route to a destination. If a host sends a packet to a router that is not the best next hop, the host will receive an ICMP redirect message from the router.

IP address preference

Every IP address advertised in RAs has a preference value. A larger preference value represents a higher preference. The IP address with the highest preference is selected as the default gateway address.

You can specify the preference for IP addresses to be advertised on a router interface.

An address with the minimum preference value (-2147483648) will not be used as a default gateway address.

Lifetime of an IP address

An RA contains a lifetime field that specifies the lifetime of advertised IP addresses. If the host does not receive a new RA for an IP address within the address lifetime, the host removes the route entry.

All the IP addresses advertised by an interface have the same lifetime.

Advertising interval

A router interface with IRDP enabled sends out RAs randomly between the minimum and maximum advertising intervals. This mechanism prevents the local link from being overloaded by a large number of RAs sent simultaneously from routers.

As a best practice, shorten the advertising interval on a link that suffers high packet loss rates

Destination address of RA

An RA uses either of the following destination IP addresses:

- Broadcast address 255.255.255.255.
- Multicast address 224.0.0.1, which identifies all hosts on the local link.

By default, the destination IP address of an RA is the multicast address. If all listening hosts in a local area network support IP multicast, specify 224.0.0.1 as the destination IP address.

Proxy-advertised IP addresses

By default, an interface advertises its primary and secondary IP addresses. You can specify IP addresses of other gateways for an interface to proxy-advertise.

VRF support

In IP-based computer networks, virtual routing and forwarding (VRF) is a technology that allows multiple instances of a routing table to co-exist within the same router at the same time. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other.

IRDP is VRF aware. As the router advertisements and solicit processing occurs on the interface, packet is through the interface and corresponding VRF.

VSX synchronization

IRDP supports VSX synchronization. For more information on using VSX, see the Virtual Switching Extension (VSX) Guide for your switch and software version

Subtopics

Configuring IRDP

Configuring IRDP

Prerequisites

A layer 3 interface.

Procedure

1. Enable IRDP on an interface with the command `ip irdp`.
2. Set the maximum hold time with the command `ip irdp holdtime`.
3. Set the maximum router advertisement interval with the command `ip irdp maxadvertinterval`.
4. Set the minimum router advertisement interval with the command `ip irdp minadvertinterval`.
5. Set the IRDP preference level with the command `ip irdp preference`.
6. Review IRDP configuration settings with the command `show ip irdp`.

Example

This example creates the following configuration:

- Enables IRDP on the layer 3 VLAN interface 2 with packet type set to broadcast.
- Sets the hold time to 5000 seconds.
- Sets the advertisement interval to 30 seconds.
- Sets the minimum advertisement interval to 25 seconds.
- Sets the IRDP preference level to 25.

```
switch(config)#  
    interface vlan 2  
switch(config-if-vlan)# ip irdp broadcast  
switch(config-if-vlan)# ip irdp holdtime 5000  
switch(config-if-vlan)# ip irdp maxadvertinterval 30  
switch(config-if-vlan)# ip irdp minadvertinterval 25  
switch(config-if-vlan)# ip irdp preference 25
```


IRDP commands

Select a command from the list in the left navigation menu.

Subtopics

[diag-dump irdp basic](#)

[ip irdp](#)

[ip irdp holdtime](#)

[ip irdp maxadvertinterval](#)

[ip irdp minadvertinterval](#)

[ip irdp preference](#)

[show ip irdp](#)

diag-dump irdp basic

Syntax

```
diag-dump irdp basic
```

Description

Displays diagnostic information for IRDP.

Example

```
switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jun  8 09:50:28 2017
=====
-----
[Start] Daemon hpe-rdiscd
-----
Interface: 1/1/1 (state : Up)
rdisd ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
Router IPs - 192.168.1.2,
Interface: 1/1/2 (state : Up)
rdisd ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
Router IPs - 192.168.2.2,
-----
[End] Daemon hpe-rdiscd
-----
=====
[End] Feature irdp
```

```

=====
Diagnostic dump captured for feature irdp
switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jan  7 04:46:25 2021
=====
-----
[Start] Daemon hpe-rdiscd
-----
Interface: vlan2 (state : Down)
rdisc ipv4 (enabled: 1, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
Interface: vlan1 (state : Down)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
-----
[End] Daemon hpe-rdiscd
-----
=====
[End] Feature irdp
=====
Diagnostic-dump captured for feature irdp

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ip irdp

Syntax

```
ip irdp [broadcast | multicast]
no ip irdp
```

Description

Enables IRDP on an interface and specifies the packet type that is used to send advertisements. By default, the packet type is set to `multicast`. IRDP is only supported on layer 3 interfaces.

The **no** form of this command disables IRDP on an interface.

Parameter	Description
<code>broadcast</code>	Advertisements are sent as broadcast packets to IP address 255.255.255.255.
<code>multicast</code>	Advertisements are sent as multicast packets to the multicast group with IP address 24.0.0.1. Default.

Examples

Enabling IRDP on interface vlan 2 with packet type set to the default value (multicast).

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# ip irdp
```

Enabling IRDP on interface vlan 2 with packet type set to broadcast.

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
```

Disabling IRDP.

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# no ip irdp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if-vlan</code>	Administrators or local user group members with execution rights for this command.

`ip irdp holdtime`

Syntax

```
ip irdp holdtime <TIME>
no ip irdp holdtime <TIME>
```

Description

Specifies the maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives. When a new advertisement arrives, hold time is reset. Hold time must be greater than or equal to the maximum advertisement interval. Therefore, if the hold time for an advertisement expires, the host can reasonably conclude that the router interface that sent the advertisement is no longer available. The default hold time is three times the maximum advertisement interval.

The **no** form of this command removes the specified maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives and update it to the default value.

Parameter	Description
<code><TIME></code>	Specifies the lifetime of router advertisements sent from this interface. Range: 4 to 9000 seconds. Default: 1800 seconds.

Example

Setting the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# ip irdp holdtime 5000
```

Removing the the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)#  
    interface vlan 2  
switch(config-if-vlan)# no ip irdp holdtime 5000
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp maxadvertinterval

Syntax

```
ip irdp maxadvertinterval <TIME>  
no ip irdp maxadvertinterval <TIME>
```

Description

Specifies the maximum router advertisement interval.

The **no** form of this command removes the specified maximum router advertisement interval and reverts to the default value.

Parameter	Description
<TIME>	Specifies the maximum time allowed between the sending of unsolicited router advertisements. Range: 4 to 1800 seconds. Default: 600 seconds.

Example

Setting the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)#  
    interface vlan 2  
switch(config-if-vlan)# ip irdp maxadvertinterval 30
```

Removing the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)#  
    interface vlan 2  
switch(config-if-vlan)# no ip irdp maxadvertinterval 30
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp minadvertinterval

Syntax

```
ip irdp minadvertinterval <TIME>  
no ip irdp minadvertinterval <TIME>
```

Description

Specifies the minimum amount of time the switch waits between sending router advertisements. By default, this value is automatically set by the switch to be 75% of the value configured for maximum router advertisement interval. Use this command to override the automatically configured value.

The **no** form of this command removes the specified minimum amount of time the switch waits between sending router advertisements and reverts to the default value.

Parameter	Description
<code><TIME></code>	Specifies the minimum time allowed between the sending of unsolicited router advertisements. Range: 3 to 1800 seconds. Default: 450 seconds (75% of the default value for maximum route advertisement interval).

Example

Setting the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# ip irdp minadvertinterval 25
```

Removing the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# no ip irdp minadvertinterval 25
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

`ip irdp preference`

Syntax

```
ip irdp preference <LEVEL>
no ip irdp preference <LEVEL>
```

Description

Specifies the IRDP preference level. If a host receives multiple router advertisement messages from different routers, the host selects the router that sent the message with the highest preference as the default gateway.

The **no** form of this command removes the specified IRDP preference level and reverts to the default value.

Parameter	Description
<LEVEL>	Specifies the IRDP preference level. Range: - 2147483648 to 2147483647. Default: 0.

Example

Setting the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# ip irdp preference 25
```

Removing the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# no ip irdp preference 25
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if-vlan</code>	Administrators or local user group members with execution rights for this command.

show ip irdp

Syntax

```
show ip irdp
```

Description

Displays IRDP configuration settings.

Example

```
switch# show ip irdp
ICMP Router Discovery Protocol
Interface Status   Advertising Minimum Maximum Holdtime Preference
          Address   Interval  Interval
-----
1/1/1     Enabled  multicast  6         8         10         10
1/1/2     Disabled multicast  450        600       1800        0
1/1/3     Enabled  broadcast  450        600       1800       115

switch# sh ip irdp
ICMP Router Discovery Protocol
Interface      Status   Advertising Minimum Maximum Holdtime Preference
              Address   Interval  Interval
-----
vlan1          Disabled multicast  450        600       1800        0
bridge_normal  Disabled multicast  450        600       1800        0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

IPv6 Router Advertisement

IPv6 RA provides a method for local IPv6 hosts to automatically configure their own IP address (and other settings such as a preferred DNS server) based on information advertised by switches/routers operating on the network.

IPv6 flags

Behavior of IPv6 hosts to IPv6 RA messages is controlled by the managed address configuration flag (M flag), and other stateful configuration flag (O flag).

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS - related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Subtopics

[Configuring IPv6 RA](#)
[IPv6 RA scenario](#)

Configuring IPv6 RA

Syntax

Procedure

1. Enable transmission of IPv6 router advertisements with the command `no ipv6 nd suppress-ra`.
2. Optionally, configure IPv6 unicast address prefixes with the command `ipv6 nd prefix`.
3. Optionally, configure support for DNS name resolution with the commands `ipv6 nd ra dns server` and `ipv6 nd ra dns search-list`.
4. For most deployments, the default values for the following features do not need to be changed. If your deployment requires different settings, change the default values with the indicated command:

IPv6 RA setting	Default value	Command to change it
Number of neighbor solicitations to be sent when performing DAD.	1	<code>ipv6 nd dad attempts</code>
Number of neighbor entries in the ND cache.	131072	<code>ipv6 nd cache-limit</code>
Hop limit to be sent in the RA messages.	64	<code>ipv6 nd hop-limit</code>
MTU value to be sent in the RA messages.	1500 bytes	<code>ipv6 nd mtu</code>
Neighbor solicitation interval	1000 milliseconds	<code>ipv6 nd ns-interval</code>
Lifetime of a default router.	1800 seconds	<code>ipv6 nd ra lifetime</code>
Retrieval of an IPv6 address by devices.	Disabled	<code>ipv6 nd ra managed-config-flag</code>

IPv6 RA setting	Default value	Command to change it
Maximum interval between transmissions of IPv6 RAs.	600 seconds	<code>ipv6 nd ra max-interval</code>
Minimum interval between transmissions of IPv6 RAs.	200 seconds	<code>ipv6 nd ra min-interval</code>
Time that an interface considers a device to be reachable.	0 milliseconds (no limit)	<code>ipv6 nd ra reachable-time</code>
Retry period between ND solicitations.	0 (Use locally configured NS - interval)	<code>ipv6 nd ra retrans-timer</code>
Default routing preference for an interface.	Medium	<code>ipv6 nd router-preference</code>

- Review IPv6 RA configuration settings with the commands `show ipv6 nd interface`, `show ipv6 nd interface prefix`, `show ipv6 nd ra dns server`, and `show ipv6 nd ra dns search-list`.

Example

This example creates the following configuration:

- Enables IPV6 RA on interface interface vlan 2.
- Sets the recursive DNS server address to `4001::1` with a lifetime of `400` seconds.
- Sets the minimum interval between transmissions to `3` seconds.
- Sets the maximum interval between transmissions to `13` seconds.
- Sets the lifetime of a default router to `1900` seconds.

```
switch(config)# interface vlan 2
switch(config-if)# no ipv6 nd suppress-ra
switch(config-if)# ipv6 nd ra dns server 4001::1 lifetime 400
switch(config-if)# ipv6 nd ra min-interval 3
switch(config-if)# ipv6 nd ra max-interval 13
switch(config-if)# ipv6 nd ra lifetime 1900
switch(config-if)# end
switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
Admin state is up
```

```

IPv6 address:
  2006::1/64 [VALID]
IPv6 link-local address: fe80::98f2:b321:368:6dc6/64 [VALID]
ICMPv6 active timers:
  Last Router-Advertisement sent: 0 Secs
  Next Router-Advertisement sent in: 13 Secs
Router-Advertisement parameters:
  Periodic interval: 3 to 13 secs
  Router Preference: medium
  Send "Managed Address Configuration" flag: false
  Send "Other Stateful Configuration" flag: false
  Send "Current Hop Limit" field: 64
  Send "MTU" option value: 1500
  Send "Router Lifetime" field: 1900
  Send "Reachable Time" field: 0
  Send "Retrans Timer" field: 0
  Suppress RA: false
  Suppress MTU in RA: true
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: vlan2
  Suppress DNS Server List: No
  DNS Server 1: 2001::1      lifetime 400

```

IPv6 RA scenario

In this scenario, two host computers are auto-configured with IP addresses using IPv6 RA. In addition, the switch provides the hosts with an address of a recursive DNS server.

Procedure

1. Configure the VLAN interfaces with IPv6 addresses.

```

switch# config switch(config)# interface vlan 2 switch(config-if-vlan)# ipv6 address 2001::1/64
switch(config)# interface vlan 3 switch(config-if-vlan)# ipv6 address 3001::1/64 switch(config)#
interface vlan 4 switch(config-if-vlan)# ipv6 address 4001::1/64

```
2. Enable transmission of all IPv6 RA messages.

```

switch(config-if-vlan)# no ipv6 nd suppress-ra

```

IGMP snooping commands

Select a command from the list in the left navigation menu.

Subtopics

- [ip igmp snooping \(config mode\)](#)
- [ip igmp snooping \(interface mode\)](#)
- [ip igmp snooping \(vlan mode\)](#)
- [ip igmp snooping all-vlans](#)
- [ip igmp snooping apply access list](#)
- [ip igmp snooping filter unknown mcast](#)
- [ip igmp snooping preprogram-starg-flow](#)
- [ip igmp snooping querier-address](#)
- [ip igmp snooping report-suppression](#)
- [ip igmp snooping static group](#)
- [ip igmp snooping vxlan replicate-all](#)
- [show ip igmp snooping](#)

ip igmp snooping (config mode)

Syntax

```
ip igmp snooping
  drop-unknown vlan-shared|vlan-exclusive
  fastlearn <PORT-LIST>
```

Description

Configures drop-unknown and fastlearn modes on the ports. While IGMP snooping is enabled, the traffic will be forwarded only to ports that made an IGMP request for the multicast. Drop unknown filters ensure that packets are not forwarded to ports that did not make a request for the traffic stream. This could either be a filter across all VLANs (**vlan-shared**) or per VLAN (**vlan-exclusive**). The default is **vlan-shared**. Fast learn enables the port to learn group information when receiving a topology change notification. By default, fast learn is not enabled on ports.

Parameter	Description
drop-unknown	Drop unknown filters ensure that packets are not forwarded to ports that did not make a request for the traffic stream.

Parameter	Description
<code>vlan-shared</code>	Enables a shared VLAN filter on the switch. Default is vlan - shared .
<code>vlan-exclusive</code>	Enables an exclusive drop unknown filter per VLAN.
<code>fastlearn <PORT-LIST></code>	Enable fast learn on ports. This parameter specifies a list of one or more ports to be configured as fast learn ports. You can specify a single port, a comma - separated list of ports or a range of ports such as 1/1/1 - 1/1/3. You may also enter an L2 LAG (1 - 128)
<code>no ...</code>	Negates any configured parameter.

Example

Configuring fast learn ports:

```
switch(config) # ip igmp snooping fastlearn 1/1/3
switch(config) # ip igmp snooping fastlearn 1/1/1-1/1/2
switch(config) # ip igmp snooping fastlearn 1/1/5,1/1/6
```

Configuring a shared VLAN filter on the switch:

```
switch(config) # ip igmp snooping drop-unknown vlan-shared
```

Configuring a exclusive drop unknown filter per VLAN:

```
switch(config) # ip igmp snooping drop-unknown vlan-exclusive
```

Disabling drop unknown on the switch:

```
switch(config) # no ip igmp snooping drop-unknown
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ip igmp snooping (interface mode)

Syntax

```
ip igmp snooping
  auto vlan <VLAN-LIST>
  blocked vlan <VLAN-LIST>
  fastleave vlan <VLAN-LIST>
  forced-fastleave vlan <VLAN-LIST>
  forward vlan <VLAN-LIST>
  no ...
```

Description

Configure IP IGMP snooping for the VLAN on the interface. When IGMP snooping is enabled, the L2 snooping switch forwards multicast packets of known multicast groups to only the receivers. When IGMP snooping is not enabled, the snooping switch floods multicast packets to all hosts on the VLAN.

Parameter	Description
<code>auto vlan <VLAN-LIST></code>	Instruct the device to monitor incoming multicast traffic on the specified ports on a VLAN or VLAN range. This is the default behavior. Enter the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10 - 40), or both (10 - 40,60).
<code>blocked vlan <VLAN-LIST></code>	Configures the specified ports in blocked mode for the specified VLAN list. In blocked mode, joins and traffic are always blocked on this port. Enter the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10 - 40), or both (10 - 40,60).
<code>fastleave vlan <VLAN-LIST></code>	IGMP fastleave is configured for ports on a per - VLAN basis. Upon receiving a Leave Group message, the querier sends an IGMP Group - Specific Query message out of the interface to ensure that no other receivers are connected to the interface. If receivers are directly attached to the switch, it is inefficient to send

Parameter

Description

the membership query as the receiver wanting to leave is the only connected host.

When a fastleave - enabled switch port is connected to a single host and receives a leave, the switch does not wait for the querier status update interval, but instead immediately removes the IGMP client from its IGMP table and ceases transmitting multicast traffic to the client. (If the switch detects multiple end nodes on the port, Fastleave does not activate regardless of whether one or more of these end nodes are IGMP clients.) This processing speeds up the overall leave process and also eliminates the CPU overhead of having to generate an IGMP Group - Specific Query message.

This parameter specifies a list of VLANs on which the port should be configured as a fastleave port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10 - 40), or both (10 - 40,60).

```
forced-fastleave vlan  
<VLAN-LIST>
```

With forced fastleave enabled, IGMP speeds up the process of blocking unnecessary multicast traffic to a switch port that is connected to multiple end nodes. When a port having multiple end nodes receives a leave group request from one end node for a given multicast group, forced fastleave activates and waits for a second to receive a join request from any other member of the same group on that port. If the port does not receive a join request for that group within the forced fastleave interval, the switch then blocks any further traffic to that group on that port.

This parameter specifies a list of VLANs on which the port should be configured as a forced fastleave port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10 - 40), or both (10 - 40,60).

This command is available in **config - if** mode.

```
forward vlan <VLAN-LIST>
```

Configures the specified ports in forward mode in the given VLAN list. In forward mode, traffic is always forwarded on this port, irrespective of joins. Specify a list of VLANs on which the port should be configured as a forward port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10 - 40), or both (10 - 40,60).

This command is available in **config - if** mode.

```
no ...
```

Negates any configured parameter.

Example

Configure auto ports for VLAN on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-if)# no shut
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 10-20
switch(config-if)# ip igmp snooping auto vlan 10
switch(config-if)# ip igmp snooping auto vlan 10-20
```

Configuring fastleave ports for the VLAN on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-if)# no shut
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 10-20
switch(config-if)# ip igmp snooping fastleave vlan 10
switch(config-if)# ip igmp snooping fastleave vlan 10-20
```

Configuring blocked ports for the VLAN on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-if)# no shut
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 10-20
switch(config-if)# ip igmp snooping blocked vlan 10
switch(config-if)# ip igmp snooping blocked vlan 10-20
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i	config-if	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
6000		
6100		

ip igmp snooping (vlan mode)

Syntax

```
ip igmp snooping
  apply access-list <ACL-NAME>
  enable|disable
  no ...
  static-group <MULTICAST-IP-ADDRESS>
  version <2-3> (vlan interface mode)
```

Description

These commands enable or disable IP IGMP snooping on the VLAN, create IGMP snooping static multicast groups, set the IGMP snooping version and configure the ACL on a particular interface.



NOTE

Disabling and enabling IGMP snooping on a VLAN causes IGMP querier re-election.

Parameter	Description
access-list	Associates an ACL with the IGMP.
enable disable	Enables or disables IGMP snooping on the VLAN. Starting with AOS - CX 10.16, IGMP snooping is enabled by default.
no ...	Negates any configured parameter.
static-group <MULTICAST-IP-ADDRESS>	This parameter configures an IGMP snooping static multicast group. Specify the IGMP static multicast group IP address in A.B.C.D format. You can configure a maximum of 32 IGMP snooping static
	Configures the IGMP snooping version on the VLAN. Select 2 for IGMPv2 (RFC2236). Select 3 for IGMPv3 (RFC3376).

Parameter	Description
<code>version <2-3></code>	

Usage

- Existing classifier commands are used to configure the ACL.
- In case an IGMPv3 packet with multiple group addresses is received, the switch only processes the permitted group addresses based on the ACL rule set. The packet is forwarded to querier and PIM router even though one of the groups present in the packet is blocked by the ACL. This avoids the delay in learning of the permitted groups. Since the access switch configured with ACL blocks the traffic for the groups which are denied, forwarding of joins has no impact. If all the groups in the packet are denied by the ACL rule, the packet is not forwarded to the querier and PIM router. Existing joins will timeout.
- In case of IGMPv2, if there is no match or if there is a deny rule match, the packet is dropped.



NOTE

If the access list is configured for both L2 VLAN and L3 VLAN, the L3 VLAN configuration will be applied.

Example

Disable IGMP snooping on a VLAN:

```
switch(config) # vlan 2
switch(config-vlan) # ip igmp snooping disable
```

Reenable IGMP snooping on a VLAN:

```
switch(config) # vlan 2
switch(config-vlan) # ip igmp snooping enable
```

Configuring an IGMP snooping static group:

```
switch(config) # vlan 2
switch(config-vlan) # ip igmp snooping static-group 239.1.1.1
switch(config-vlan) # no ip igmp snooping static-group 239.1.1.1
```

Configuring IGMP snooping version on the VLAN:

```
switch(config) # vlan 2
switch(config-vlan) # ip igmp snooping version 2
```

Command History

Release	Modification
10.16	Starting with this release, IGMP snooping is enabled on VLANs by default, including the Management VLAN (vlan 1) and VLANs with a associated VNI.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-vlan- <VLAN-ID></code>	Administrators or local user group members with execution rights for this command.

ip igmp snooping all-vlans

Syntax

```
ip igmp snooping all-vlans
no ip igmp snooping all-vlans
```

Description

Enable IGMP snooping on all VLANs. This configuration setting enables IGMP snooping on all existing VLANs, as well as on any VLANs that are created after the **ip igmp snooping all-vlans** command is issued. When IGMP snooping is enabled globally, IGMP snooping can not be configured on individual VLANs within the VLAN context of the command-line interface.

The **no** form of this command disables global IGMP snooping, allows IGMP snooping to remain enabled at the VLAN level, and allows the configuration of VLAN-level IGMP snooping settings.

Example

Configure IGMP snooping globally:

```
switch(config)# ip igmp snooping all-vlans
```

Disable global IGMP snooping:

```
switch(config)# no ip igmp snooping all-vlans
```

Command History

Release	Modification
10.16.1000	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip igmp snooping apply access list

Syntax

```
ip igmp snooping apply access list <ACL-NAME>  
no ip igmp snooping apply access list <ACL-NAME>
```

Description

Configures the access list (ACL) in a particular interface to filter IGMP join or leave packets based on rules set in a particular access list name. The **no** form of this command removes the configuration.

Parameter	Description
<ACL-NAME>	Specifies the access list name.

Usage

Existing classifier commands are used to configure ACL. In case of IGMPv3 packets with multiple group addresses received, only permitted group addresses based on the ACL rule set are processed. The packet is forwarded to querier and PIM router even though one of the groups present in the packet is blocked by

ACL to avoid the delay in learning of the permitted groups because the access switch configured with the ACL blocks the traffic for the groups which are denied forwarding of joins have no impact. If all of the groups in a packet are denied by the ACL rule packet, it is not forwarded to the querier and PIM router. If the ACE has the source address configured, the source address in the IGMPv3 report is matched against the ACL and corresponding action is taken. Existing joins timeout.

With IGMPv2, if there is no match or if there is a deny rule match, the packet is dropped.



NOTE

If the access list is configured for both L2 VLAN and L3 VLAN, then the L3 VLAN configuration is applied.

Example

Configure the access list:

```
switch(config) # vlan 2
switch(config-vlan-2) # ip igmp snooping apply access-list mygroup
```

Remove the access list:

```
switch(config) # vlan 2
switch(config-vlan-2) # no ip igmp snooping apply access-list mygroup
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platform	config-vlan	Administrators or local user group members with execution rights for this command.

ip igmp snooping filter unknown mcast

Syntax

```
ip igmp snooping filter-unknown-mcast
no ip igmp snooping filter-unknown-mcast
```

Description

Enables the avoidance of initial flooding of unknown multicast traffic on IGMP-snooping-enabled VLANs.

The no form of this command returns to the default behavior of initial flooding of unknown multicast traffic.

Usage

In the default behavior, the unknown multicast traffic is flooded until the IP Multicast Flow programming is done on the hardware. This is known as initial flooding of unknown multicast. Use this command to filter unknown multicast instead of flooding.



NOTE

Initial flooding of multicast traffic is observed for a few seconds after the device comes up from a reboot. This issue is only seen when the multicast source connected device is rebooted. Once the device is up after a reboot, it takes a few seconds for the CPU Rx rule to be programmed during the timeframe that the initial flooding is observed. This is an expected behavior.

Example

Configure the unknown multicast to steal globally on IGMP snooping enabled VLANs.

```
switch# configure terminal
switch(config)# ip igmp snooping filter-unknown-mcast
```

Removing the configuration of the unknown multicast to steal globally on IGMP snooping enabled VLANs.

```
switch# configure terminal
switch(config)# no ip igmp snooping filter-unknown-mcast
```

Command History

Release	Modification
10.14	Command introduced on the 4100i, 6000, and 6100 Switch series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ip igmp snooping preprogram-starg-flow

Syntax

```
ip igmp snooping preprogram-starg-flow {enable | disable}
```

Description

Configures preprogramming of the starg (*,G) flow feature on the IGMP snooping enabled VLAN.

Parameter	Description
enable	Enables preprogramming starg flows on the VLAN.
disable	Disables preprogramming starg flows on the VLAN.

Usage

When this feature is enabled, a summarized multicast bridge entry is programmed into the hardware table when an IGMPv2 or IGMPv3 join is received on the IGMP snooping enabled VLAN. This enables multicast flow to be programmed in the hardware before the data packet arrives for a multicast flow. If an unknown packet is received for a multicast flow, having this feature enabled triggers programming of starg entry in the hardware on selected platforms, which is helpful in optimizing hardware resource utilization and PIM registration in deployments where a L2 device is connected along the PIM registration path.

This feature is currently supported for IGMPv2 and IGMPv3 joins, so IGMPv3 joins that are sent for a specific source are treated as IGMPv2 joins and summarized entry is programmed in the corresponding hardware.

Preprogramming of (*,G) flows is supported only on the IGMP snooping enabled VLANs. If IGMP snooping is disabled on a VLAN, this feature is auto-disabled.

This feature is currently supported for IGMPv2 and IGMPv3 joins, as a result, summarized multicast flow is programmed in advance when an IGMPv2 join or IGMPv3 join for a specific group is received. For IGMPv3 deployments, traffic from all sources for a specific multicast group is sent to all clients, regardless of whether they send IGMPv2 or IGMPv3 joins for this group. Keeping this feature disabled is recommended on VLANs where traffic from the specific source is only expected for the IGMPv3 clients.

On the HPE Aruba Networking 4100i, 6000, and 6100 Switch series, a single (*,G) entry is programmed in advance for each join received and data driven programming of SG entries occurs when the traffic is received from a specific source for this group. As a result, the (*,G) entry is used to forward the initial traffic and the SG entries programmed in the multicast forwarding table are used to forward subsequent traffic to all of the clients (that have sent either IGMPv2 joins or IGMPv3 joins) for all of the active joins in the feature enabled VLANs.

When an unknown multicast packet is received on a VLAN where this feature is enabled, it triggers programming of a (*,G) entry in the hardware instead of the SG.

It is highly recommended to not enable this feature on devices where PIM or L3 multicast routing is enabled as it can lead to issues like permanent traffic loss.

Configuring this feature on devices where there are multiple sources sending traffic for the same group address is recommended.

This feature is mutually exclusive with the IGMP snooping static group feature.

Example

Enable preprogramming multicast (*,G) flows:

```
switch(config) # vlan 2
switch(config-vlan-2) # ip igmp snooping preprogram-starg-flow enable
```

Disable preprogramming multicast (*,G) flows:

```
switch(config) # vlan 2
switch(config-vlan-2) # ip igmp snooping preprogram-starg-flow disable
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-vlan	Administrators or local user group members with execution rights for this command.

ip igmp snooping querier-address

Syntax

```
ip igmp snooping querier-address <QUERIER-IP-ADDRESS>  
no ip igmp snooping querier-address <QUERIER-IP-ADDRESS>
```

Description

Configures the querier IP address for the IGMP-MLD proxy feature in EVPN.

The **no** form of this command removes the configuration.

Parameter	Description
<QUERIER-IP-ADDRESS>	Specifies the proxy querier IP address.

Usage

Fabrics running the IGMP-MLD Proxy feature must perform distributed querier functionality on every VTEP where the VLAN is extended. Typically, a valid IP address must be configured on the SVI for IGMP querier functionality. However, in centralized routing scenarios within the fabric, most VTEPs only have Layer 2 VLANs. In such cases, IGMP querier functionality needs to be enabled on those VTEPs. This configuration provides a valid querier IP address for those Layer 2 VLANs.

This command is used in conjunction with the IGMP-MLD proxy feature of EVPN.



NOTE

This is an optional command applicable to VTEPs that have Layer 2 VLANs (without an IP address) and requires IGMP-MLD proxy querier functionality.

If a Layer 3 SVI with an IP address is configured for the VLAN, it takes precedence. In the absence of a valid IP address or SVI, the given IP address is used for querier operations. This configuration is global and applies to all tenants' Layer 2 VLANs where the IGMP-MLD proxy feature is enabled. Fabrics running the **igmp-mlD proxy** feature must perform distributed querier functionality on every VTEP where the VLAN is extended. Typically, a valid IP address must be configured on the SVI for IGMP querier functionality needs to be enabled on those VTEPs. This configuration provides a valid querier IP address for those Layer 2 VLANs. This configuration is applicable only when **igmp-mlD-proxy** configuration is configured under EVPN.

In the event of underlay tunnel fluctuations, the querier state reinitializes and transitions back to the querier state only after the default querier wait of 260 seconds.

This configuration is not applicable to VLANs that are not mapped to a VNI.



NOTE

The specified IP address does not need to be assigned to any interface.

Example

Configure the querier IP address for the IGMP-MLD proxy feature in EVPN.

```
switch(config) # ip igmp snooping querier-address 10.10.10.1
```

Remove the configuration of the querier IP address for the IGMP-MLD proxy feature in EVPN:

```
switch(config) # no ip igmp snooping querier-address 10.10.10.1
```

Command History

Release	Modification
10.16	Command introduced.

Command Information

Platforms	Command context	Authority
All platform	config	Administrators or local user group members with execution rights for this command.

ip igmp snooping report-suppression

Syntax

```
ip igmp snooping report-suppression enable|disable
```

Description

Enable IGMP report suppression to minimize unnecessary multicast control traffic by aggregating and suppressing duplicate reports for the same group from multiple clients within a VLAN. IGMP Report Suppression only works with IGMPv2 reports, not IGMPv3. However, if IGMPv3 is explicitly configured, this feature still will suppress reports for IGMPv2 join and leave messages.

Usage

This feature is enabled by default.

Example

Enabling IGMP report suppression:

```
switch(config)# vlan 100
switch(config-vlan)# ip igmp snooping enable
switch(config-vlan)# ip igmp snooping report-suppression enable
```

Disabling IGMP report suppression:

```
switch(config-vlan)# ip igmp snooping report-suppression disable
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ip igmp snooping static group

Syntax

```
ip igmp snooping static group <GROUP-NAME>
no ip igmp snooping static group <GROUP-NAME>
```

Description

Configures static multicast group. The **no** form of this command removes the configuration.

Parameter	Description
<code><GROUP-NAME></code>	Specifies the group name.

Example

Configure static multicast group on group 239.1.1.1:

```
switch(config) # vlan 2
switch(config-vlan-2) # ip igmp snooping static-group 239.1.1.1
```

Remove static multicast group on group 239.1.1.1:

```
switch(config) # vlan 2
switch(config-vlan-2) # no ip igmp snooping static-group 239.1.1.1
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	<code>config-vlan</code>	Administrators or local user group members with execution rights for this command.

ip igmp snooping vxlan replicate-all

Syntax

```
ip igmp snooping vxlan replicate-all
no ip igmp snooping vxlan replicate-all
```

Description

This command enables AOS-CX leaf devices (access-layer switches that connect directly to servers, storage, and other endpoints) to unconditionally forward multicast traffic to all VTEPs that do not support the exchange of IGMP reports over the fabric. By default, when IGMP snooping is enabled in AOS-CX, multicast traffic is selectively forwarded only to VTEPs connected to have interested hosts. This is achieved by exchanging IGMP reports over VXLAN tunnels where each VTEP processes the remote reports and uses the membership information to forward multicast traffic efficiently, conserving fabric bandwidth. However, in a multi-vendor deployment, some VTEPs may not support the exchange of IGMP reports, which can result in the switch failing to learn the complete group membership, and can cause subsequent multicast traffic loss. This command ensures that multicast traffic sourced from the leaf switch is always replicated to all VTEPs, regardless of report exchange capability, thereby maintaining service continuity in mixed-vendor environments.

Usage

When this configuration is applied, multicast traffic sourced from hosts attached to the local VTEP is unconditionally replicated to third-party VTEPs. This ensures that receivers connected to those remote VTEPs can receive the multicast streams. It is expected that the remote VTEP also replicates its locally sourced multicast traffic back to the local VTEP. Based on the local IGMP reports, the traffic will then be forwarded to the corresponding host ports. Therefore, before you issue the **ip igmp snooping vxlan replicate-all** command, ensure that an IGMP querier is configured in the same VTEP to maintain group memberships.



NOTE

While this command supports multicast traffic exchange across VTEPs in deployments where the source and receiver VLANs reside within the same subnet, it does not enable inter-VLAN multicast routing. Routing multicast traffic between VLANs still requires valid group membership reports from the peer VTEPs, and in the absence of such reports, traffic will not be routed from the source VLAN to the receiver VLAN. The configuration created by this command is not applicable when operating in standards-based IGMP-MLD Proxy mode and is supported only in the AOS-CX native IGMP/MLD mode over VXLANs.

Example

```
switch(config)# ip igmp snooping vxlan replicate-all
switch(config)# no ip igmp snooping vxlan replicate-all
```

Release

Modification

10.17.1000

Command introduced

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show ip igmp snooping

Syntax

```
show ip igmp snooping
  counters
  detail
  groups [vlan <vlan-id>]
  no ...
  packet-exceptions
  static-groups
  statistics
  vlan <vlan-id> [group {<ip-addr> [client_details]}|{port <IF-NAME>}|
{vtep-peer <A.B.C.D>}]
  vsx-peer
```

Description

Shows IGMP snooping configuration information and status for all VLANs. Specify a VLAN ID or a VLAN and a group to display details for only that VLAN or VLAN group.

Parameter	Description
counters	Shows IGMP query packets transmitted (Tx), received (Rx), and error packet counters.
detail	Shows IGMP Snooping details for all VLANs, including joined ports or VXLAN tunnel endpoints (VTEPs) for each group in the VLAN.
groups	Shows IGMP snooping groups information. Include the optional vlan <vlan - id> parameter to display information for groups on a specific VLAN.

Parameter	Description
<code>no ...</code>	Negates any configured parameter.
<code>packet-exceptions</code>	Troubleshoot issues in L2 multicast bridge entries for data packets forwarded to the CPU.
<code>static-groups</code>	Shows MLD snooping static group details, including the number of static groups joined.
<code>statistics</code>	Shows MLD snooping statistics.
<code>vlan <vlan-id></code>	Shows IGMP snooping protocol information and number of different groups joined for the VLAN.
<code>group</code>	Shows IGMP snooping group information for the specified VLAN, including the number of different groups joined for the VLAN. Identify the group by IP address or interface name.
<code><ip-addr> [client-details]</code>	Shows IGMP snooping group address information. Include the optional client details parameter to display IGMP snooping client details.
<code>port <IF-NAME></code>	Shows IGMP snooping group information for the interface name in member/slot/port format.
<code>vtep-peer <A.B.C.D></code>	Shows IGMP snooping info for the specified VTEP.

Examples

Showing IGMP snooping configuration and status:

```
switch# show ip igmp snooping
IGMP Snooping Protocol Info
Total VLANs with IGMP enabled           : 1
IGMP Drop Unknown Multicast             : Global
Configured IGMP Proxy Querier address    :10.10.10.1
VLAN ID : 1
VLAN Name : DEFAULT_VLAN_1
IGMP Snooping is not enabled
VLAN ID : 2
VLAN Name : VLAN2
IGMP Configured Version : 3
IGMP Operating Version : 3
IGMP preprogram-starg-flow is operational
Querier Address [this switch] : 20.1.1.1
Querier Port :
```

```
Querier UpTime :0m 21s
Querier Expiration Time :0m 2s
```

Include the **detail** parameter for additional information on joined ports or VTEPs, as shown in the example below:

```
switch# show ip igmp snooping detail
IGMP Snooping Protocol Info
Total VLANs with IGMP enabled           : 1
Current count of multicast groups joined : 4
IGMP Drop Unknown Multicast             : Global
VLAN ID : 100
VLAN Name : VLAN100
IGMP Configured Version : 3
IGMP Operating Version : 3
IGMP preprogram-starg-flow is not operational
Querier Address [this switch] : 15.1.1.1
Querier Port :
Querier UpTime :9m 32s
Querier Expiration Time :0m 10s
Router Detected Port(s) :
Active Group Address   Tracking   Vers  Mode  Uptime   Expires   Ports/Vteps
-----
225.1.1.1              Filter    3     EXC   1m 2s    0s
200.1.1.1,200.1.1.2
1/6/22
225.1.1.2              Filter    3     EXC   1m 2s    0s
200.1.1.1,200.1.1.2
1/6/22
226.1.1.1              Filter    3     EXC   1m 4s    0s
200.1.1.3
226.1.1.2              Filter    3     EXC   1m 4s    0s
200.1.1.3
```



NOTE

With standard based IGMP-MLD Proxy mode, the membership reports from remote VTEPs does not have an expiry timer associated and shall be shown as 0 seconds.

Command History

Release	Modification
10.16	Updated example to include new IGMP proxy querier address.
10.13	Programming starg flow is now supported.

Release	Modification
10.10	The packet - exceptions parameter is introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

sFlow

sFlow is a technology for monitoring traffic in switched or routed networks. The sFlow monitoring system is comprised of:

- An sFlow Agent that runs on a network device, such as a switch. The agent uses sampling techniques to capture information about the data traffic flowing through the device and forwards this information to an sFlow collector.
- An sFlow Collector that receives monitoring information from sFlow agents. The collector stores this information so that a network administrator can analyze it to understand network data flow patterns.



NOTE

The sFlow UDP datagrams sent to a collector are not encrypted, therefore any sensitive information contained in an sFlow sample is exposed.

Subtopics

[sFlow agent](#)

[Configuring the sFlow agent](#)

[sFlow scenario](#)

[sFlow scenario 2](#)

[sFlow agent commands](#)

sFlow agent

The sFlow agent on the switch provides ingress sampling of all forwarded layer 2 and layer 3 traffic on LAG and Ethernet ports. High-availability is supported (packet sampling continues to work after switch-over).

The sFlow agent can communicate with up to three sFlow collectors at the same time. The agent communicates with collectors on the default VRF and non-default VRF.

Although you can configure very high sampling rates, the switch may drop samples if it cannot handle the rate of sampled packets. High sampling rates may also cause high CPU usage resulting in control plane performance issues.

A single sFlow datagram sent to a collector contains multiple flow and counter samples. The total number of samples an sFlow datagram can contain varies depending on the settings for header size and maximum datagram size.

Default settings

- sFlow is disabled on all interfaces.
- Collector port: UDP port 6343.
- sflow sampling mode: Ingress
- Sampling rate: 4096.
- Polling interval: 30 seconds.
- Header size: 128 bytes.
- Max datagram size: 1400 bytes.

Supported features

- Global sampling rate
- Interface counters polling
- Agent IP configuration for IPv4 and IPv6
- Header size configuration
- Max datagram size configuration
- Ingress sampling for all forwarded traffic (L2, L3)
- Enable/Disable sFlow per interface
- Support for three remote collectors
- An out-of-band collector can be defined on the management VRF
- A collector can be defined on the default and non-default VRF

- Sampling on Ethernet and LAG interfaces
- High availability support (sampling continues to work after switch-over)
- Source IP support (setting source IP for sFlow datagrams sent to a remote collector)



NOTE

For the 4100i, 6000, and 6100 switch series, collector can be defined only on the default VRF. Also, there is no sampling of egress traffic.

Limitations

- Sampling rate cannot be set per interface (global only)
- sFlow is not configurable through SNMP.

Configuring the sFlow agent

Procedure

1. Configure one or more sFlow collectors with the command `sflow collector`. This determines where the sFlow agent sends sFlow information.
2. Enable the sFlow agent on all interfaces, or on a specific interface, with the command `sflow`.
3. Define the address of the sFlow agent with the command `sflow agent-ip`.
4. By default, the source IP address for sFlow datagrams is set to the IP address of the outgoing switch interface on which the sFlow client is communicating with a collector. Since the switch can have multiple routing interfaces, datagrams can potentially be sent on different paths at different times, resulting in different source IP addresses for the same client. To resolve this issue, define a single source IP address. For details, see Single source IP address in the Fundamentals Guide.
5. For most deployments, the default values for the following settings do not need to be changed. If your deployment requires different settings, change the default values with the indicated commands:

sFlow setting	Default value	Command to change it
Rate at which packets are sampled.	1 in every 4096 packets	<code>sflow sampling</code>

sFlow setting	Default value	Command to change it
Rate at which the switch sends data to an sFlow collector.	30 seconds	<code>sflow polling</code>
Size of the sFlow header.	128 bytes	<code>sflow header-size</code>
Maximum size of an sFlow datagram.	1400 bytes	<code>sflow max-datagram-size</code>

6. Review sFlow configuration settings with the command `show sflow`.

Example

This example creates the following configuration:

- Configures an sFlow collector with the IP address **10.10.20.209**.
- Enables the sFlow agent on all interfaces.
- Defines the sFlow agent IP address to be **10.10.1.5**.

```
switch(config)# sflow collector 10.10.20.209
switch(config)# sflow
switch(config)# sflow agent-ip 10.0.0.1
```

sFlow scenario

In this scenario, two hosts send sFlow traffic through a switch to an sFlow collector.

Procedure

1. Enable sFlow globally.
switch# **config** switch(config)# **sflow**
2. Set the sFlow agent IP address to **10.10.12.1**.
switch(config)# **sflow agent-ip 10.10.12.1**
3. Set the sFlow collector IP address to **10.10.12.2**.
switch(config)# **sflow collector 18.2.2.2**
4. Configure sFlow sampling rate and polling interval.
switch(config)# **sflow sampling 5000** switch(config)# **sflow polling 20**

5. Configure interface **vlan 2** with IP address **10.10.10.1/24**.
switch(config)# **interface vlan 2** switch(config-if-vlan)# **no shutdown** switch(config-if-vlan)# **ip address 10.10.10.1/24** switch(config)# **quit**
6. Configure interface **vlan 3** with IP address **10.10.11.1/24**.
switch(config)# **interface vlan 3** switch(config-if-vlan)# **no shutdown** switch(config-if-vlan)# **ip address 10.10.11.1/24** switch(config)# **quit**
7. Configure interface **vlan 4** with IP address **10.10.12.1/24**.
switch(config)# **interface vlan 4** switch(config-if-vlan)# **no shutdown** switch(config-if-vlan)# **ip address 10.10.12.1/24** switch(config)# **quit**
8. Verify sFlow configuration

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                               enabled
Collector IP/Port/Vrf               10.10.10.2/6343/default
Agent Address                       10.0.0.1
Sampling Rate                       1024
Polling Interval                    30
Header Size                         128
Max Datagram Size                   1400
sFlow Status
```

```
-----
Running - Yes
sFlow enabled on Interfaces:
```

```
-----
lag100
sFlow Statistics
```

```
-----
Number of Samples                   200
```

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                               enabled
Collector IP/Port/Vrf               10.10.10.2/6343/default
Agent Address                       10.0.0.1
Sampling Rate                       1024
Polling Interval                    30
Header Size                         128
Max Datagram Size                   1400
Sampling Mode                       both
```

```
sFlow Status
-----
Running - Yes
sFlow enabled on Interfaces:
-----
lag100
sFlow Statistics
-----
Number of Ingress Samples      200
Number of Egress Samples      0
```

sFlow scenario 2

In this scenario, two hosts connected to different switches send sFlow traffic to a collector. A LAG is used to connect the two switches.

Procedure

1. Configure switch 1.

a. Enable sFlow globally.

```
switch# config
switch(config)# sflow
```

b. Set the sFlow agent IP address to **10.10.12.1**.

```
switch(config)# sflow agent-ip 10.10.12.1
```

c. Set the sFlow collector IP address to **10.10.12.2**.

```
switch(config)# sflow collector 10.10.12.2
```

d. Configure sFlow sampling rate and polling interval.

```
switch(config)# sflow sampling 5000
switch(config)# sflow polling 10
```

e. Create VLAN **8**.

```
switch(config)# vlan 8
switch(config-vlan-8)# no shutdown
switch(config)# exit
```

f. Define LAG **100** and assign VLAN **vlan 8** to it.


```
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown

switch(config-lag-if)# vlan access 8
switch(config-lag-if)# lacp mode active
```

- g. Configure interface **vlan 5**.

```
switch(config)#
                    interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-lag-if)# no routing
switch(config-if-vlan)# vlan access 8
```

- h. Configure VLANs **4** and **5** as members of LAG **100**.

```
switch# (config)#
                    interface vlan 4
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# lag 100
switch(config-if-vlan)# exit
switch(config)#
                    interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# lag 100
switch(config-if-vlan)# exit
```

- i. Configure interface **vlan 5** with IP address **10.10.12.1/24**.

```
switch# (config)#
                    interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.12.1/24
switch(config-if-vlan)# quit
```

- j. Verify sFlow configuration.

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                      enabled
Collector IP/Port/Vrf      10.10.10.2/6343/default
Agent Address              10.0.0.1
Sampling Rate              1024
```

```

Polling Interval          30
Header Size              128
Max Datagram Size        1400
sFlow Status
-----
Running - Yes
sFlow enabled on Interfaces:
-----
lag100
sFlow Statistics
-----
Number of Samples        200
switch# show sflow
sFlow Global Configuration
-----
sFlow                    enabled
Collector IP/Port/Vrf    10.10.10.2/6343/default
Agent Address            10.0.0.1
Sampling Rate            1024
Polling Interval         30
Header Size              128
Max Datagram Size        1400
Sampling Mode            both
sFlow Status
-----
Running - Yes
sFlow enabled on Interfaces:
-----
lag100
sFlow Statistics
-----
Number of Ingress Samples 200
Number of Egress Samples  0

```

2. Configure switch 2.

a. Create VLAN **8**.

```

switch(config)# vlan 8
switch(config-vlan-8)# no shutdown
switch(config)# exit

```

b. Define LAG **100** and assign VLAN **vlan 8** to it.

```
switch(config)# interface lag 100  
switch(config-lag-if)# no shutdown  
  
switch(config-lag-if)# vlan access 8  
switch(config-lag-if)# lacp mode active
```

- c. Configure interface **1/1/1**.

```
switch(config)# interface 1/1/1  
switch(config-if)# no shutdown  
  
switch(config-if)# vlan access 8
```

- d. Configure interface **1/1/2** and **1/1/3** as members of LAG **100**.

```
switch# (config)#interface 1/1/2  
switch(config-if)# no shutdown  
switch(config-if)# lag 100  
switch(config-if)# exit  
switch(config)-if# interface 1/1/3  
switch(config-if)# no shutdown  
switch(config-if)# lag 100  
switch(config-if)# exit
```

sFlow agent commands

Select a command from the list in the left navigation menu.

Subtopics

[clear sflow statistics](#)

[sflow](#)

[sflow agent-ip](#)

[sflow collector](#)

[sflow disable](#)

[sflow header-size](#)

[sflow max-datagram-size](#)

[sflow polling](#)

[sflow sampling](#)

[show sflow](#)

clear sflow statistics

Syntax

```
clear sflow statistics {global | interface <INTERFACE-NAME>}
```

Description

This command clears the sFlow sample statistics counter to 0 either globally or for a specific interface.

Parameter	Description
global	Specifies all interfaces on the switch.
interface <INTERFACE-NAME>	Specifies the name of an interface on the switch.

Examples

Clearing the global sFlow sample statistics counter to 0 globally:

```
switch(config)# clear sflow statistics global
```

Clearing the global sFlow sample statistics counter to 0 for interface 1/1/1:

```
switch(config)# clear sflow statistics interface 1/1/1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow

Syntax

```
sflow  
no sflow
```

Description

Enables the sFlow agent.

- In the **config** context, this command enables the sFlow agent globally on all interfaces.
- In an **config-if** context, this command enables the sFlow agent on a specific interface. sFlow cannot be enabled on a member of a LAG, only on the LAG.

The sFlow agent is disabled by default.

The **no** form of this command disables the sFlow agent and deletes all sFlow configuration settings, either globally, or for a specific interface.

Examples

Enabling sFlow globally on all interfaces:

```
switch(config) # sflow
```

Disabling sFlow globally on all interfaces:

```
switch(config) # no sflow
```

Enabling sFlow on interface **1/1/1**:

```
switch(config) # interface 1/1/1  
switch(config-if) # sflow
```

Disabling sFlow on interface **1/1/1**:

```
switch(config) # interface 1/1/1  
switch(config-if) # no sflow
```

Enabling sFlow on interface **lag100**:

```
switch(config) # interface lag100  
switch(config-if) # sflow
```

Disabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# no sflow
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config config-if	Administrators or local user group members with execution rights for this command.

sflow agent-ip

Syntax

```
sflow agent-ip <IP-ADDR>
no sflow agent-ip [<IP-ADDR>]
```

Description

Defines the IP address of the sFlow agent to use in sFlow datagrams. This address must be defined for sFlow to function. HPE recommends that the address:

- can uniquely identify the switch
- is reachable by the sFlow collector
- does not change with time

The **no** form of this command deletes the IP address of the sFlow agent. This causes sFlow to stop working and no datagrams will be sent to the sFlow collector.

Parameter	Description
	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number f

Parameter	Description
<IP-ADDR>	rom 0 to F. The agent address is used to identify the switch in all sFlow datagrams sent to sFlow collectors. It is usually set to an IP address on the switch that is reachable from an sFlow collector.

Examples

Setting the agent address to **10.10.10.100**:

```
switch(config)# sflow agent-ip 10.0.0.100
```

Setting the agent address to **2001:0db8:85a3:0000:0000:8a2e:0370:7334**:

```
switch(config)# sflow agent-ip 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Removing the address configuration from the switch, which results in sFlow being disabled:

```
switch(config)# no sflow agent-ip
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow collector

Syntax

```
sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
no sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

Description

Defines a collector to which the sFlow agent sends data. Up to three collectors can be defined. At least one collector should be defined, and it must be reachable from the switch for sFlow to work.

Parameter	Description
<code>collector <IP-ADDR></code>	Specifies the IP address of a collector in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F.
<code>port <PORT></code>	Specifies the UDP port on which to send information to the sFlow collector. Range: 0 to 65536. Default: 6343.
<code>vrf <VRF></code>	Specifies the VRF on which to send information to the sFlow collector. The VRF must be defined on the switch. If no VRF is specified, the default VRF (default) is used.

Example

Defining a collector with IP address **10.10.10.100** on UDP port **6400**:

```
switch(config)# sflow collector 10.0.0.1 port 6400
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

`sflow disable`

Syntax

```
sflow disable
```

Description

Disables the sFlow agent, but retains any existing sFlow configuration settings. The settings become active if the sFlow agent is re-enabled.

Example

Disabling sFlow support:

```
switch(config)# sflow disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow header-size

Syntax

```
sflow header-size <SIZE>  
no sflow header-size [<SIZE>]
```

Description

Sets the sFlow header size in bytes.

The **no** form of this command sets the header size to the default value of 128.

Parameter	Description
header-size <SIZE>	Specifies the sFlow header size in bytes. Range: 64 to 256. Default: 128.

Examples

Setting the header size to **64** bytes:

```
switch(config) # sflow header-size 64
```

Setting the header size to the default value of **128** bytes:

```
switch(config) # no sflow header-size
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sf flow max-datagram-size

Syntax

```
sf flow max-datagram-size <SIZE>
no sf flow max-datagram-size [<SIZE>]
```

Description

Sets the maximum number of bytes that are sent in one sFlow datagram.

The **no** form of this command sets maximum number of bytes to the default value of 1400.

Parameter	Description
<code>max-datagram-size <SIZE></code>	Specifies the maximum datagram size in bytes. Range: 1 to 9000. Default: 1400.

Examples

Setting the datagram size to **1000** bytes:

```
switch(config) # sflow max-datagram-size 1000
```

Setting the header size to the default value of **1400** bytes:

```
switch(config) # no sflow max-datagram-size
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

sflow polling

Syntax

```
sflow polling <INTERVAL>
no sflow polling [<INTERVAL>]
```

Description

Defines the global polling interval for sFlow in seconds.

The **no** form of this command sets the polling interval to the default value of 30 seconds.

Parameter	Description
<code><INTERVAL></code>	Specifies the polling interval in seconds. Range: 10 to 3600. Default: 30.

Examples

Setting the polling interval to 10:

```
switch(config)# sflow polling 10
```

Setting the polling interval to the default value.

```
switch(config)# no sflow polling
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow sampling

Syntax

```
sflow sampling <RATE>
no sflow sampling [<RATE>]
```

Description

Defines the global sampling rate for sFlow in number of packets. The default sampling rate is 4096, which means that one in every 4096 packets is sampled. A warning message is displayed when the sampling rate is set to less than 4096 and proceeds only after user confirmation.

The **no** form of this command sets the sampling rate to the default value of 4096.

Parameter	Description
sampling <RATE>	Specifies the sampling rate. Range: 1 to 16777215. Default: 4096.

Examples

Setting the sampling rate to **5000**:

```
switch(config)# sflow sampling 5000
```

Setting the sampling rate to the default:

```
switch(config)# no sflow sampling
```

Setting the sampling rate to **1000**:

```
switch(config)# sflow sampling 1000
Setting the sFlow sampling rate lower than 4096 is not recommended and might
affect system performance.
Do you want to continue [y/n]? y
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show sflow

Syntax

```
show sflow [interface <INTERFACE-NAME>]
```

Description

Shows sFlow configuration settings and statistics for all interfaces, or for a specific interface. It also displays the current status of sFlow on the device and reports any errors that require attention.



NOTE

If sFlow is enabled on the interfaces associated with a lag interface, then the interfaces will not be shown as separate entries under `sFlow enabled on Interface` in the output. Only the associated lag interface will have an entry in the column.

Parameter

Description

interface <INTERFACE-NAME>

Specifies the name of an interface on the switch.

Examples

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                      enabled
Collector IP/Port/Vrf      10.0.0.2/6343/default
                           10.0.0.3/6400/default
Agent Address              10.0.0.1
Sampling Rate              1024
Polling Interval           30
Header Size                128
Max Datagram Size          1400
Sampling Mode              ingress
sFlow Status
-----
Running - Yes
sFlow enabled on Interfaces:
-----
1/1/2
1/1/3
```

```

lag100
sFlow Statistics
-----
Number of Ingress Samples      200
Number of Egress Samples      120

```

Showing sFlow information for all interfaces:

```

switch# show sflow
sFlow Global Configuration
-----
sFlow                          enabled
Collector IP/Port/Vrf          10.10.10.2/6343/default
Agent Address                  10.0.0.1
Sampling Rate                  1024
Polling Interval               30
Header Size                    128
Max Datagram Size              1400
sFlow Status
-----
Running - Yes
sFlow enabled on Interfaces:
-----
lag100
sFlow Statistics
-----
Number of Samples              200

```

```

switch# show sflow interface 1/1/1
sFlow configuration - Interface 1/1/1
-----
sFlow                          enabled
Sampling Rate                  1024
Sampling Mode                  both
Number of Ingress Samples      81
Number of Egress Samples       20
sFlow Sampling Status          success

```

Showing sFlow information for interface **1/1/1**:

```

switch# show sflow interface 1/1/1
sFlow configuration - Interface 1/1/1
-----
sFlow                          enabled
Sampling Rate                  1024
Number of Samples              30
sFlow Sampling Status          success

```

```

switch# show sflow interface lag 10
sFlow Configuration - Interface lag10
-----
sFlow                               enabled
Sampling Rate                       4096
Sampling Mode                       both
Number of Ingress Samples           0
Number of Egress Samples            0
sFlow Sampling Status               error
Sampling Status on LAG members
-----
Intf 1/1/2                          no agent
Intf 1/1/3                          no agent

```

Showing sFlow information for interface **lag 10**:

```

switch# show sflow interface lag 10
sFlow Configuration - Interface lag10
-----
sFlow                               enabled
Sampling Rate                       4096
Number of Samples                   0
sFlow Sampling Status               error
Sampling Status on LAG members
-----
Intf 1/1/2                          no agent
Intf 1/1/3                          no agent

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

DHCP

The Dynamic Host Configuration Protocol (DHCP) enables the automatic assignment of IP addresses and other configuration settings to network devices.

On 4100i, 6000 and 6100 series switches, only SVIs are supported (ROP is not supported on these platforms). Hence, you need to first configure the dependent DHCP feature on a vlan, and then configure the vlan as an access vlan for a physical interface.

Example to configure the vlan as an access vlan for a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config-if)# interface vlan 100
switch(config-if-vlan)# <DHCP feature CLI>
```

Subtopics

[**DHCP client**](#)

[**DHCP relay agent**](#)

[**FAQ**](#)

DHCP client

By default, the switch operates as a DHCP client on VLAN 1 allowing it to automatically obtain an IP address from a DHCP server on the network to which it is connected.

Subtopics

[**Protocol and feature details**](#)

[**Supported platform and standards**](#)

[**Configuration task list**](#)

[**Considerations and best practices**](#)

[**Use case**](#)

[**DHCP client commands**](#)

Protocol and feature details

DHCP Client is supported on IPv4.

The IP DHCP assignment with HPE Aruba Networking AOS-CX switch is to allow the following:

- IP switch assignment.
- Managing the initial IP access to the switch.
- Prepare for Zero Touch Provisioning (ZTP) with HPE Aruba Networking Central.



NOTE

Zero Touch Provisioning (ZTP) is only supported for IPV4 in HPE Aruba Networking Central

- Prepare for Zero Touch Provisioning (ZTP) by using DHCP options along with a TFTP server.

Supported platform and standards

The following table list the supported VLANs and ports for the AOS-CX Switch.

Table 1. Supported VLAN and Ports for DHCP
Platform VLAN 1

4100i	Yes
6000	Yes

6100	Yes
------	-----

Configuration task list

To run the DHCP Client do the following:

- Enter the interface VLAN or Management VRF context
- Enable DHCP

Considerations and best practices

The IP DHCP can only be configured on one VLAN per switch.

If an OOB Management port is available, you can configure the DHCP service simultaneously for the VLAN IN Band and the OOB Management port. When both the IN Band data port and the OOB Management port receive a DHCP address, then the DHCP address received on the OOBM port is preferred over the IN Band data port for ZTP.

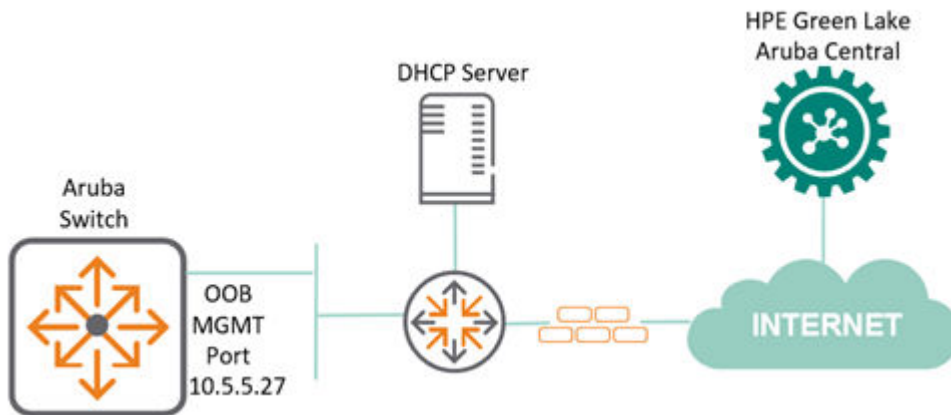
Use case

About this task

IP DHCP assignment with HPE Aruba Networking AOS-CX switches is to allow IP switch assignment and Zero Touch Provisioning using HPE Aruba Networking Central, as shown below.

In the following DHCP client scenario the OOB Management port is in use. The IN Band data ports can also be used for Zero Touch Provisioning (ZTP).

For more information related to the supported ports for the different switches, see [Table 1, Supported VLAN and Ports for DHCP](#).



Procedure

1. Switch configuration from factory on OOB Management port.

```
switch#  
!  
interface mgmt  
    no shutdown  
    ip dhcp  
!
```

2. OOB Management port gets assigned with an IPv4 address 10.5.5.27 from DHCP server.

```
switch# show interface mgmt  
Address Mode                : dhcp  
Admin State                  : up  
Link State                    : up  
Mac Address                   : 88:3a:30:9a:39:01  
IPv4 address/subnet-mask     : 10.5.5.27/24  
Default gateway IPv4         : 10.5.5.254  
IPv6 address/prefix          :  
2a00:23c5:ac8a:3e01:8a3a:30ff:fe9a:3901/64  
IPv6 link local address/prefix: fe80::8a3a:30ff:fe9a:3901/64
```

```
Default gateway IPv6      : fe80::f286:20ff:fe0b:fe5
Primary Nameserver        : 10.5.19.69
Secondary Nameserver      : 10.5.19.79
Tertiary Nameserver       : 8.8.4.4
```

Results

The following output shows a switch connected to HPE Aruba Networking Central allocated with an IP address of 10.5.5.27 on its OOB MGMT port by a DHCP server. Based on its designated configuration in the public cloud, the switch can get its configuration parameters directly from HPE Aruba Networking Central.

```
switch# show hpe_anw-central
  Central admin state      : enabled
  Central location         : device-prod2-
d2.central.arubanetworks.com
  VRF for connection      : mgmt
  Central connection status : connected
  Central source          : activate
  Central source connection status : connected
  Central source last connected on : Wed Sep 22 18:39:05 UTC 2021
  System time synchronized from Activate : True
  Activate Server URL      : devices-v2.arubanetworks.com
  CLI location             : N/A
  CLI VRF                  : N/A
  Source IP                : 10.5.5.27
  Source IP Overridden     : False
  Central support mode     : disabled
```

DHCP client commands

Select a command from the list in the left navigation menu.

Subtopics

[**ip dhcp**](#)

[**ipv6 dhcp**](#)

[**ip dhcp option**](#)

[**ip dhcp preferred-vlan**](#)

[**show ip dhcp**](#)

[**show ipv6 dhcp**](#)

[**show ip dhcp preferred-vlan**](#)

ip dhcp

Syntax

```
ip dhcp
no ip dhcp
```

Description

Enables the DHCP client on any interface VLAN to automatically obtain an IP address from a DHCP server on the network. By default, the DHCP client is enabled on VLAN 1.

The **no** form of the command disables DHCP mode and is supported only on interface VLANs.

Examples

Enabling the DHCP client on the interface vlan 1:

```
switch(config) # interface vlan 1
switch(config-if-vlan) # ip dhcp
switch(config-if-vlan) # no shutdown
```

Disabling the DHCP client on the interface vlan 1:

```
switch(config) # interface vlan 1
switch(config-if-vlan) # no ip dhcp

switch(config) # interface vlan 4
switch(config-if-vlan) # vrf attach red
switch(config-if-vlan) # ip dhcp
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.



NOTE

DHCP client can be enabled on one or more user VLANs.

General Limitations

1. ZTP (Zero Touch Provisioning) options from a DHCP server are only supported on VLAN 1. These options will be ignored if received on any other VLAN.
2. CoP and HTTPS-proxy DHCP options should be received on only one DHCP client-enabled VLAN. If multiple VLANs receive the same option, the selection of the VLAN from which the options are applied is non-deterministic. Therefore, configuring multiple VLANs to receive these options is not recommended.

3. It is recommended to limit the number of DHCP client VLANs in a system to 32. This total includes both DHCPv4 and DHCPv6 client instances.
4. The ip dhcp option broadcast-flag configuration must be enabled to successfully obtain DHCP IP addresses across multiple VLANs when the default gateway route is present.

Command History

Release	Modification
10.17	Support enabled on one or more user VLANs.
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms		Administrators or local user group members with execution rights for this command.

ipv6 dhcp

Syntax

```
ipv6 dhcp
no ipv6 dhcp
```

Description

Configure to enable DHCPv6 mode on a VLAN. In DHCPv6 Mode, the VLAN will get a DHCP-Server allocated IPv6 address along with DNS and NTP attributes.

The **no** form of the command disables DHCP mode and is supported only on interface VLANs.



NOTE

ZTPv6 attributes are included in the request packet, but they are not utilized by ZTP on the data port. Currently, ZTPv6 attributes are only supported on the management port. Also, IPv6 addresses assigned through DHCPv6 only support a /64 prefix.

Examples

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 dhcp
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 dhcp
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 dhcp
```

Enabling the DHCPv6 client on the interface vlan 4 under non-default VRF:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# vrf attach red
switch(config-if-vlan)# ipv6 dhcp
```

Disabling the DHCPv6 client on the interface vlan 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 dhcp
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

General Limitations

1. ZTP (Zero Touch Provisioning) options from a DHCP server are only supported on VLAN 1. These options will be ignored if received on any other VLAN.
2. CoP and HTTPS-proxy DHCP options should be received on only one DHCP client-enabled VLAN. If multiple VLANs receive the same option, the selection of the VLAN from which the options are applied is non-deterministic. Therefore, configuring multiple VLANs to receive these options is not recommended.
3. It is recommended to limit the number of DHCP client VLANs in a system to 32. This total includes both DHCPv4 and DHCPv6 client instances.
4. The `ip dhcp option broadcast-flag` configuration must be enabled to successfully obtain DHCP IP addresses across multiple VLANs when the default gateway route is present.

Command History

Release	Modification
10.17	Support added for DHCPv6 on multiple VLANs.
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.15	Command was introduced.

Command Information

Platforms	Command context	Authority
All platforms		Administrators or local user group members with execution rights for this command.

ip dhcp option

Syntax

```
ip dhcp option [host-name | broadcast-flag]
no ip dhcp option [host-name | broadcast-flag]
```

Description

This command enables the DHCP client host name and broadcast flag globally.

If the **ip dhcp option broadcast-flag** command is enabled, then the DHCP offer and ack packets in the DHCP requests will be treated as broadcast packets. These packets will not be forwarded due to the presence of a default static route.

The **no** form of this command globally disables the host name and DHCP client broadcast flag options.



NOTE

The **ip dhcp option broadcast-flag** command should be configured before configuring the **ip dhcp** command.

Example

Enabling the DHCP client broadcast flag globally:

```

switch(config)# ip dhcp option
broadcast-flag  DHCP Client broadcast-flag option (Default:disabled)
host-name       DHCP Client hostname option
switch(config)# ip dhcp option
% Command incomplete.
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# ip dhcp op
% There is no matched command.
switch(config-if-vlan)# ip dhcp op
Invalid input: ip
switch(config-if-vlan)#

```

Enabling the DHCP client host name globally:

```

switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp option host-name

```

Disabling the DHCP client broadcast flag globally:

```

switch(config)# interface vlan 1
switch(config-if-vlan)# no ip dhcp option broadcast-flag

```

Disabling the DHCP client host name globally:

```

switch(config)# ip dhcp option
broadcast-flag  DHCP Client broadcast-flag option (Default:disabled)
host-name       DHCP Client hostname option
switch(config)# ip dhcp option
% Command incomplete.
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no ip dhcp option host-name
% There is no matched command.
switch(config-if-vlan)# ip dhcp op
Invalid input: ip
switch(config-if-vlan)#

```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.13.1000	Command Introduced

Command Information

Platforms	Command context	Authority
All platforms		Administrators or local user group members with execution rights for this command.

ip dhcp preferred-vlan

Syntax

```
ip dhcp preferred-vlan <VLAN-ID> [vrf <VRF-NAME>]  
no ip dhcp preferred-vlan vlan <VLAN-ID> [vrf <VRF-NAME>]
```

Description

Specifies the preferred VLAN for DHCP client options, including DNS, NTP, and Default Gateway. The preferred VLAN is the VLAN used to select DHCP client options within a specified VRF when multiple VLANs have DHCP client enabled in the same VRF. If no VRF is specified, the configuration applies to the default VRF. In cases where the preferred VLAN is not explicitly configured, the lowest VLAN with DHCP client enabled will be automatically selected.

The **no** form of the command removes the preferred VLAN configuration and restores the default behavior.

Parameter	Description
<VLAN-ID>	VLAN ID to set as preferred VLAN. VLAN identifier range: 1 to 4094.
<VRF-NAME>	Specifies the VRF in which the preferred VLAN should be configured.

General Considerations

1. If the preferred VLAN configuration status is not success, the DHCP client-enabled VLAN with the lowest VLAN ID will be used for DHCP attribute selection.

Use the command `show ip dhcp preferred-vlan` to check the configuration status.

2. The preferred or lowest VLAN will be inactive in the following scenarios:
 - a. When the DHCP interface is in a shutdown state and corresponds to the preferred or lowest VLAN ID.
 - b. When a static IP is configured on an interface that also has DHCP enabled — the static IP takes precedence over DHCP.
 - c. When DHCP attributes are not received from the server on the preferred or lowest VLAN.

Examples

Selecting VLAN 1 as the preferred VLAN:

```
switch(config)# ip dhcp preferred-vlan 1
```

Selecting VLAN 10 as the preferred VLAN with VRF name:

```
switch(config)# ip dhcp preferred-vlan 10 vrf red
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show ip dhcp

Syntax

```
show ip dhcp
```

Description

Displays DHCP IPv4 information on the ports.

Examples

Displaying the DHCP IPv4 information on the ports:

```

switch# show ip dhcp
DHCP Options: Broadcast-flag, Hostname
INTERFACE-NAME    ADDRESS                DEFAULT_GATEWAY    DOMAIN_NAME    VRF
DNS-SERVERS
-----
vlan1              10.254.239.10/27          domain.com         default
50.0.0.2, 50.0.0.3, 50.0.0.4

```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.13.1000	The output parameters, Broadcast - flag and Hostname were introduced.
10.09 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 dhcp

Syntax

```
show ipv6 dhcp
```

Description

This command displays DHCP IPv6 information on the ports.

Examples

Displaying the DHCP IPv6 information on the ports:

```

6410(config)# show ipv6 dhcp
VRF default
-----
Interface vlan1
-----
IPv6 address       : 3013::1050/64
Domain search      : paramv6.org, test1.in, test2.in, test3.in, test4.in,
test5.in
DNS servers        : 3013::100, 3013::101, 3013::102
NTP servers        : 3013::201, 3013::202

```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.15	Command was introduced.

Command Information

Platforms	Command context	Authority
All platforms		Administrators or local user group members with execution rights for this command.

show ip dhcp preferred-vlan

Syntax

```
show ip dhcp preferred-vlan [{vrf <VRF-NAME> | all-vrfs}]
```

Description

Displays the preferred DHCP client VLANs configured on the system.

Parameter	Description
vrf <VRF-NAME>	Show information for a specified VRF.
all-vrf	Show information for all VRFs.

Examples

Displaying the DHCP information for the preferred VLAN and VRF red:

```
switch# show ip dhcp preferred-vlan vrf red
VRF                Preferred-VLAN        Config-status
-----
red                 10                      L3 interface not found
```

Displaying the DHCP information for the all VRFs:

```
switch# show ip dhcp preferred-vlan all-vrfs
VRF                Preferred-VLAN        Config-status
-----
default            1                      Success
red                 9                      VRF Mismatch
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

DHCP relay agent

The function of the DHCP relay agent is to forward the DHCP messages to other subnets so that the DHCP server does not have to be on the same subnet as the DHCP clients. The DHCP relay agent transfers DHCP messages from the DHCP clients located on a subnet without a DHCP server, to other subnets. It also relays answers from DHCP servers to DHCP clients.

Subtopics

- [**Supported platform and standards**](#)
- [**Protocol and feature details**](#)
- [**DHCPv4 relay agent**](#)
- [**DHCPv6 relay agent**](#)

Supported platform and standards

The following table list the supported platforms for DHCPv4 relay and DHCPv6 relay.

Table 1. Support platforms for DHCPv4 relay and DHCPv6 relay

Platform	DHCPv4 Relay
----------	--------------

4100i	Yes
6000	Yes
6100	Yes

Table 2. Scale
Platform

DHCP v4/v6 enabled SVI

4100i	16
6000	16
6100	16

DHCP relay behaviors are as follows:

- DHCPv6 relay is disabled by default.
- DHCP relay hop count is enabled by default.
- In DHCPv4 relay the option 82 policy is replace by default.
- The MAC address of the switch is used as the option 82 remote ID by default.
- In DHCPv6 relay the option 79 policy is disabled by default.
- In DHCPv4 relay the source-interface is disabled by default.

Protocol and feature details

Supported interfaces

The DHCP relay agent is supported on layer 3 VLAN interfaces, and LAG interfaces. DHCP relay is not supported on the management interface.

DHCP server interoperation

Both DHCP relay and DHCP server can be configured on the same VRF.

DHCPv4 relay agent

Hop count in DHCP requests

When a DHCP client broadcasts request, the DHCP relay agent in the switch receives the packets and forwards them to the DHCP server as unicast requests. During this process, the DHCP relay agent increments the hop count before forwarding DHCP packets to the server. The DHCP server, in turn, includes the hop count in the DHCP header in the response sent back to a DHCP client.

DHCP relay option 82

Option 82 is called the relay agent information option. When a DHCP relay agent forwards client-originated DHCP packets to a DHCP server, the option 82 field is inserted/replaced, or the packet with this option is dropped. Servers recognizing the relay agent information option may use the information to implement policies for the assignment of IP addresses and other parameters. The relay agent relays the server-to-client replies to the client.

If a second relay agent is configured to add its own option 82 information, it can encapsulate option 82 information in messages from a first relay agent. The DHCP server uses the option 82 information from both relay agents to decide the IP address for the client.



NOTE

A DHCP relay option 82 source-interface needs to be enabled when a source interface is used for an Intra-VRF deployment. In this deployment type, the client and DHCP server are in same VRF, but the relay will use the source interface.

Configuring a BOOTP/DHCP relay gateway

The DHCP relay agent selects the lowest-numbered IP address on the interface to use for DHCP messages. The DHCP server then uses this IP address when it assigns client addresses. However, this IP address may not be the same subnet as the one on which the client needs the DHCP service. This feature provides a way to configure a gateway address for the DHCP relay agent to use for relayed DHCP requests, rather than the DHCP relay agent automatically assigning the lowest-numbered IP address.



NOTE

DHCP smart relay

The DHCP Smart Relay feature first attempts to use the primary IP address from the client-connected interfaces as a gateway IP address (giaddr) and as an IP address for pool selection. If the DHCP server does not reply to the DHCP discover messages with the primary IP address, the feature attempts to use secondary IP addresses in sequential order from the lowest to the highest. The DHCP Smart Relay forwards

three client discover messages with every IP address configured on the client interface until it receives a response from the server. If the list of IP addresses from the client-connected interface is exhausted or the client times out, the DHCP Smart Relay uses the primary IP address. If the DHCP Smart Relay is not configured, the giaddr is the lowest IP address on the interface.

The DHCP Smart Relay maintains the client cache which includes the information about the client, giaddr, retry count for giaddr, port, the total number of processed discovery messages the timestamp of the discover packets. This client cache is rebuilt upon high availability and VSF switchover, VSX-MM (redundancy) switchover, and when the switch reboots.

Subtopics

Configuring the DHCPv4 relay agent

Use Case

DHCPv4 relay commands

Configuring the DHCPv4 relay agent

Prerequisites

- An enabled layer 3 VLAN

Procedure

1. The DHCPv4 relay agent is enabled by default. If it was previously disabled, enable it with the command `dhcp-relay`.
2. Configure one or more IP helper addresses with the command `ip helper-address`. This determines where the DHCPv4 relay agent forwards DHCP requests. IP helper addresses can be configured on layer 3 interfaces, layer 3 VLAN interfaces, and LAG interfaces.
3. If you want to modify the content of forwarded DHCP packets or drop DHCP packets, configure option 82 support with the command `dhcp-relay option 82`.
4. Define the gateway address that the DHCPv4 relay agent will use with the command `ip bootp-gateway`.
5. If required, enable the hop count increment feature with the command `dhcp-relay hop-count -increment`.
6. Review DHCPv4 relay agent configuration settings with the commands `show dhcp-relay`, `show ip helper-address`, and `show dhcp-relay bootp-gateway`.

Example

This example creates the following configuration:

- Enables the DHCPv4 relay agent.
- Enables interface **1/1/1** and assigns an IPv4 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **10.10.20.209** on the interface.
- Enables DHCP option 82 support and replaces all option 82 information with the values from the switch with the switch MAC address as the remote ID.



NOTE

On 4100i, 6000 and 6100 series switches, only SVIs are supported.

```
switch(config)# dhcp-relay
switch(config)# interface 1/1/1
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface vlan 100
switch(config-if-vlan)# ip address 198.51.100.1/24
switch(config-if-vlan)# ip helper-address 10.10.20.209
switch(config-if-vlan)# exit
switch(config)# dhcp-relay option 82 replace mac
switch# show dhcp-relay
```

DHCP Relay Agent	:	enabled
DHCP Request Hop Count Increment	:	enabled
Option 82	:	disabled
Source-Interface	:	disabled
Response Validation	:	disabled
Option 82 Handle Policy	:	replace
Remote ID	:	mac

```
DHCP Relay Statistics:
```

Valid Requests	Dropped Requests	Valid Responses	Dropped Responses
60	10	60	10

```
DHCP Relay Option 82 Statistics:
```

Valid Requests	Dropped Requests	Valid Responses	Dropped Responses
50	8	50	8

Use Case

This section explain the use cases for DHCPv4 relay agent.

Subtopics

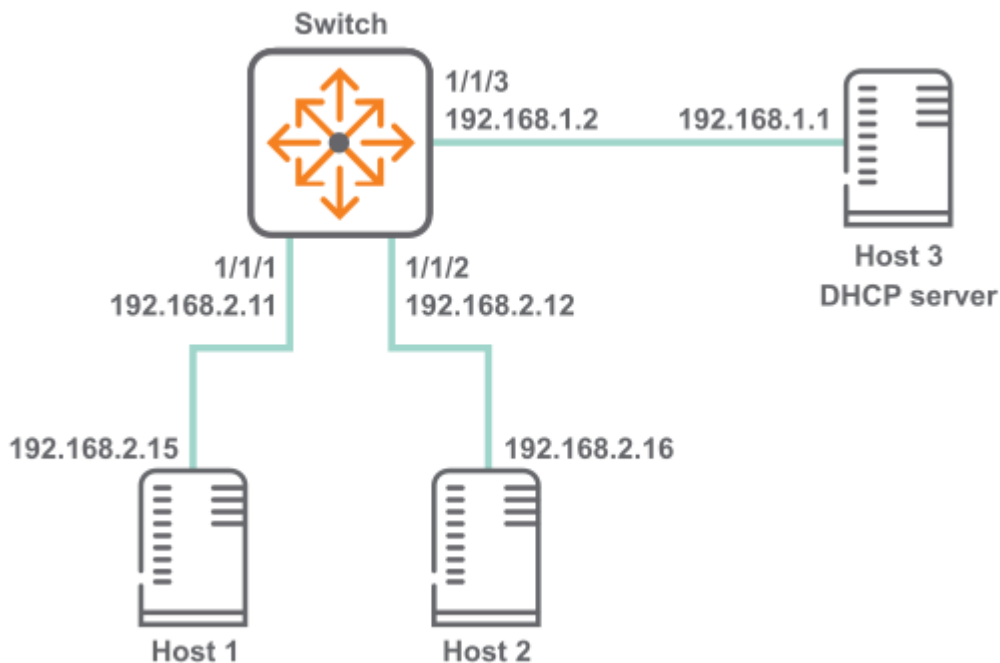
[DHCPv4 relay scenario 1](#)

[DHCPv4 relay scenario 2](#)

[DHCPv4 relay scenario 3](#)

DHCPv4 relay scenario 1

In this scenario, DHCP relay on the switch enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



Procedure

1. DHCP relay is enabled by default. If it was previously disabled, enable it.

```
switch# config
switch(config)# dhcp-relay
```

2. Define an IPv4 helper address on interface vlan 100

```
switch(config)# interface 1/1/1
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# vlan access 100
switch(config-if)# exit
```

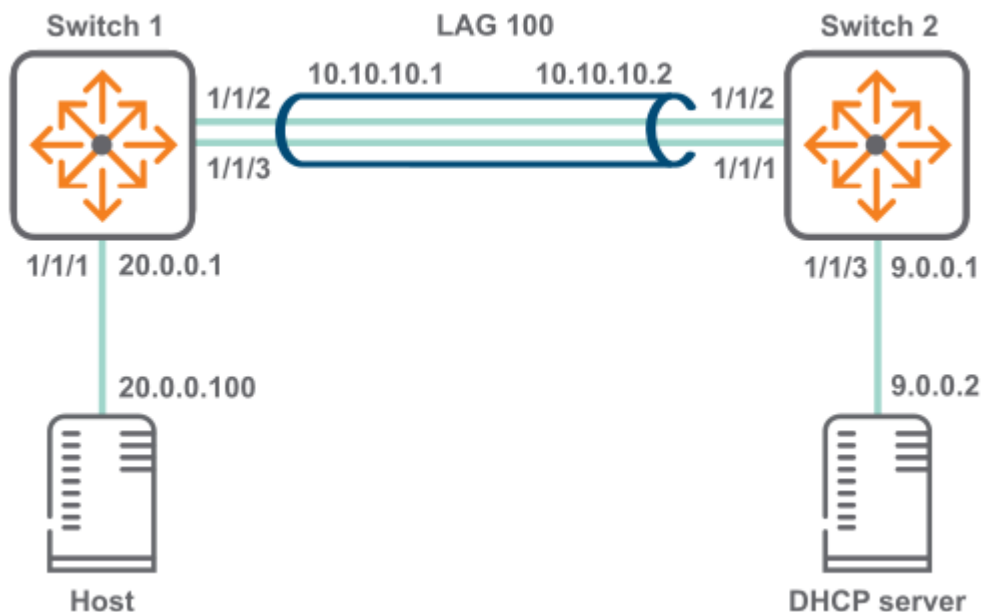
```
switch(config)# interface vlan 100
switch(config-if-vlan)# ip address 192.168.2.11/24
switch(config-if-vlan)# ip helper-address 192.168.1.1
```

3. Verify DHCP relay configuration.

```
switch# show dhcp-relay
DHCP Relay Agent           : enabled
DHCP Request Hop Count Increment : enabled
Option 82                  : disabled
Source-Interface           : disabled
Response Validation        : disabled
Option 82 Handle Policy    : replace
Remote ID                  : mac
DHCP Relay Statistics:
Valid Requests Dropped Requests Valid Responses Dropped Responses
-----
60             10             60             10
DHCP Relay Option 82 Statistics:
Valid Requests Dropped Requests Valid Responses Dropped Responses
-----
50             8             50             8
switch(config)# show ip helper-address
IP Helper Addresses
Interface: vlan100
IP Helper Address          VRF
-----
192.168.1.1               default
```

DHCPv4 relay scenario 2

In this scenario, host on switch 1 reaches the DHCP server on switch two via a LAG. The physical topology of the network looks like this:



Procedure

1. On switch 1:
Create LAG 100 and assign an IP address to it.

```
switch# config
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 200
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# exit
switch(config)# interface vlan 200
switch(config-if-vlan)# ip address 10.0.10.1/24
```

Assign an IP address to interface 1/1/1 and an IP helper address to reach the DHCP server.

Assign an IP address to interface vlan 100 and an IP helper address to reach the DHCP server.

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface vlan 100
switch(config-if-vlan)# ip address 20.0.0.1/8
switch(config-if-vlan)# ip helper-address 9.0.0.2
```

Assign interfaces 1/1/2 and 1/1/3 to LAG 100.

```
switch(config-if) # interface 1/1/2  
switch(config-if) # lag 100  
switch(config-if) # interface 1/1/3  
switch(config-if) # lag 100
```

Create a route between 10.0.10.2 and 9.0.0.0.

```
switch(config) # ip route 9.0.0.0/24 10.0.10.2
```

2. On switch 2:

Create LAG 100 and assign an IP address to it.

```
switch# config  
switch(config) # interface lag 100  
switch(config-lag-if) # no shutdown  
switch(config-lag-if) # vlan access 200  
switch(config-lag-if) # lacp mode active  
switch(config-lag-if) # exit  
switch(config) # interface vlan 200  
switch(config-if-vlan) # ip address 10.0.10.2/24
```

Assign an IP address to interface vlan 300.

```
switch(config) # interface 1/1/3  
switch(config-if) # no shutdown  
switch(config-if) # vlan access 300  
switch(config-if) # exit  
switch(config) # interface vlan 300  
switch(config-if-vlan) # ip address 9.0.0.1/24
```

Assign an IP address to interface 1/1/3.

Assign interfaces 1/1/1 and 1/1/2 to LAG 100.

```
switch(config-if) # interface 1/1/1  
switch(config-if) # lag 100  
switch(config-if) # interface 1/1/2  
switch(config-if) # lag 100
```

Create a route between 20.0.0.0 and 10.0.10.1.

```
switch(config) # ip route 20.0.0.0/8 10.0.10.1
```

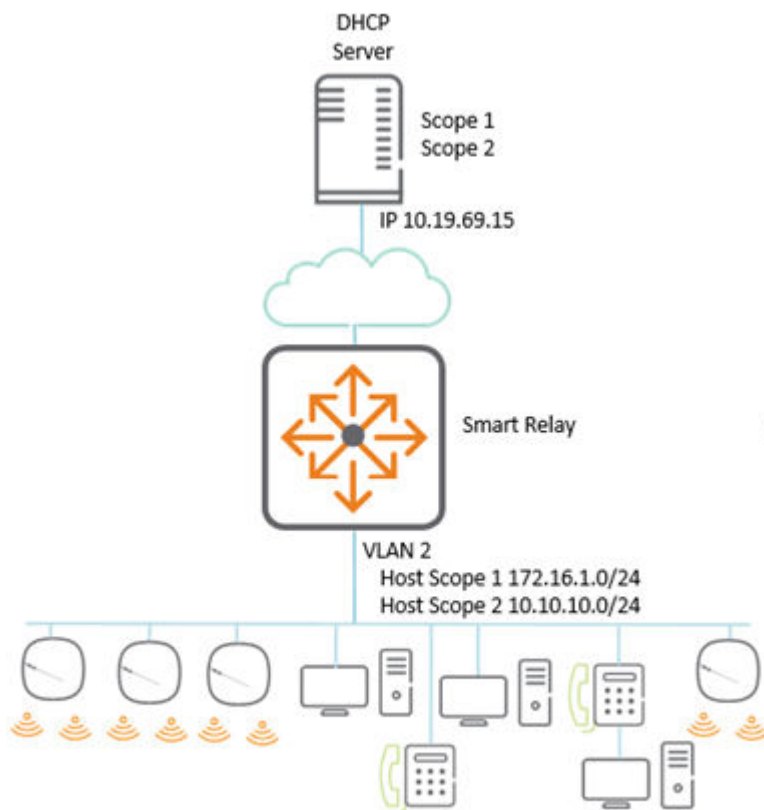

DHCPv4 relay scenario 3

About this task

In the following scenario of the DHCP smart relay, the primary address is used as the GIADDR; if the server is unavailable, then the secondary address is used. From AOS-CX 10.10.1000 and later releases, the GIADDR use the secondary address in an ascending order when there is more than one secondary address.

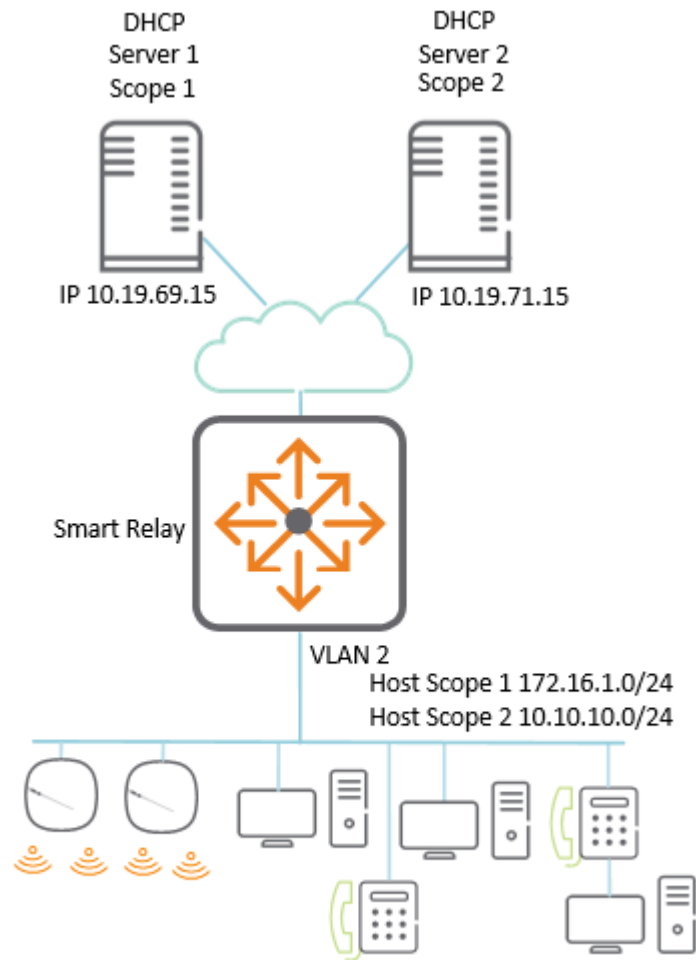
Procedure

1. If DHCP scope exhaustion happens at a site, an additional scope can be added to increase the IP address capacity while keeping the existing IP scope and subnet. The physical topology of the network looks like



this:

2. If the DHCP server is resilient or scope is required, then whole scopes can be placed on separate servers. If a server or scope becomes unavailable, then for address assignment, the next available scope is



attempted.

Results

In both the above topology, the access switch is set up the same way. The only difference is in the server which is used for helper addresses.

```
!  
dhcp-smart-relay                                <-- Enable DHCP smart relay  
!  
interface vlan 2  
    ip address 172.16.1.254/24                    <-- Add the local IPS which  
will act as GIADDR  
    ip address 10.10.610.254/24 secondary  
    ip helper-address 10.19.69.15                 <-- Add the IP helpers to  
reach the required DHCP servers  
    ip helper-address 10.19.71.15
```

DHCPv4 relay commands

Select a command from the list in the left navigation menu.

Subtopics

- [dhcp-relay](#)
- [dhcp-relay hop-count-increment](#)
- [dhcp-relay option 82](#)
- [diag-dump dhcp-relay basic](#)
- [ip bootp-gateway](#)
- [ip helper-address](#)
- [show dhcp-relay](#)
- [show dhcp-relay bootp-gateway](#)
- [show ip helper-address](#)

dhcp-relay

Syntax

```
dhcp-relay
no dhcp-relay
```

Description

Enables DHCP relay support. DHCP relay is enabled by default. DHCP relay is not supported on the management interface.

The **no** form of this command disables DHCP relay (and DHCP relay option 82) support.

Examples

This example enables DHCP relay support.

```
switch(config) # dhcp-relay
```

This example removes DHCP relay support.

```
switch(config) # no dhcp-relay
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-relay hop-count-increment

Syntax

```
dhcp-relay hop-count-increment  
no dhcp-relay hop-count-increment
```

Description

Enables the DHCP relay hop count increment feature, which causes the DHCP relay agent to increment the hop count in all relayed DHCP packets. Hop count is enabled by default.

The **no** form of this command disables the hop count increment feature.

Examples

Enabling the hop count increment feature.

```
switch(config) # dhcp-relay hop-count-increment
```

Disabling the hop count increment feature.

```
switch(config) # no dhcp-relay hop-count-increment
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-relay option 82

Syntax

```
dhcp-relay option 82 [replace {ip | mac | validate} | drop {ip | mac |  
validate} | keep {ip | mac} |  
    source-interface | circuit-id port-id | validate {drop {ip | mac} |  
replace {ip | mac}}]  
dhcp-relay option 82 [replace {ip | mac | validate} | drop {ip | mac |  
validate} | keep {ip | mac} |  
    source-interface | circuit-id port-id | validate {drop {ip | mac} |  
replace {ip | mac}}]
```

Description

Configures the behavior of DHCP relay option 82. A DHCP relay agent can receive a message from another DHCP relay agent having option 82. The relay information from the previous relay agent is replaced by default.

The **no** form of this command disables the DHCP relay option 82 configurations. Option 82 is disabled when DHCP relay is disabled globally. When DHCP relay is re-enabled, option 82 also needs to be re-enabled using the **dhcp-relay option 82** command.



NOTE

DHCP Relay is supported over VXLAN with both IPv4 and IPv6 underlay.

Parameter	Description
replace	Replace the existing option 82 field in an inbound client DHCP packet with the information from the switch. The remote ID and circuit ID information from the first relay agent is lost. Default.
drop	Drop any inbound client DHCP packet that contains option 82 information.
keep	Keep the existing option 82 field in an inbound client DHCP packet. The remote ID and circuit ID information from the first relay agent is preserved.
source-interface	Configures the DHCP relay to use a configured source IP address for inter-VRF server reachability. Set the source IP address with the command <code>ip source-interface</code> .
circuit-id port-id	Configures the circuit ID and remote ID options. Client connected port information to be used in Option 82 circuit ID and remote ID.
 NOTE When enabled, DHCP Relay drops packets from single-homed clients connected to VSX Secondary, unless the <code>dhcp-relay vsx active-active</code> command is enabled. To avoid this, connect single-homed clients to the VSX Primary or enable <code>dhcp-relay vsx active-active</code>.	
validate	Validate option 82 information in DHCP server responses and drop invalid responses.
ip	Use the IP address of the interface on which the client DHCP packet entered the switch as the option 82 remote ID.
mac	Use the MAC address of the switch as the option 82 remote ID. Default.

Example

This example enables DHCP option 82 support and replaces all option 82 information with the values from the switch, with the switch MAC address as the remote ID.

```
switch(config)# dhcp-relay option 82 replace mac
```

Command History

Release	Modification
10.17	The circuit - id port - id parameter was introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

diag-dump dhcp-relay basic

Syntax

```
diag-dump dhcp-relay basic
```

Description

Dumps DHCP relay configurations for all interfaces.

Examples

This example enables DHCP relay support.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ip bootp-gateway

Syntax

```
ip bootp-gateway <IPV4-ADDR>  
no ip bootp-gateway <IPV4-ADDR>
```

Description

Configures a gateway address for the DHCP relay agent to use for DHCP requests. By default DHCP relay agent picks the lowest-numbered IP address on the interface.

The **no** form of this command removes the gateway address.

Parameter	Description
<IPV4-ADDR>	Specifies the IP address of the gateway in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

Examples

Setting the IP address of the gateway for interface vlan 100 to **10.10.10.10**:

```
switch(config)# interface vlan100  
switch(config-if)# ip bootp-gateway 10.10.10.10
```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

ip helper-address

Syntax

```
ip helper-address <IPv4-ADDR> [vrf <VRF-NAME>]  
no ip helper-address <IPv4-ADDR> [vrf <VRF-NAME>]
```

Description

Defines the address of a remote DHCP server or DHCP relay agent. Up to eight addresses can be defined. The DHCP relay agent forwards DHCP client requests to all defined servers.

If IP helper address is defined with VRF argument then this command requires you define a source IP address for DHCP relay with the command `ip source-interface`. The configured source IP on the VRF is used to forward DHCP packets to the server.

A helper address cannot be defined on the OOBM interface.

The **no** form of this command removes an IP helper address.

Parameter	Description
helper-address <IPv4-ADDR>	Specifies the helper IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
vrf <VRF-NAME>	Specifies the name of a VRF. Default: default .

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# ip helper-address 10.10.10.209

switch(config-if)# no ip helper-address 10.10.10.209
```

Defining the IP helper address 10.10.10.209 on interface vlan 100:

```
switch(config)# interface vlan100
switch(config-if-vlan)# ip helper-address 10.10.10.209
```

Removing the IP helper address 10.10.10.209 on interface vlan 100:

```
switch(config-if-vlan)# no ip helper-address 10.10.10.209
switch(config)# interface 1/1/2
switch(config-if)# ip helper-address 10.10.10.209 vrf myvrf
switch(config-if)# no ip helper-address 10.10.10.209 vrf myvrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

show dhcp-relay

Syntax

```
show dhcp-relay
```

Description

Shows DHCP relay configuration settings.

Example

Showing DHCP relay settings:

```
switch(config)# show dhcp-relay
DHCP Relay Agent           : enabled
DHCP Smart Relay           : enabled
DHCP Request Hop Count Increment : enabled
Option 82                  : enabled
Source-Interface           : disabled
Response Validation        : disabled
Option 82 Handle Policy    : replace
Remote ID                  : mac
DHCP Relay Statistics:
  Valid Requests Dropped Requests Valid Responses Dropped Responses
  -----
  60             10              60             10
DHCP Relay Option 82 Statistics:
  Valid Requests Dropped Requests Valid Responses Dropped Responses
  -----
  50             8               50             8
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show dhcp-relay bootp-gateway

Syntax

```
show dhcp-relay bootp-gateway [interface <INTERFACE-NAME>]
```

Description

Shows the bootp gateway defined for all interfaces or a specific interface.

Parameter	Description
<code><INTERFACE-NAME></code>	Specifies an interface. Format: <code>member/slot/port</code> .

Examples

Showing the designated bootp gateway for all interfaces:

```
switch(config)# show dhcp-relay bootp-gateway
BOOTP Gateway Entries
Interface          Source IP
-----
vlan10             1.1.1.1
vlan20             2.2.2.2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip helper-address

Syntax

```
show ip helper-address [interface <INTERFACE-ID>]
```

Description

Shows the IP helper addresses defined for all interfaces or a specific interface.

Parameter	Description
interface <INTERFACE-ID>	Specifies an interface. Format: member/slot/port.

Examples

Showing the IP helper addresses for all interfaces:

```
switch(config)# show ip helper-address
IP Helper Addresses
Interface: vlan11
  IP Helper Address      VRF
  -----
  10.10.10.209           default
Interface: vlan20
  IP Helper Address      VRF
  -----
  3.3.3.3                default
  20.20.20.20            default
```

Showing the IP helper addresses for interface vlan20:

```
switch(config)# show ip helper-address interface vlan20
IP Helper Addresses
Interface: vlan20
  IP Helper Address      VRF
  -----
  3.3.3.3                default
  20.20.20.20            default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

DHCPv6 relay agent

DHCPv6 relay VRF support

DHCPv6 relay is VRF aware. DHCPv6 client requests received on an interface are forwarded to the configured servers via that interface's VRF. DHCPv6 server responses received on an interface are forwarded to the client which is reachable via that interface's VRF.



NOTE

DHCPv6 relay does not have inter-VRF support.

Limitations

DHCPv6 Relay requires an egress interface to be configured for the IPv6 multicast helper address. DHCPv6-Relay multicast helper configuration is not allowed over VxLAN tunnels. The following configuration is required in order to use source interface for DHCPv6 relay.

Configuring the IPv6 source-interface interface to use for DHCPv6 relay:

```
switch(config)# ipv6 source-interface dhcp_relay interface loopback3 vrf test1
```

Enabling DHCPv6 relay to use the configured source interface:

```
switch(config)# dhcpv6-relay source-interface
```

Subtopics

Configuring the DHCPv6 relay agent

Use Case

DHCP relay (IPv6) commands

Configuring the DHCPv6 relay agent

Prerequisites

- An enabled layer 3 VLAN

Procedure

1. Enable the DHCPv6 relay agent with the command `dhcpv6-relay`.
2. Configure one or more IP helper addresses with the command `ipv6 helper-address`. This determines where the DHCPv6 agent forward DHCP requests.
3. If you want to enable DHCP option 79 support to forward client link-layer addresses, use the command `dhcpv6-relay option 79`.
4. Review DHCPv6 relay agent configuration settings with the commands `show dhcpv6-relay` and `show ipv6 helper-address`.

Example

This example creates the following configuration:

- Enables the DHCPv6 relay agent.
- Enables interface **1/1/2** and assigns an IPv6 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **FF01::1:1000** on interface **1/1/2**.
- Enables DHCP option 79.

```
switch(config)# dhcpv6-relay
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 10
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# vlan access 20
```

```

switch(config-if)# exit
switch(config)# interface vlan 20
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::22/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000 egress
vlan 50
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-servers
egress vlan50
switch(config-if-vlan)# exit
switch(config)# interface vlan 20
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::21/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000 egress
vlan 50
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-servers
egress vlan50
switch(config-if-vlan)# exit
switch(config)# end

```

Use Case

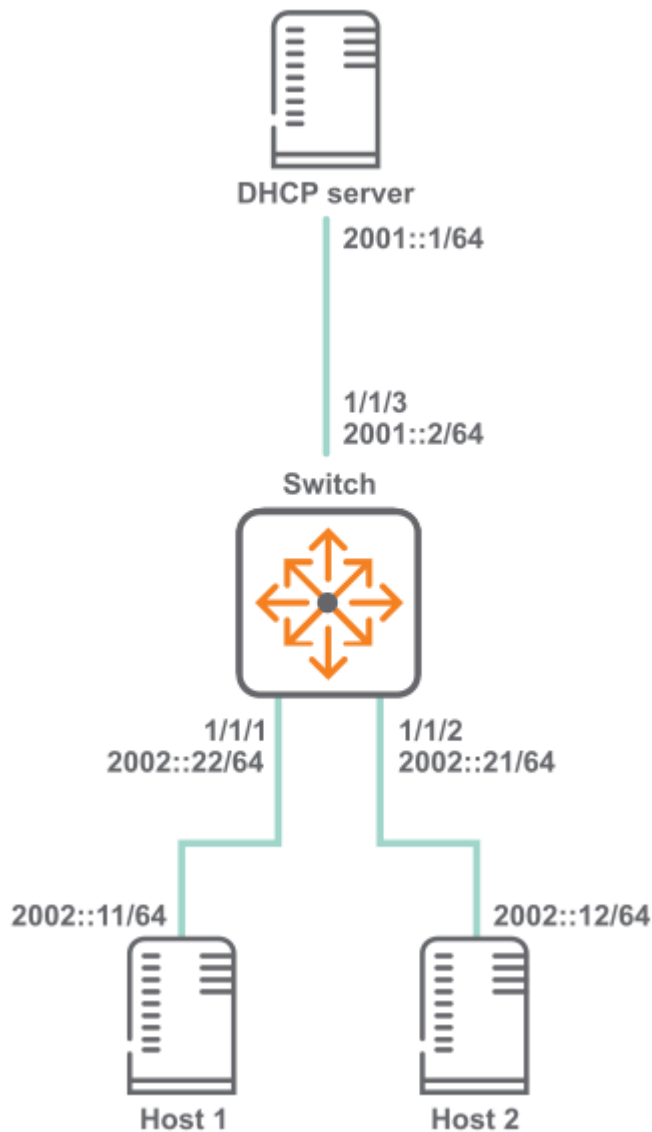
This section explain the use cases for DHCPv6 relay agent.

Subtopics

DHCPv6 relay scenario 1

DHCPv6 relay scenario 1

In this scenario, DHCP relay on the switch enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



Procedure

1. Enable DHCP relay.

```
switch# config  
switch(config)# dhcpv6-relay
```

2. Define an IPv6 helper address.

- On interface `vlan 10` and `vlan 20`.

```
switch(config)# dhcpv6-relay  
switch(config)# interface 1/1/1  
switch(config)# no shutdown  
switch(config-if)# vlan access 10  
switch(config-if)# exit  
switch(config)# interface 1/1/2  
switch(config)# no shutdown
```

```

switch(config-if)# vlan access 20
switch(config-if)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::22/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000
egress vlan30
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-
servers egress vlan50
switch(config-if-vlan)# exit
switch(config)# interface vlan 20
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::21/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000
egress vlan30
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-
servers egress vlan50
switch(config-if-vlan)# exit
switch(config)# end

```

3. Verify DHCP relay configuration.

- When interface `vlan 10` and `vlan 20` is configured as relay.

```

switch# show dhcpv6-relay
  DHCPv6 Relay Agent : enabled
  Option 79           : disabled
switch# show ipv6 helper-address
IP Helper Addresses
Interface: vlan10
IP Helper Address                                Egress Port
-----
all-dhcp-servers                                vlan50
ff01::1:1000                                    vlan30
2001::1                                          -
Interface: vlan20
IP Helper Address                                Egress Port
-----
all-dhcp-servers                                vlan50
ff01::1:1000                                    vlan30
2001::1                                          -

```

DHCP relay (IPv6) commands

Select a command from the list in the left navigation menu.

Subtopics

[dhcpv6-relay](#)

[dhcpv6-relay option 79](#)

[diag-dump dhcpv6-relay basic](#)

[ipv6 helper-address](#)

[show dhcpv6-relay](#)

[show ipv6 helper-address](#)

dhcpv6-relay

Syntax

```
dhcpv6-relay [l2vpn-clients | source-interface]
no dhcpv6-relay [l2vpn-clients | source-interface]
```

Description

Enables DHCPv6 relay support. DHCPv6 relay is disabled by default. DHCP relay is not supported on the management interface. Best practices is to disable this configuration on all the VXLAN tunnel endpoints (VTEPs), to avoid forwarding duplicate DHCPv6 requests to the server.

The **no** form of this command disables DHCP relay support.

DHCPv6 Relay requires that you configure the egress interface using the [ipv6 helper-address](#) command. The egress interface of a VTEP is used as an underlay, so a DHCPv6 Relay Multicast ipv6 address is not supported in a VXLAN topology.

Parameter	Description
<code>l2vpn-clients</code>	Enables packets from l2vpn clients to be forwarded to configured servers. Enabled by default.
<code>source-interface</code>	Enables DHCPv6 relay to use the configured source interface.

Usage

In Asymmetric/Symmetric Integrated Routing and Bridging (IRB) VXLAN deployments with a VLAN extension in subset of VTEPs, client DHCPv6 broadcast requests are received by all the VTEPs where a client VLAN is configured. A DHCPv6 relay agent on those VTEPs forward DHCPv6 packets to configured DHCPv6 server(s). As DHCPv6 requests are forwarded by multiple DHCPv6 relay agents, the DHCPv6

server receives duplicate copies of the same packet. When this configuration is disabled, the DHCPv6 relay agent on VTEPs ignores DHCPv6 request packets that are received from client MACs addresses learned via EVPN.

Examples

Enables DHCPv6 relay support.

```
switch(config) # dhcpv6-relay
```

Removes DHCPv6 relay support.

```
switch(config) # no dhcpv6-relay
```

Command History

Release	Modification
10.12.1000	l2vpn - clients and source - interface added.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-relay option 79

Syntax

```
dhcpv6-relay option 79  
no dhcpv6-relay option 79
```

Description

Enables support for DHCP relay option 79. When enabled, the DHCPv6 relay agent forwards the link-layer address of the client. This option is disabled by default.

The **no** form of this command disables support for DHCP relay option 79.

Examples

Enables DHCP option 79 support.

```
switch(config)# dhcpv6-relay option 79
```

Disables DHCP option 79 support.

```
switch(config)# no dhcpv6-relay option 79
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

diag-dump dhcpv6-relay basic

Syntax

```
diag-dump dhcpv6-relay basic
```

Description

Dumps DHCPv6 relay configurations for all interfaces.

Examples

This example enables DHCP relay support.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ipv6 helper-address

Syntax

```
ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
no ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>}
egress <PORT-NUM>
no ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>}
egress <PORT-NUM>
```

Description

Defines the address of a remote DHCPv6 server or DHCPv6 relay agent. Up to eight addresses can be defined. The DHCPv6 agent forwards DHCPv6 client requests to all defined servers.

Not supported on the OOBM interface.

The **no** form of this command removes an IP helper address.

Parameter	Description
	Specifies the unicast helper IP address in IPv6 format (<code>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</code>), where x is a hexadecimal number from 0 to F.

Parameter	Description
<code><UNICAST-IPV6-ADDR></code>	
<code><MULTICAST-IPV6-ADDR></code>	Specifies the multicast helper IP address in IPv6 format (<code>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</code>), where x is a hexadecimal number from 0 to F.
<code>all-dhcp-servers</code>	Specifies all the DHCP server IPv6 addresses for the interface.
<code>egress <PORT-NUM></code>	Specifies the port number on which DHCPv6 service requests are relayed to a multicast destination. The egress port must be different than the one on which the multicast helper address is configured. Format: <code>member/slot/port</code> .
<code>vrf <VRF-NAME></code>	Specifies the name of the VRF from which the specified protocol sets its source IP address.

Examples

Defining a multicast IPv6 helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if) # ipv6 helper-address multicast 2001:DB8:0:0:0:0:0:1
egress 1/1/2
```

Removing the IP helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if) # no ipv6 helper-address multicast 2001:DB8:0:0:0:0:0:1
egress 1/1/2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000	<code>config-if</code>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
6100		

show dhcpv6-relay

Syntax

```
show dhcpv6-relay
```

Description

Shows DHCP relay configuration settings.

Example

```
switch(config)# show dhcpv6-relay
DHCPv6 Relay Agent : enabled
Option 79           : disabled
L2vpn-clients       : enabled
Source-interface    : enabled
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 helper-address

Syntax

```
show ipv6 helper-address [interface <INTERFACE-ID>]
```

Description

Shows the helper IP addresses defined for all interfaces or a specific interface.

Parameter

Description

interface <INTERFACE-ID>

Specifies an interface. Format: **member/slot/port**.

Examples

```
switch# show ipv6 helper-address
```

```
Interface: 1/1/1
```

```
IPv6 Helper Address
```

```
Egress Port
```

```
-----  
2001:db8:0:1::
```

```
-----  
-
```

```
FF01::1:1000
```

```
1/1/2
```

```
Interface: 1/1/2
```

```
IPv6 Helper Address
```

```
Egress Port
```

```
-----  
2001:db8:0:1::
```

```
-----  
-
```

```
switch# show ipv6 helper-address interface 1/1/1
```

```
Interface: 1/1/1
```

```
IPv6 Helper Address
```

```
Egress Port
```

```
-----  
2001:db8:0:1::
```

```
-----  
-
```

```
FF01::1:1000
```

```
1/1/2
```

```
switch# show ipv6 helper-address interface vlan20
```

```
Interface: vlan20
```

```
IP Helper Address
```

```
Egress Port
```

```
-----  
2001::1
```

```
-----  
-
```

```
ff01::1:1000
```

```
vlan30
```

```
default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

FAQ

1. What ports use DHCP for switch assignments?

You can set up a DHCP switch with a management VRF port; if the management VRF port is unavailable, use an IN-Band data port.

2. Is DHCP assignment available on boot up from the factory?

Yes, DHCP assignment is available on boot up from the factory. This assists Zero Touch depending on the platform; this can be a management VRF or an IN-Band data port on the default VLAN 1.

3. Is DHCP assignment available on both management VRF and IN-Band data ports simultaneously?

Yes, DHCP assignment is possible on both management VRF and IN Band data ports, but it depends on the platform.

If both ports can get a DHCP address, the DHCP address from the management VRF port is used for Zero Touch Provisioning instead of the one from the IN-Band data port. The data ports will wait 30 seconds before taking ownership of Zero Touch Provisioning (ZTP).

4. Which VLAN is assigned a DHCP address?

The factory default VLAN to which the DHCP address is assigned is VLAN1.

Users can configure the VLAN for DHCP allocation on platforms supporting multiple VLAN configurations. Only a single VLAN can support DHCP assignments.

5. Can DHCP assignments be disabled in an HPE Aruba Networking AOS-CX switch?

Use the **no ip dhcp** command on the required VLAN interface or management VRF port you can disable the DHCP assignments in an HPE Aruba Networking AOS-CX switch.

A static configuration option is available on both ports.

6. Can IP DHCP switch assignment be carried out on a Routed only Port (RoP)?

No, RoP is not supported. Only VLAN and OOBM ports can be allocated an address with DHCP.

7. Can an HPE Aruba Networking AOS-CX switch support DHCP IPv6 Allocation?

Yes, an HPE Aruba Networking AOS-CX switch can support DHCP IPv6 allocation. However, for Zero Touch Provisioning (ZTP), IPv4 is required.

8. Do HPE Aruba Networking AOS-CX switches support DHCP server?

Since the software release of AOS-CX 10.10, all platforms support DHCP server, except for 6100, 6000, and 4100i.

9. Can HPE Aruba Networking AOS-CX switches support DHCP address exclusion from a subnet?

Not formally; however, you can use static bindings to unused MAC addresses as a workaround on a smaller scale.

10. Do all HPE Aruba Networking AOS-CX switches support DHCP helper?

Since the software release of AOS-CX 10.10, all platforms support DHCP helper for IPv4 and IPv6.

11. Can DHCP Server and Relay simultaneously be supported on a switch?

Yes, since the software release AOS-CX 10.08, the platforms that support DHCP Server and Relay both can be set up simultaneously.

12. How many helper addresses can be used?

The Switched Virtual Interface can use up to eight helper addresses for IPv4 and IPv6.

13. Is DHCP Relay supported for multi-netted addresses?

Yes, for both IPv4 and IPv6 DHCP Relay supports multi-netted addresses. The Gateway IP Address (GIADDR) used in the DHCP request will be the lowest IP address configured on the interface.

You can override the lowest IP address using the `bootp-gateway` command. IPV6 does not support bootp.

14. Can a multi-netted address support GIADDR from a different address?

Yes, for IPv4 only. For more information, see [DHCP relay agent](#).

15. Can the binding of DHCP and IP remain after a switch reboot?

Yes, the binding of DHCP and IP remains by using an external storage option `dhcpv4-snooping external-storage`.

16. If multiple 'ip-helper' IPs are configured for DHCP relay on an interface or SVI, which switch will forward the DHCP requests?

In case of multiple 'ip-helper' IPs configured for DHCP relay on an interface or SVI, the client will receive multiple offers from different DHCP servers, and the client will choose which one to accept.

17. What happens when global *dhcp-smart-relay* is enabled in an interface or SVI with multiple IP addresses (multi-netting) that services DHCP clients in different subnets?

On enabling global **dhcp-smart-relay** in an interface or SVI with multiple IP addresses (multi-netting) that services DHCP clients in different subnets, the AOS-CX device will first use the primary interface or SVI IP address as the unicast source to the DHCP server. If no response is received the AOS-CX device will use the secondary IPs from the lowest-numbered IP to the highest until a DHCP response is received.

18. What are the troubleshooting commands for DHCP relay agent?

The initial troubleshooting for DHCP relay agent can be done using the following commands:

- `show dhcp-relay`
- `show dhcp-relay bootp-gateway`
- `show ip helper-address`

Next level of troubleshooting can be done by taking packets captured at the DHCP server side and AOS-CX side. For more information about AOS-CX packet capture, see Mirroring chapter of AOS-CX Monitoring Guide.

Perform additional AOS-CX side DHCP debugging using **debug dhcprelay** and **debug dhcpv6relay** commands. For more information, see Debug logging chapter of AOS-CX Diagnostics and Supportability Guide.

DHCP snooping

DHCP is a protocol used by DHCP servers in IP networks to dynamically allocate network configuration data to client devices (DHCP clients). Possible network configuration data includes user IP address, subnet mask, default gateway IP address, DNS server IP address, and lease duration. The DHCP protocol enables DHCP clients to be dynamically configured with such network configuration data without any manual setup process.

DHCP snooping is a security feature that helps avoid problems caused by an unauthorized DHCP server on the network that provides invalid configuration data to DHCP clients. A user without malicious intent may cause this problem by unknowingly adding to the network a switch or other device that includes a DHCP server enabled by default. In some cases, a user with malicious intent adds a DHCP server to the network as part of their Denial of Service or Man in the Middle attack.

DHCP snooping helps prevent such problems by distinguishing between trusted ports connected to legitimate DHCP servers and untrusted ports connected to general users. DHCP packets are forwarded between trusted ports without inspection. DHCP packets received on other switch ports are inspected before being forwarded. DHCP Packets from untrusted sources are dropped.



NOTE

DHCP Snooping and DHCP relay can be configured on the same switch.

When DHCP snooping and DHCP relay are both enabled on a VLAN, the following actions occur:

- Received packet: DHCP snooping processes the DHCP packet before (possibly) handing it to DHCP relay.
- Transmitted packet: DHCP packets sent by DHCP relay are intercepted by DHCP snooping to learn IP bindings.



NOTE

For even more rigorous security that is applied in hardware on a packet-by-packet basis, you can use IP source lockdown feature as described in `IP source lockdown`.

DHCP server interoperation

DHCP server can not be configured with DHCP snooping.

DHCPv4 snooping conditions for dropping DHCPv4 packets

Applies only to DHCPv4 snooping.

Packet types that are dropped

DHCPOFFER, DHCPACK, DHCPNACK

Conditions for dropping the packets

- A packet from a DHCP server is received on an untrusted port.
- The switch is configured with a list of authorized DHCP server addresses and a DHCP response received on a trusted port, but the source IP address is not an authorized DHCP server.

DHCPRELEASE, DHCPDECLINE

- A broadcast packet that has a MAC address in the DHCP binding database, but the port in the DHCP binding database does not match the port on which the packet is received.

Packet types that are dropped

Conditions for dropping the packets

All DHCP packet types

- A DHCP packet received on an untrusted port in which the DHCP client hardware MAC address does not match the source MAC address in the packet.
- A DHCP packet containing DHCP relay information (option 82) is received on an untrusted port.

Subtopics

[Protocol details](#)

[Supported platform and standards](#)

[DHCPv4 Snooping Use case](#)

[RFC 7610 Compliance for DHCPv6 Guard](#)

[DHCPv4 snooping commands](#)

[DHCPv6 snooping commands](#)

[Troubleshooting](#)

Protocol details

Procedure

1. When a port is configured as a trusted port, all the dynamic IP binding entries learned on that port will be deleted.
2. When a client is connected on a trusted port, the dynamic IP binding entries will not be learned on the switch, even though the client gets an IP address.
3. If DHCPv4 snooping is enabled on two back-to-back access switches, DHCP packets will be dropped, Since by default option 82 is enabled on DHCPv4 snooping and the default policy is drop. The second switch with DHCPv4 snooping enabled drops the packets. In this scenario the user should enable DHCPv4 snooping option 82 on one switch, or else you can disable on both.

Supported platform and standards

The following table list the supported platforms for DHCP snooping.

Table 1. Support platforms for DHCP snooping

Platform	DHCP snooping
4100i	Yes
6000	Yes
6100	Yes

Table 2. Scale

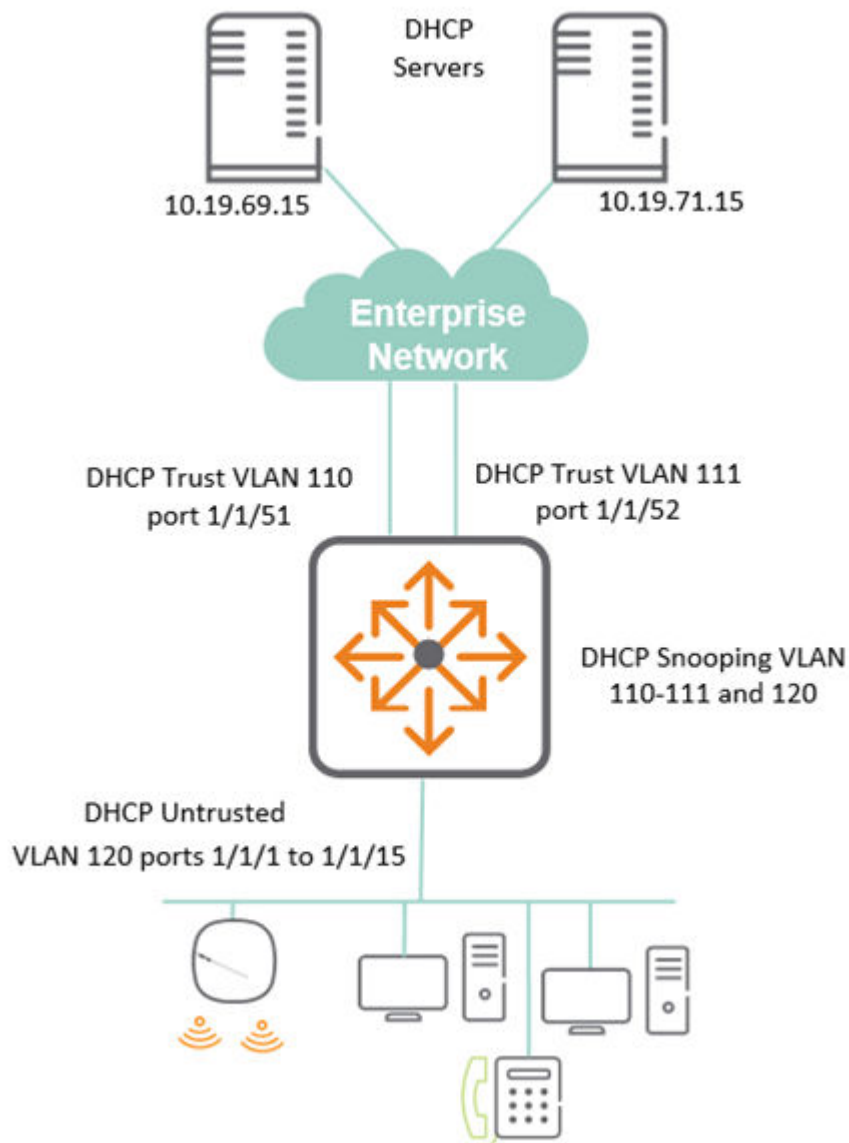
Platform	IP Bindings per port
4100i	256
6000	256
6100	256

DHCPv4 Snooping Use case

DHCP Snooping is used to protect from rogue or unwanted attacks from posing DHCP services.

In the following use case:

- Set the switch for DHCPv4 snooping.
- Allow DHCP communication from the uplinks in the Enterprise network port 1/1/51 and 1/1/52.
- For communication to occur, identify the authorized DHCP servers 10.19.69.15 and 10.19.71.15.
- LAN facing user ports 1/1/1 to 1/1/15 is not configured as trusted.



Key configuration portion for above set up is shown with the relevant DHCPv4 snooping parameters:

```
dhcpv4-snooping                                <--enable DHCPv4 snooping
dhcpv4-snooping authorized-server 10.19.69.15  <--allow authorized
```



```

servers
dhcpv4-snooping authorized-server 10.19.71.15
vlan 110
    description uplink
    dhcpv4-snooping                                <--enable DHCPv4
snooping on required VLANs
vlan 111
    description uplink
    dhcpv4-snooping
vlan 120
    description User
    dhcpv4-snooping
interface 1/1/1
    no shutdown
    description User
    vlan access 120
...
interface 1/1/15
    no shutdown
    description User
    vlan access 120
interface 1/1/51                                <--Unlinks on 1/1/51 and 52 trusted
    no shutdown
    vlan trunk native 1
    vlan trunk allowed 110
    dhcpv4-snooping trust
interface 1/1/52
    vlan trunk native 1
    vlan trunk allowed 111
    dhcpv4-snooping trust
interface vlan 110
    description uplink
    ip address 172.16.69.1/30
interface vlan 111
    description uplink
    ip address 172.16.71.1/30
interface vlan 120
    description User
    ip address 172.16.10.1/254
    ip helper-address 10.19.69.15
    ip helper-address 10.19.71.15
...
ip route 0.0.0.0/0 172.16.69.2 distance 10

```

```
ip route 0.0.0.0/0 172.16.71.2 distance 20
...
```

Using the show dhcpv4-snooping command, you can see some of the following DHCPv4 information:

- Applied VLANs for snooping, 110-112 and 120
- The untrusted policy is set to drop packets
- The list of authorized DHCP servers are set to 10.19.69.15 and 10.19.71.15
- Set the trusted or untrusted ports and any DHCP bindings for those ports. Port 1/1/1 and 1/1/2 offer DHCP.

```
show dhcpv4-snooping
DHCPv4-Snooping Information
  DHCPv4-Snooping      : Yes          Verify MAC Address   : Yes
  Allow Overwrite Binding : No          Enabled VLANs       : 110-111,120
  IP Binding Disabled VLANs :
  Static Attributes    : No
  Client Event Logs    : No
Option 82 Configurations
  Untrusted Policy      : drop          Insertion            : Yes
  Option 82 Remote-id   : mac
External Storage Information
  Volume Name           : --
  File Name              : --
  Inactive Since         : --
  Error                  : --
Flash Storage Information
  File Write Delay       : --
  Active Storage         : --
Authorized Server Configurations
  VRF                    Authorized Servers
  -----
default                  10.19.69.15
default                  10.19.71.15
Port Information
  Port      Trust  Max   Static  Dynamic
              Bindings Bindings Bindings
  -----
1/1/51     Yes    0      0        0
1/1/52     Yes    0      0        0
1/1/1      No     1024   0        1
1/1/2      No     1024   0        1
!
```

```
!output omitted
!
1/1/14      No      1024      0          0
1/1/15      No      1024      0          0
```

RFC 7610 Compliance for DHCPv6 Guard

DHCPv6 Guard implements **RFC 7610** and enforces all must requirements for filtering rogue DHCPv6 server messages at Layer 2. The following sections describes the compliance and implementation details.

Subtopics

DHCPv6 Guard Extension Header Handling

DHCPv6 Guard Fragmentation Handling

DHCPv6 Guard Unknown Next Header Handling

DHCPv6 Guard Statistics and Logging

DHCPv6 Guard Extension Header Handling

Extension Header Handling in DHCPv6 Guard

The DHCPv6 Guard implementation processes IPv6 extension headers as follows:

A. Supported Extension Headers (fully parsed and inspected):

- Hop-by-Hop Options Header: DHCPv6 Guard inspects the header chain for DHCPv6 Server packets.
- Destination Options Header: DHCPv6 Guard inspects the header chain for DHCPv6 Server packets.
- Routing Header: DHCPv6 Guard inspects the header chain for DHCPv6 Server packets.
- Authentication Header (AH): Parsed. If DHCPv6 header is found after AH, DHCPv6 Guard policy is enforced.
- Mobility Header: Packets with this header are dropped and not forwarded.

The following table shows the next headers values and the behavior in DHCPv6 Guard:

Extension Header	Next Header Value	Behavior in DHCPv6 Guard
Hop-by-Hop Options	0	Parsed and inspected
Destination Options	60	Parsed and inspected

Extension Header	Next Header Value	Behavior in DHCPv6 Guard
Routing	43	Parsed and inspected
Authentication Header	51	Parsed and inspected
ESP	50	Not inspected, always allowed
Mobility	135	Parsed and dropped
No Next Header	59	Not inspected, allowed
Unknown/Custom Header	(others)	Forwarded by default

B. Upper-Layer Protocols (not inspected for DHCPv6, always allowed):

Packet / Payload	Next Header Value	Behavior in DHCPv6 Guard
Encapsulating Security Payload (ESP)	50	Treated as an upper-layer protocol. Packets with ESP are not inspected for DHCPv6 and are forwarded as per RFC 7610.
Unsupported or Unrecognized Extension Headers	-	If an unknown or unsupported Next Header value is encountered in the extension header chain, the default behavior is to forward the packet. This is configurable.
No Next Header	59	Not inspected for DHCPv6. Such packets are forwarded as per standard IPv6 behavior.
Custom/Experimental Headers	-	Any custom or experimental extension headers. For example, Next Header values not defined in RFC 8200 or not listed above, are forwarded by default, but can be dropped.

Packet / Payload	Next Header Value	Behavior in DHCPv6 Guard
		d and logged if the relevant configuration is enabled.



NOTE

The implementation ensures that all supported extension headers are parsed in the order they appear, and the full header chain is inspected to locate DHCPv6 Server packets. Packets are dropped and logged for deviations from the expected header chain or the presence of unsupported or unknown headers only if the relevant configuration is enabled, as per RFC 7610 recommendations.

Client Packet Handling with Extension Headers

DHCPv6 client packets (Solicit, Request, Confirm, Renew, Rebind, Release, Decline, Information-Request) containing IPv6 extension headers are handled based on the **dhcpv6-snooping allow-ipv6-ext-headers** configuration:

- When **allow-ipv6-ext-headers** is **enabled**: Client packets with extension headers are correctly parsed and processed. The DHCPv6 message is correctly located past all extension headers in the chain.
- When **allow-ipv6-ext-headers** is **disabled** (default): Client packets containing valid extension headers are dropped.

DHCPv6 Guard Fragmentation Handling

The DHCPv6 Guard implementation complies with RFC 7610 for fragmented packets. The behavior depends on the fragment type, header completeness, and the **allow-ipv6-ext-headers** configuration.

First Fragment (Fragment Offset = 0)

- **Complete headers + allow-ext-hdrs=yes**: If the first fragment contains the complete IPv6 header chain (including all extension headers, the UDP header, and the DHCPv6 header), the packet is processed by DHCPv6 Guard as per the configured policy and allowed.
- **Complete headers + allow-ext-hdrs=no**: The packet is dropped because extension headers are present (the Fragment header itself is an IPv6 extension header). The **V6_PACKETS_DROP** counter is incremented. No event log is generated.
- **Incomplete UDP/DHCPv6 header + allow-ext-hdrs=yes**: If the first fragment contains the Fragment header and reaches the UDP header (ports 546/547 visible for ASIC classification), but the DHCPv6 header is incomplete, the packet is dropped. An event log is generated and the **V6_PACKETS_DROP** counter is incremented.

- **Incomplete UDP/DHCPv6 header + allow-ext-hdrs=no:** The packet is dropped due to the extension header policy before DHCPv6 header validation. The **V6_PACKETS_DROP** counter is incremented. No event log is generated.

Non-First Fragment (Fragment Offset > 0)

- Non-first fragments do not contain the IPv6 header chain and therefore cannot be classified as DHCPv6 packets by the ASIC. These fragments are **hardware forwarded** regardless of the **allow-ipv6-ext-headers** configuration and are never inspected by DHCPv6 Guard.

Summary:

Only first fragments where the full IPv6 header chain and the DHCPv6 header are present and complete are subject to DHCPv6 Guard inspection. First fragments with incomplete headers are dropped. Non-first fragments are hardware forwarded.

Fragmentation Behavior Summary

Scenario	allow-ext-hdrs	Result
First fragment with complete headers (ext hdrs + frag)	yes	ALLOW
First fragment with complete headers (ext hdrs + frag)	no	DROP
First fragment with complete headers (frag only)	yes	ALLOW
First fragment with complete headers (frag only)	no	DROP
Non-first fragment (offset > 0)	yes	ALLOW
Non-first fragment (offset > 0)	no	ALLOW
First fragment + incomplete UDP/DHCPv6 header	yes	DROP
First fragment + incomplete UDP/DHCPv6 header	no	DROP

DHCPv6 Guard Unknown Next Header Handling

If the packet contains an unrecognized Next Header value, the default behavior is to forward the packet. The **dhcpv6-snooping drop-unknown-ipv6-ext-headers** configuration enables dropping and logging of such packets, as per RFC 7610 recommendations.

When **drop-unknown-ipv6-ext-headers** is configured, the behavior depends on whether the ingress port is trusted or untrusted:

- **Untrusted ports:** Packets with unknown Next Header values are **reforwarded** instead of dropped, maintaining backward compatibility.

- **Trusted ports:** Packets with unknown Next Header values are **dropped** and logged, as per RFC 7610 compliance.

Port Type	Drop-unknown-ipv6-ext-headers Configured	Behavior for Unknown NH
Untrusted	Yes	Reforwarded
Untrusted	No (default)	Forwarded
Trusted	Yes	Dropped and logged
Trusted	No (default)	Forwarded

DHCPv6 Guard Statistics and Logging

- **Drop Counters:**

The `show dhcpv6-snooping statistics` command provides a summary of all DHCPv6 packet processing statistics, including drop counters. The drop counter is incremented whenever a packet is dropped due to validation failure, DHCPv6 Guard policy, or RFC 7610 compliance.

- **Event Logging:**

Whenever a packet is dropped by DHCPv6 Guard (including drops due to RFC 7610 compliance), an event log is generated. These logs include details such as the reason for the drop, source/destination addresses, VLAN, and port.

Example event log:

```
2025-06-24:12:34:56.123456|ipsavd|LOG_ERR|DHCPV6GUARD|DHCPv6 server
packet dropped on port:1/1/21 vlan:100. Reason: first fragment missing
complete IPv6 header chain (RFC 7610 compliance)
```

Refer to the **IPSAV event logs** table in the Supportability section for all possible drop log messages and their descriptions.



NOTE

Event logs are critical for troubleshooting and auditing. Every drop action (including those triggered by extension header or fragmentation checks) is logged for operator visibility.

DHCPv4 snooping commands

Subtopics

DHCP snooping commands

DHCP snooping commands

Select a command from the list in the left navigation menu.

Subtopics

- [clear dhcp-snooping binding](#)
- [clear dhcp-snooping statistics](#)
- [dhcp-snooping](#)
- [dhcp-snooping \(in config-vlan context\)](#)
- [dhcp-snooping allow-overwrite-binding](#)
- [dhcp-snooping authorized-server](#)
- [dhcp-snooping event-log client](#)
- [dhcp-snooping external-storage](#)
- [dhcp-snooping flash-storage](#)
- [dhcp-snooping max-bindings](#)
- [dhcp-snooping option 82](#)
- [dhcp-snooping static-attributes](#)
- [dhcp-snooping trust](#)
- [dhcp-snooping verify mac](#)
- [show dhcp-snooping](#)
- [show dhcp-snooping binding](#)
- [show dhcp-snooping statistics](#)

clear dhcp-snooping binding

Syntax

```
clear dhcp-snooping binding {all | ip <IP-ADDR> vlan <VLAN-ID> | port <PORT-  
NUM> | vlan <VLAN-ID>}
```

Description

Clears DHCP snooping binding entries.

Parameter	Description
<code>all</code>	Specifies that all DHCP binding information is to be cleared.
<code>ip <IP-ADDR> vlan <VLAN-ID></code>	Specifies the IP address and VLAN for which all DHCP binding information is to be cleared.
<code>port <PORT-NUM></code>	Specifies the port number for which all DHCP binding information is to be cleared.
<code>vlan <VLAN-ID></code>	Specifies the VLAN for which all DHCP binding information is to be cleared.

Examples

Clearing all DHCP binding information for IP address 192.168.2.4 and VLAN 5:

```
switch(config)# clear dhcp-snooping binding ip 192.168.2.4 vlan 5
```

Clearing all DHCP binding information for port 1/1/1:

```
switch(config)# clear dhcp-snooping binding port 1/1/1
```

Clearing all DHCP binding information for VLAN 10:

```
switch(config)# clear dhcp-snooping binding vlan 10
```

Clearing all DHCP binding information:

```
switch(config)# clear dhcp-snooping binding all
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping . The ipv4 parameter is deprecated and replaced with ip .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

clear dhcp-snooping statistics

Syntax

```
clear dhcp-snooping statistics
```

Description

Clears all DHCP snooping statistics.

Examples

Clear all DHCP snooping statistics:

```
switch# clear dhcp-snooping statistics
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

dhcp-snooping

Syntax

```
dhcp-snooping  
no dhcp-snooping
```

Description

Enables DHCP snooping. DHCP snooping is disabled by default. DHCP snooping is not supported on the management interface.

The **no** form of the command disables DHCP snooping, flushing all the IP bindings learned since DHCP snooping was enabled.

Examples

Enabling DHCP snooping:

```
switch(config)# dhcp-snooping
```

Disabling DHCP snooping:

```
switch(config)# no dhcp-snooping
```

Command History

Release	Modification
10.14	The dhcpx4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping (in config-vlan context)

Syntax

```
dhcp-snooping
no dhcp-snooping
```

Description

Enables DHCP snooping for the specified VLAN in the **config-vlan** context. DHCP snooping is disabled by default for all VLANs.

The no form of the command disables DHCP snooping on the specified VLAN, flushing all the IP bindings learned for this VLAN since DHCP snooping was enabled for this VLAN.

Examples

Enabling DHCP snooping on VLAN 100:

```
switch(config) # vlan 100
switch(config-vlan-100) # dhcp-snooping
switch(config-vlan-100) # exit
switch(config) #
```

Disabling DHCP snooping on VLAN 100:

```
switch(config) # vlan 100
switch(config-vlan-100) # no dhcp-snooping
switch(config-vlan-100) # exit
switch(config) #
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	<code>config-vlan</code>	Administrators or local user group members with execution rights for this command.

dhcp-snooping allow-overwrite-binding

Syntax

```
dhcp-snooping allow-overwrite-binding  
no dhcp-snooping allow-overwrite-binding
```

Description

Allows binding to be overwritten for the same IP address. When enabled, and a DHCP server offers a host an IP address that is already bound to an existing host in the binding table, the existing binding is overwritten for the new host if the new host is successfully able to acquire the same IP address. This overwriting is disabled by default, causing the DHCP server offers to be dropped.

The **no** form of the command disables DHCP snooping overwrite binding.

Examples

Enabling DHCP snooping overwrite binding:

```
switch(config) # dhcp-snooping allow-overwrite-binding
```

Disabling DHCP snooping overwrite binding:

```
switch(config)# no dhcp-snooping allow-overwrite-binding
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping authorized-server

Syntax

```
dhcp-snooping authorized-server <IP-ADDR> [vrf <VRF-NAME>]  
no dhcp-snooping authorized-server <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Adds an authorized (trusted) DHCP server to a list of authorized servers for use by DHCP snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default,

with an empty list of authorized servers, all DHCP servers are considered to be trusted for DHCP snooping purposes.



NOTE

The **mgmt** VRF cannot be used with this command.

The no form of this command deletes the specified DHCP server from the authorized list.

Parameter	Description
<code><IP-ADDR></code>	Specifies the IP address of the trusted DHCP server.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.

Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

Examples

Adding DHCP servers 192.168.2.2, 192.168.2.3, and 192.168.2.10 to the authorized server list:

```
switch(config) # dhcp-snooping authorized-server 192.168.2.2
switch(config) # dhcp-snooping authorized-server 192.168.2.3 vrf default
switch(config) # dhcp-snooping authorized-server 192.168.2.10 vrf default
```

Removing DHCP server 192.168.2.3 from the authorized server list:

```
switch(config) # no dhcp-snooping authorized-server 192.168.2.3 vrf default
```

Command History

Release	Modification
10.14	The dhcipv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.09	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping event-log client

Syntax

```
dhcp-snooping event-log client  
no dhcp-snooping event-log client
```

Description

This command enables or disables dhcp-snooping client level event logs that help with client telemetry on a remote management station such as HPE Aruba Networking Central. By default, client level event logs are disabled. The **no** form of this command disables client-level event logs for DHCP snooping after they are enabled. View these logged DHCP snooping events by issuing the command **show events -c dhcp-snooping**.



NOTE

For additional information on DHCP-related event logging, please refer to the Event Log Message Reference Guide.

Examples

Enabling DHCP client level event logs:

```
switch(config)# dhcp-snooping event-log client
```

Disabling external storage:

```
switch(config)# no dhcp-snooping event-log client
```


Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping external-storage

Syntax

```
dhcp-snooping external-storage volume <VOL-NAME> file <FILE-NAME>  
no dhcp-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

Description

Configures external storage to be used for backing up IP bindings (used by DHCP snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IP bindings from the configured external storage file to populate its local cache.



NOTE

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in an external storage file.

Parameter	Description
<code>volume <VOL-NAME></code>	Specifies the name of the existing external storage volume where the IP bindings file will be saved. Before running the dhcp - snooping external - storage volume command, first create the external storage volume using command external - storage <VOLUME - NAME> . See External storage commands in the Command-Line Interface Guide .
<code>file <FILE-NAME></code>	Specifies the file name to use for storing IP bindings. Maximum 255 characters.

Configuring IP bindings storage in file **dsnoop_ipbindings** on existing volume **dhcp_snoop**:

```
switch(config)# dhcp-snooping external-storage volume dhcp_snoop file
dsnoop_ipbindings
```

Disabling external storage:

```
switch(config)# no dhcp-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)# no dhcp-snooping external-storage volume dhcp_snoop
dhcp-snooping will use flash storage to store IP Binding database
switch(config)#
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Updated example with flash storage information.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping flash-storage

Syntax

```
dhcp-snooping flash-storage [delay <DELAY>]  
no dhcp-snooping flash-storage [delay <DELAY>]
```

Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCP snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting **delay <DELAY>** sets the default delay of 900 seconds.



CAUTION

To reduce switch flash aging it is recommended that you use external storage (command **dhcp-snooping external-storage**) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.



NOTE

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in flash storage.

Parameter	Description
delay <DELAY>	Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400.

Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcp-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to
use an external-storage.
Do you want to continue (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcp-snooping flash-storage
```

Command History

Release	Modification
10.14	The dhcpx4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping max-bindings

Syntax

```
dhcp-snooping max-bindings <MAX-BINDINGS>
no dhcp-snooping max-bindings <MAX-BINDINGS>
```

Description

Sets the maximum number of DHCP bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max binding is the maximum value of the range.

The no form of the command reverts max bindings for the selected interface to its default.

Parameter	Description
<code><MAX-BINDINGS></code>	Specifies the maximum number of DHCP bindings. Range: 1 to 256.

Examples

Set the DHCP max bindings to 256 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# dhcp-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Revert DHCP max bindings to its default on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no dhcp-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

dhcp-snooping option 82

Syntax

```
dhcp-snooping option 82 [remote-id {mac | subnet-ip}][untrusted-policy {drop | keep | replace}]
no dhcp-snooping option 82 [remote-id {mac | subnet-ip}][untrusted-policy {drop | keep | replace}]
```

Description

Configures the addition of option 82 DHCP relay information to DHCP client packets that are being forwarded on trusted ports. DHCP relay is enabled by default.

In the switch default state and when this command is entered without parameters (**dhcp-snooping option 82**), this default configuration is used:

```
dhcp-snooping option 82 remote-id mac untrusted-policy drop
```

When **remote-id** is omitted, its default (**mac**) is used. When **untrusted-policy** is omitted, its default (**drop**) is used.

The no form of this command disables DHCP snooping option 82.

Parameter	Description
remote-id	Specifies what address to use as the remote ID for the <code>replace</code> option of <code>untrusted-policy</code> . Specify one of these address types:

Parameter	Description
<code>mac</code>	The default. Uses the switch MAC address as the remote ID.
<code>subnet-ip</code>	Uses the IP address of the client VLAN as the remote ID.
<code>untrusted-policy</code>	Specifies what action to take for DHCP packets (with option 82) that are received on untrusted ports. Specify one of these actions:
<code>drop</code>	The default. Drop DHCP packets (with option 82) without forwarding them.
<code>keep</code>	Forward DHCP packets (with option 82).
<code>replace</code>	Replace the option 82 information in the DHCP packets with whatever is set for remote-id (one of: mac , subnet-ip , or management-ip) and forward the packets.

Examples

Configuring DHCP snooping option 82 with the keep action:

```
switch(config)# dhcp-snooping option 82 untrusted-policy keep
```

Disabling DHCP snooping option 82:

```
switch(config)# no dhcp-snooping option 82 untrusted-policy keep
```

Command History

Release	Modification
10.14	The dhcipv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping static-attributes

Syntax

```
dhcp-snooping static-attributes  
no dhcp-snooping static-attributes
```

Description

Enables storage of static attributes provided to the DHCP client by DHCP server during DHCP packet exchange. Disabled by default. When enabled, the following attributes are stored in OVSDB along with the client IP binding entry:

1. Name server IP addresses: DNS server IPs provided by the DHCP server to the client. Maximum: 3 per client.
2. Default gateway IP address: Router IP addresses provided by DHCP server to the client. Maximum: 3 per client.
3. Server IP address: IP address of the DHCP server that leased the IP to the client.

The **no** form of the command disables storing of client static attributes. After disabling, existing client static attributes will be flushed.

Examples

Enabling the storage of DHCP snooping static attributes:

```
switch(config)# dhcp-snooping static-attributes
```

Disabling the storage of DHCP snooping static attributes:

```
switch(config)# no dhcp-snooping static-attributes
```


Command History

Release	Modification
10.14	The dhcpx4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping trust

Syntax

```
dhcp-snooping trust  
no dhcp-snooping trust
```

Description

Enables DHCP snooping trust on the selected port. Only server packets received on trusted ports are forwarded. All the ports are untrusted by default.

The **no** form of the command disables DHCP snooping trust on the selected port.

Examples

Enabling DHCP snooping trust on interface 2/2/1:

```
switch(config) # interface 2/2/1  
switch(config-if) # dhcp-snooping trust  
switch(config-if) # exit  
switch(config) #
```

Disabling DHCP snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcp-snooping trust
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

dhcp-snooping verify mac

Syntax

```
dhcp-snooping verify mac
no dhcp-snooping verify mac
```

Description

This command enables verification of the hardware address field in DHCP client packets. When enabled, the DHCP client hardware address field and the source MAC address must be the same for packets received on untrusted ports or else the packet is dropped. This DHCP snooping MAC verification is enabled by default.

The **no** form of the command disables DHCP snooping MAC verification.

Examples

Enabling DHCP snooping MAC verification:

```
switch(config) # dhcp-snooping verify mac
```

Disabling DHCP snooping MAC verification:

```
switch(config) # no dhcp-snooping verify mac
```

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show dhcp-snooping

Syntax

```
show dhcp-snooping
```

Description

Shows the DHCP snooping configuration.

Examples

Showing the DHCP snooping configuration:

```
switch# show dhcp-snooping
dhcp-snooping Information
dhcp-snooping                : Yes                Verify MAC Address      : Yes
Allow Overwrite Binding      : No                Enabled VLANs           :
1-100
Static Attributes            : Yes
Client Event Logs            : Yes
Option 82 Configurations
Untrusted Policy              : replace              Insertion                 : Yes
Option 82 Remote-id          : mac
External Storage Information
Volume Name                   : ipbinding
File Name                     : ipv4Bindings
Inactive Since                : 01:23:20 09/10/2021
Error                         : File Write Failure
Flash Storage Information
File Write Delay              : 300 seconds
Active Storage                : External
Authorized Server Configurations
VRF                            Authorized Servers
-----
default                        1.1.10.3
default                        10.10.10.1
default                        10.10.10.56
default                        200.10.10.3
green                          1.1.10.3
green                          1.10.10.3
green                          10.10.100.3
red                             192.168.122.53
red                             192.168.122.121
Port Information
Port      Trust      Max      Static      Dynamic
-----  -
1/1/2     Yes       5000     50          0
1/1/3     Yes       8192     0           0
1/1/5     Yes       8192     0           22
1/1/16    No        100      0           0
10/10/10  No       8100     320         200
lag120    No        512      0           0
```

Command History

Release	Modification
10.14	The dhcpx4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Updated example with flash storage information.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show dhcp-snooping binding

Syntax

```
show dhcp-snooping binding
```

Description

Shows the DHCP snooping binding configuration.

Parameter	Description
detail	Shows detailed information for active IP bindings on the system.

Examples

Showing the DHCP snooping binding configuration:

```
switch(config)# show dhcp-snooping binding
```

MacAddress	IP	VLAN	Interface	Time-Left
aa:b1:c1:dd:ee:ff	10.2.3.4	1	1/1/2	582
aa:b2:c2:dd:ee:ff	10.2.3.5	1	1/1/2	584

Showing detailed information for active IP bindings:

```
switch(config)# show dhcp-snooping binding detail
```

VLAN Id : 2, MAC : 00:50:56:96:74:46

IP	Interface	Time-Left
100.1.2.100	1/1/23	194

Static Attributes:

Default Router : 100.1.2.1, 192.1.1.1, 1.1.1.2

Server IP : 10.1.84.2

Name Servers : 192.1.1.2, 2.2.2.2, 1.1.1.1

VLAN Id : 3, MAC : 00:50:56:96:e5:8e

IP	Interface	Time-Left
100.1.3.100	2/1/22	145

Static Attributes:

Default Router : 100.1.3.1, 192.1.1.1, 1.1.1.2

Server IP : 10.1.84.2

Name Servers : 192.1.1.2, 2.2.2.2, 1.1.1.1

VLAN Id : 3, MAC : 00:11:01:00:00:03

IP	Interface	Time-Left
100.1.3.99	2/1/24	137

Static Attributes:

Default Router : 100.1.3.1, 192.1.1.1, 1.1.1.2

Server IP : 10.1.84.2

Name Servers : 192.168.0.1, 192.168.1.1, 192.168.2.1

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.10	Detail parameter added.

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show dhcp-snooping statistics

Syntax

```
show dhcp-snooping statistics
```

Description

Shows the DHCP snooping statistics.

Examples

Showing the DHCP snooping statistics:

```
switch(config) # show dhcp-snooping statistics
Packet-Type  Action  Reason                                     Count
-----
server       forward  from trusted port                        5425
client       forward  to trusted port                         3895
server       drop     received on untrusted port              117
server       drop     unauthorized server                     214
client       drop     destination on untrusted port           78
client       drop     untrusted option 82 field                85
client       drop     bad DHCP release request                 0
```

client	drop	failed verify MAC check	5
client	drop	failed on max-binding limit	15

Command History

Release	Modification
10.14	The dhcpv4 - snooping keyword is deprecated and replaced with dhcp - snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

DHCPv6 snooping commands

Subtopics

[DHCPv6 snooping commands](#)

DHCPv6 snooping commands

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clear dhcpv6-snooping binding

Syntax

```
clear dhcpv6-snooping binding {all | ip <IPV6-ADDR> vlan <VLAN-ID> |
interface <IFNAME> | vlan <VLAN-ID>}
```

Description

Clears DHCPv6 snooping binding entries.

Parameter	Description
all	Specifies that all DHCPv6 binding information is to be cleared.
ip <IPV6-ADDR> vlan <VLAN-ID>	Specifies the IPv6 address and VLAN for which all DHCPv6 binding information is to be cleared.
interface <IFNAME>	Specifies the interface for which all DHCPv6 binding information is to be cleared.
vlan <VLAN-ID>	Specifies the VLAN for which all DHCPv6 binding information is to be cleared. Range: 1 to 4094.

Examples

Clearing all DHCPv6 binding information for 5000::1 vlan 1:

```
switch(config) # clear dhcpv6-snooping binding ip 5000::1 vlan 1
```

Clearing all DHCPv6 binding information for interface 1/1/10:

```
switch(config) # clear dhcpv6-snooping binding interface 1/1/10
```

Clearing all DHCPv6 binding information for VLAN 10:

```
switch(config) # clear dhcpv6-snooping binding vlan 10
```

Clearing all DHCPv6 binding information:

```
switch(config) # clear dhcpv6-snooping binding all
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

clear dhcpv6-snooping statistics

Syntax

```
clear dhcpv6-snooping statistics
```

Description

Clears all DHCPv6 snooping statistics.

Examples

Clear all DHCPv6 snooping statistics:

```
switch# clear dhcpv6-snooping statistics
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

dhcpv6-snooping

Syntax

```
dhcpv6-snooping  
no dhcpv6-snooping
```

Description

Enables DHCPv6 snooping. DHCPv6 snooping is disabled by default. DHCPv6 snooping is not supported on the management interface.

The no form of the command disables DHCPv6 snooping, flushing all the IP bindings learned since DHCPv6 snooping was enabled.

Examples

Enabling DHCPv6 snooping:

```
switch(config) # dhcpv6-snooping
```

Disabling DHCPv6 snooping:

```
switch(config) # no dhcpv6-snooping
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping allow-ipv6-ext-headers

Syntax

```
[no] dhcpv6-snooping allow-ipv6-ext-headers
```

Description

This command enables or disables forwarding of DHCPv6 packets containing any IPv6 extension headers. By default, such packets are **parsed and dropped** for DHCPv6 Guard policy (RFC 7610 compliance). When enabled, these packets are **forwarded** instead of dropped.

Examples

Enabling forwarding:

```
switch# configure terminal
switch(config)# dhcpv6-snooping allow-ipv6-ext-headers
```

Disabling forwarding (default as per RFC 7610):

```
switch# configure terminal
switch(config)# no dhcpv6-snooping allow-ipv6-ext-headers
```

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
	Manager (#)	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping (in config-vlan context)

Syntax

```
dhcpv6-snooping
no dhcpv6-snooping
```

Description

Enables DHCPv6 snooping in the `config-vlan` context. DHCPv6 snooping is disabled by default for all VLANs.

The no form of the command disables DHCPv6 snooping on the specified VLAN, flushing all the IPv6 bindings learned for this VLAN since DHCPv6 snooping was enabled for this VLAN.

Examples

Enabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100  
switch(config-vlan-100)# no dhcpv6-snooping  
switch(config-vlan-100)# exit  
switch(config)#
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-vlan	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping authorized-server

Syntax

```
dhcpv6-snooping authorized-server <IPV6-ADDR> [vrf <VRF-NAME>]  
no dhcpv6-snooping authorized-server <IPV6-ADDR> [vrf <VRF-NAME>]
```

Description

Adds an authorized (trusted) DHCPv6 server to a list of authorized servers for use by DHCPv6 snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By

default, with an empty list of authorized servers, all DHCPv6 servers are considered to be trusted for DHCPv6 snooping purposes.



NOTE

The `mgmt` VRF cannot be used with this command.



NOTE

Configure the link local IPv6 address instead of global IPv6 address of the DHCPv6 server as the authorized-server. For example:

```
switch(config)# dhcpv6-snooping authorized-server  
fe80::2ca4:fa40:d4cd:bc2f
```

The no form of this command deletes the specified DHCPv6 server from the authorized list.

Parameter	Description
<code><IPv6-ADDR></code>	Specifies the IPv6 address of the trusted DHCPv6 server.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.

Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

Examples

Adding DHCP servers ABCD:5ACD::2000, and ABCD:5ACD::2010 to the authorized server list:

```
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf  
default  
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2010 vrf  
default
```

Removing DHCP server ABCD:5ACD::2000 from the authorized server list:

```
switch(config)# no dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf  
default
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping drop-unknown-ipv6-ext-headers

Syntax

```
[no] dhcpv6-snooping drop-unknown-ipv6-ext-headers
```

Description

This command enables or disables dropping of DHCPv6 packets containing **unknown or unsupported IPv6 extension headers** (i.e., Next Header values not defined in RFC 8200 or not supported by the implementation). By default, such packets are forwarded. When this command is configured, these packets are **dropped** and logged (RFC 7610 compliance).

Examples

Dropping packets with unknown IPv6 extension headers:

```
switch# configure terminal
switch(config)# dhcpv6-snooping drop-unknown-ipv6-ext-headers
```

Forwarding packets with unknown IPv6 extension headers (default):

```
switch# configure terminal
switch(config)# no dhcpv6-snooping drop-unknown-ipv6-ext-headers
```


Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
	Manager (#)	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping event-log client

Syntax

```
dhcpv6-snooping event-log client
no dhcpv6-snooping event-log client
```

Description

This command enables or disables DHCPv6 snooping client level event logs that help with client telemetry on a remote management station such as HPE Aruba Networking Central. By default, client level event logs are disabled. The **no** form of this command disables client-level event logs for DHCPv6 snooping after they are enabled. View these logged DHCPv6 snooping events by issuing the command `show events -c dhcpv6-snooping`.



NOTE

For additional information on DHCP-related event logging, please refer to the Event Log Message Reference Guide.

Examples

Enabling DHCPv6 client level event logs:

```
switch(config)# # dhcpv6-snooping event-log client
```

Disabling external storage:

```
witch(config)# # no dhcpv6-snooping event-log client
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping external-storage

Syntax

```
dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>  
no dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

Description

Configures external storage to be used for backing up IPv6 bindings (used by DHCPv6 snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IPv6 bindings from the configured external storage file to populate its local cache.



NOTE

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IPv6 bindings in an external storage file.

Parameter	Description
<code>volume <VOL-NAME></code>	Specifies the name of the existing external storage volume where the IPv6 bindings file will be saved. Before running the <code>dhcpv6-snooping external-storage volume</code> command, first create the external storage volume using command <code>external-storage <VOLUME-NAME></code> . See External storage commands in the Command-Line Interface Guide.
<code>file <FILE-NAME></code>	Specifies the file name to use for storing IPv6 bindings. Maximum 255 characters.

Examples

Configuring IPv6 bindings storage in file `ipv6Bindings` on existing volume `dhcp_snoop`:

```
switch(config)# dhcpv6-snooping external-storage volume dhcp_snoop file
ipv6Bindings
```

Disabling external storage:

```
switch(config)# no dhcpv6-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)# no dhcpv6-snooping external-storage volume dhcp_snoop
DHCPv6-Snooping will use flash storage to store IP Binding database
switch(config)#
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Updated example with flash storage information.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping flash-storage

Syntax

```
dhcpv6-snooping flash-storage [delay <DELAY>]  
no dhcpv6-snooping flash-storage [delay <DELAY>]
```

Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCPv6 snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting `delay <DELAY>` sets the default delay of 900 seconds.



CAUTION

To reduce switch flash aging it is recommended that you use external storage (command `dhcpv6-snooping external-storage`) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.



NOTE

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in flash storage.

Parameter	Description
delay <DELAY>	Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400.

Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcpv6-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to
use an external-storage.
Do you want to continue (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcpv6-snooping flash-storage
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.
6000		
6100		

dhcpv6-snooping max-bindings

Syntax

```
dhcpv6-snooping max-bindings <MAX-BINDINGS>
no dhcpv6-snooping max-bindings <MAX-BINDINGS>
```

Description

Sets the maximum number of DHCPv6 bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max binding is the maximum value of the range.

The no form of the command reverts max bindings for the selected interface to its default.

Parameter	Description
<code><MAX-BINDINGS></code>	Specifies the maximum number of DHCP bindings. Range: 1 to 256.

Examples

Set the DHCPv6 max bindings to 256 on interface 1/1/1:

```
switch(config) # interface 1/1/1
switch(config-if) # dhcpv6-snooping max-bindings 256
switch(config-if) # exit
switch(config) #
```

Revert DHCPv6 max bindings to its default on interface 1/1/1:

```
switch(config) # interface 1/1/1
switch(config-if) # no dhcpv6-snooping max-bindings 256
switch(config-if) # exit
switch(config) #
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping trust

Syntax

```
dhcpv6-snooping trust  
no dhcpv6-snooping trust
```

Description

Enables DHCPv6 snooping trust on the selected interface. Only server packets received on trusted interfaces are forwarded. All the interfaces are untrusted by default.

The no form of the command disables DHCPv6 snooping trust on the selected interface.

```
config-if
```

Examples

Enabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1  
switch(config-if)# dhcpv6-snooping trust  
switch(config-if)# exit  
switch(config)#
```

Disabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1  
switch(config-if)# no dhcpv6-snooping trust  
switch(config-if)# exit  
switch(config)#
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show dhcpv6-snooping

Syntax

```
show dhcpv6-snooping
```

Description

Shows the DHCPv6 snooping configuration.

Examples

Showing the DHCPv6 snooping configuration:

```
switch# show dhcpv6-snooping
DHCPv6-Snooping Information
  DHCPv6-Snooping      : Yes   Enabled VLANs      : 1,5,7,100-110
  Trusted Port Bindings Enabled VLANs :
  Client Event Logs    : Yes
External Storage Information
  Volume Name          : dhcp_snoop
  File Name            : ip_binding
  Inactive Since       : 01:23:20 09/10/2021
  Error                : Failed to write external storage
```


Flash Storage Information

File Write Delay : 300 seconds

Active Storage : External

Authorized Server Configurations

VRF

Authorized Servers

default

2001:0db8:85a3:0000:0000:8a2e:0370:7334

default

2002::2

default

2004::1

red

2002::1

red

2002::2

red

2002::9

green

5000::1

green

5000::2

green

5000::3

green

5000::7

green

5000::8

Port Information

Port	Trust	Max Bindings	Static Bindings	Dynamic Bindings
-----	-----	-----	-----	-----
1/1/2	Yes	0	0	0
1/1/3	Yes	0	3	0
1/1/5	Yes	0	22	0
1/1/16	No	256	0	20
10/10/10	No	256	12	7
lag120	No	256	3	0

Command History

Release

Modification

10.09 Command introduced for the 6000 and 6100 Switch Series.

10.08 Updated example with flash storage information.

10.07 or earlier Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show dhcpv6-snooping binding

Syntax

```
show dhcpv6-snooping binding
```

Description

Shows the DHCPv6 snooping binding configuration.

Examples

Showing the DHCPv6 snooping binding configuration:

```
switch# show dhcpv6-snooping binding
IP Binding Information
=====
MAC-ADDRESS      IPV6-ADDRESS      VLAN
INTERFACE      TIME-LEFT
-----
00:50:56:96:e4:cf  aaaa:bbbb:cccc:dddd:eeee:1234:5678:abcd      1
1/1/1            584
00:50:56:96:04:4d  1000::3      134
1/1/2            435
00:50:56:96:d8:3d  2000:1000::4      2002
lag123          21234
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show dhcpv6-snooping statistics

Syntax

```
show dhcpv6-snooping statistics
```

Description

Shows the DHCPv6 snooping statistics.

Examples

Showing the DHCPv6 snooping statistics:

```
switch(config)# show dhcpv6-snooping statistics
Packet-Type  Action   Reason                                     Count
-----
server       forward  from trusted port                         12
client       forward  to trusted port                           20
server       drop     received on untrusted port                5
server       drop     unauthorized server                       4
client       drop     destination on untrusted port             2
```

client	drop	bad DHCP release request	5
server	drop	relay reply on untrusted port	2
client	drop	failed on max-binding limit	5

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

Troubleshooting

DHCP client not receiving IP

1. Trusted port configuration

Check if the server reachable port is configured as trusted.

2. Validate Packet drop counters

Use `show dhcpv4-snooping statistics` and `show dhcpv6-snooping statistics` commands to view the packet drop counters. This helps to identify a possible reason for the termination of DHCP packet exchange between client and server. A few possible reasons are listed below:

- Maximum binding limit reached
 - System-wide limit—The maximum binding limit is shared between DHCPv4-Snooping, DHCPv6-Snooping, ND-Snooping, and IP-Source-Bindings. On reaching the limit, DHCP client requests get dropped. To check the limit, use the `show capacities ip-bindings` command.

- Per port limit: This is a per port configurable value for a protocol. On reaching the limit, the subsequent client requests are dropped.
 - Conflicts with IP-Source-Bindings—User-configured IP-Source-Bindings are given priority over dynamically learned bindings. Validate if the dynamic client is conflicting with any of the existing IP-Source-Binding.
 - Authorized server check failures—Authorized server configuration is global. When one or more authorized servers are set up, all server replies should come from one of the authorized server IP addresses.
 - CoPP packet drops—If the rate of the DHCP packets entering is higher than the configured value, excessive packets are dropped by CoPP. Check CoPP statistics for the DHCP class and configure the value accordingly. Higher CoPP values could cause other protocols to misbehave depending on the platform.
3. Server configuration and reachability
Validate the following to ensure that the DHCP Server is reachable and configured correctly.
 - a. Check that the server is reachable over one of the trusted ports.
 - b. Check the server logs to understand the DHCP exchange status between client and server.
 - c. Check if DHCP Server pools are exhausted.
 4. Option-82 configuration issues (Specific to DHCP Snooping IPv4)
 - a. This configuration usually arises when DHCP Snooping and Relay are configured on the same VLAN. Option 82 at DHCP Snooping IPv4 should be disabled when configured for Inter-VRF DHCP-Relay. Use `no dhcpv4-snooping option 82` command to disable the functionality.
 - b. Option 82 is enabled by default, and the drop policy is set. If client packets are received with option 82, then they are dropped with this configuration.
 5. Server and client are connected to same the VLAN
 - a. Check if the server is reachable on the same VLAN as the client.
 - b. If the server is on a different L3 network, configure DHCP-Relay to forward the packets between the client and server.
 6. Authorized server configurations—When configured, the server assigning IP address should be one of the IP addresses from the authorized server list.
In DHCP snooping IPv6, if DHCP Relay is between a test device and a server, the authorised server configuration should include a link to the local IP address of the relay interface. This is because, while forwarding the server's response to the client, it will stamp a link to the local IP address as the source IP of the packet.

IP-Binding mismatch in VSX topologies

1. Both primary and secondary VSX peers should have the same clock value. If not, there is a possibility of inconsistent ip-bindings that could result in ip-binding sync in a loop. There is no straight-forward way to check the inconsistencies but by looking at the number of bindings and debug log analysis
2. Ensure the ISL state is up and operational.

DHCP client IP traffic is dropped at data path

1. This can happen when IP-Source-Lockdown is used along with DHCP snooping. Ensure IP-Bindings are learned for clients that are experiencing traffic failure issues
2. Check the hardware programming state of the IP-Binding entry by IP-Source-Lockdown. Use `show ipv4 source-lockdown bindings` or `show ipv6 source-lockdown bindings` commands.

DHCP Options

A number of deployments provide access to WAN or public network only through a proxy server for the network security. This section provides information about the various methods of HTTP proxy configuration on the switch. The HTTP proxy configured is used to connect to HPE Aruba Networking Central as per the zero-touch provisioning requirement. When HTTP proxy location and VRF are configured, they override any existing HTTP proxy location and VRF.

HTTP proxy location can be configured using the CLI or REST interface or auto-configured through the DHCPv4/DHCPv6 server connected to the switch.

There are four sources for HTTP proxy location:

- User configured HTTP proxy via CLI or REST interface.
- DHCPv4 options received via the management/OOBM port
- DHCP options received via VLAN 1 on supported switch platforms.
- DHCPv6 options received via management/OOBM port

Operational configuration for HTTP proxy location is determined by the source with the highest priority. The source priority is as follows:

1. User configured.
2. DHCPv4 options received via the management/OOBM port.
3. DHCP options received via VLAN 1.

4. DHCPv6 options received via the management/OOBM port



NOTE

- DHCPv6 options support FQDN and IPv6 addresses, while DHCPv4 options support FQDN and IPv4 addresses.
- DHCPv6 support is available only on the management/OOBM interface and provides the lowest priority in the arbitration hierarchy.

An **HTTP proxy** FQDN or IP address can be sent with DHCPv4 option 43 sub-option 148 (which supports an FQDN or IPv4 address) or DHCPv6 option 17 sub-option 148 (which supports an FQDN or IPv6 address).

An **HPE Aruba Networking Central** FQDN or IP address can be sent using DHCPv4 option 43 sub-option 146 (which supports an FQDN or IPv4 address) or DHCPv6 option 17 sub-option 146 (which supports an FQDN or IPv6 address).

When the switch uses zero-touch provisioning with HPE Aruba Networking Central, the HPE Aruba Networking Central location, shared secret and alternate location sent using DHCPv6 option 17 suboption 146 is in the format **<group-details>,<aruba-central-fqdn-or-ipv6>,<shared-secret>**, or for dual cluster support, **<group-details>,<aruba-central-fqdn-or-ipv6>,<shared-secret>,<alt-aruba-central-fqdn-or-ipv6>**.

Subtopics

DHCP options commands

DHCP options commands

Select a command from the list in the left navigation menu.

Subtopics

http-proxy

http-proxy

Syntax

```
http-proxy {<FQDN | {IPV4-ADDR|IPV6-ADDR[:PORT]}} [vrf <VRF-NAME>]  
no http-proxy [<FQDN | {IPV4-ADDR|IPV6-ADDR[:PORT]}} [vrf <VRF-NAME>]
```

Description

Specifies HTTP proxy location and VRF.

When HTTP proxy location and VRF are configured on the switch, it overrides any existing HTTP proxy location and VRF as this has the highest priority over the values obtained from other sources.

Following locations can be used for the HTTP proxy location:

- A fully qualified domain name (FQDN).
- An IPv4 address with colon separated port number
- An IPv6 address with colon separated port number



NOTE

When configuring an IPv6 address with a port number, the address must be specified inside square brackets. An example - [2001:0db8:85a3:0000:0000:8a2e:0370:7334]:8080.

If the command is entered without the VRF parameter, then the VRF used will be 'default' VRF.

The **no** form of this command removes a specified HTTP proxy location.

Parameter	Description
<FQDN>	Specifies FQDN for HTTP proxy location.
<IPV4-ADDR>	Specifies IPV4 address for HTTP proxy location.
<IPV6-ADDR>	Specifies IPV6 address for HTTP proxy location.
<VRF-NAME>	Specifies VRF for HTTP proxy.



NOTE

A FQDN or IPV4 address is optional in the **no** form of the command.

Examples

Specifying a FQDN for HTTP proxy location and **MGMT** VRF with an IPv6 address:

```
switch(config) # http-proxy http-proxy.example.com vrf mgmt
switch(config) # http-proxy [2000::100]:8080 vrf mgmt
```


Specifying a FQDN for HTTP proxy location and **MGMT** VRF with an IPv4 address:

```
switch(config) # http-proxy http-proxy.example.com vrf mgmt
switch(config) # http-proxy 192.168.1.3:8080 vrf mgmt
```

Removing HTTP proxy location

```
switch(config) # no http-proxy
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

IP Source Lockdown

IP source lockdown provides added security by preventing IP source address spoofing on a per-port basis. Every packet is inspected for this purpose in hardware. When IP source lockdown is enabled, IP traffic received on an interface (port) is forwarded only if the VLAN, IP address, MAC address, interface (port) matches the IP binding database entry.



NOTE

It is best to configure IP source lockdown during a switch maintenance period as enabling it may cause client traffic to be dropped for 10 to 15 seconds.

To use IPv4 source lockdown, the IPv4 binding database must be populated. The binding database is typically dynamically populated by DHCPv4 snooping that learns and saves the binding information. Alternatively, the IPv4 binding database can be statically populated with the `ipv4 source-binding` command described in this chapter. Often DHCPv4 snooping is used to dynamically populate most of the IP binding database along with the `ipv4 source-binding` command that is used to add the binding

information for several known and trusted clients, typically administrators. For dynamic IP binding database population with DHCPv4 snooping, see [DHCP snooping](#).

To use IPv6 source lockdown, the IPv6 binding database must be populated. The binding database is typically dynamically populated by DHCPv6 snooping that learns and saves the binding information. Alternatively, the IPv6 binding database can be statically populated with the `ipv6 source-binding` command described in this chapter. Often DHCPv6 snooping is used to dynamically populate most of the IPv6 binding database along with the `ipv6 source-binding` command that is used to add the binding information for several known and trusted clients, typically administrators. For dynamic IPv6 binding database population with DHCPv6 snooping, see [DHCP snooping](#).



NOTE

IP source lockdown should not be configured on ISL (inter-switch link) ports.

Subtopics

[IP source lockdown commands](#)

[IPv4 source lockdown commands](#)

[IPv6 source lockdown commands](#)

IP source lockdown commands

IPv4 source lockdown commands

Subtopics

[IPv4 source lockdown commands](#)

IPv4 source lockdown commands

Select a command from the list in the left navigation menu.

Subtopics

[ipv4 source-binding](#)

[ipv4 source-lockdown](#)

[ipv4 source-lockdown hardware retry](#)

[show ipv4 source-binding](#)

[show ipv4 source-lockdown](#)

ipv4 source-binding

Syntax

```
ipv4 source-binding <VLAN-ID>
    <IPV4-ADDR>
    <MAC-ADDR>
    <IFNAME>
no ipv4 source-binding <VLAN-ID>
    <IPV4-ADDR>
    <MAC-ADDR>
    <IFNAME>
```

Description

Adds static IPv4 client source binding information to the switch IP binding database. Although DHCPv4 snooping is often used to dynamically populate the binding database, this command is available for manually adding entries to the switch IP binding database.



NOTE

Statically configured IP binding information supersedes any dynamically collected binding information for the same client.

The no form of this command removes the specified binding that was statically configured with the **ipv4 source-binding** command. The no form has no effect on bindings that were dynamically configured with DHCPv4 snooping.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPV4-ADDR>	Specifies the client IPv4 unicast address.
	Specifies the client MAC address.

Parameter	Description
<MAC-ADDR>	
<IFNAME>	Specifies the interface on which the client is connected.

Examples

Adding a static IPv4 binding:

```
switch(config) # ipv4 source-binding 1 10.2.1.4 00:50:56:96:e4:cf 1/1/1
```

Removing a IPv4 binding:

```
switch(config) # no ipv4 source-binding 1 10.2.1.4 00:50:56:96:e4:cf 1/1/1
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ipv4 source-lockdown

Syntax

```
ipv4 source-lockdown
no ipv4 source-lockdown
```

Description

Enables IPv4 source lockdown for all VLANs on the selected interface (port).

The no form of this command disables IPv4 source lockdown for the selected interface (port).



CAUTION

This configuration will disable flow tracking statistics collection.

Examples

Enabling IPv4 source lockdown on interface 1/1/1:

```
switch(config) # interface 1/1/1
switch(config-if) # ipv4 source-lockdown
```

Enabling IPv4 source lockdown on interface lag112:

```
switch(config) # interface lag112
switch(config-if) # ipv4 source-lockdown
```

Disabling IPv4 source lockdown on interface 1/1/1:

```
switch(config) # interface 1/1/1
switch(config-if) # no ipv4 source-lockdown
```

Command History

Release	Modification
10.14	Added information related to role based IPFIX.
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

ipv4 source-lockdown hardware retry

Syntax

```
ipv4 source-lockdown hardware retry <VLAN-ID>  
                                <IPV4-ADDR>
```

Description

Retries the IPv4 source lockdown hardware programming for a client identified by VLAN and IPv4 address.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPV4-ADDR>	Specifies the client IPv4 unicast address.

Example

Configure IPv4 source lockdown hardware retry for the client on VLAN 10.

```
switch(config) # ipv4 source-lockdown hardware retry 10 1.1.2.1
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show ipv4 source-binding

Syntax

```
show ipv4 source-binding
```

Description

Shows all IPv4 static source binding information irrespective of source lockdown configuration..

Examples

Showing all IPv4 source binding information:

```
switch# show ipv4 source-binding
  PORT          VLAN      MAC-ADDRESS      HW-STATUS  FROM      IPv4-
ADDRESS
-----
1/1/1          2        aa:bb:cc:dd:ee:ff  Yes        static    1.2.3.4
1/1/2          12       aa:ab:cc:dd:ee:ff  Yes        static
10.20.30.40
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv4 source-lockdown

Syntax

```
show ipv4 source-lockdown [binding [interface <IFNAME> | ip <IPV4-ADDR> |  
mac <MAC-ADDR> | vlan <VLAN-ID>] | interface <IFNAME>]
```

Description

Shows summary or detailed IPv4 source lockdown information. When entered without parameters, summary status information for all interfaces (ports) in the binding database is shown.

Parameter	Description
binding	Specifies that detailed lockdown binding record information is to be displayed. The binding database record can be identified by any one of interface (port), ip , mac , or vlan .
interface <IFNAME>	Specifies the client interface (port). When entered without the binding parameter, the summary status information is displayed for the specified interface.
ip <IPV4-ADDR>	Specifies the client IPv4 unicast address.

Parameter	Description
mac <MAC-ADDR>	Specifies the client MAC address.
vlan <VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

Examples

Showing the summary status information for all interfaces in the binding database:

```
switch# show ipv4 source-lockdown
  INTERFACE  LOCKDOWN  HW-STATUS
  -----
  1/1/1      Yes      Yes
  1/1/2      Yes      No
  lag112     Yes      Yes
```

Showing the summary status information for the specified interface in the binding database:

```
switch# show ipv4 source-lockdown interface 1/1/2
  INTERFACE  LOCKDOWN  HW-STATUS
  -----
  1/1/2      Yes      No
```

Showing the detailed binding record and related information for all interfaces in the binding database:

```
switch# show ipv4 source-lockdown binding
Interface Name      : 1/1/1
VLAN Id             : 2000
MAC Address         : 00:50:56:96:e4:cf
IP Address          : 192.168.142.113
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --
Interface Name      : 1/1/2
VLAN Id             : 100
MAC Address         : 00:50:56:96:04:4d
IP Address          : 120.168.43.52
Time Remaining      : 115 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable
Interface Name      : lag112
VLAN Id             : 12
MAC Address         : 00:50:56:96:d8:3d
```

```

IP Address      : 120.168.76.182
Time Remaining  : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --
Interface Name  : 1/1/1
VLAN Id         : 2000
MAC Address     : 00:50:56:96:e4:cf
IP Address      : 192.168.142.113
Time Remaining  : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --
Interface Name  : 1/1/2
VLAN Id         : 100
MAC Address     : 00:50:56:96:04:4d
IP Address      : 120.168.43.52
Time Remaining  : 115 seconds
Lockdown Status : Yes
Hardware Status : No
Hardware Error Reason : Resource unavailable
Interface Name  : lag112
VLAN Id         : 12
MAC Address     : 00:50:56:96:d8:3d
IP Address      : 120.168.76.182
Time Remaining  : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --

```

Showing the detailed binding record and related information for interface 1/1/2:

```
switch# show ipv4 source-lockdown binding interface 1/1/2
```

```

Interface Name  : 1/1/2
VLAN Id         : 100
MAC Address     : 00:50:56:96:04:4d
IP Address      : 120.168.43.52
Time Remaining  : 115 seconds
Lockdown Status : Yes
Hardware Status : No
Hardware Error Reason : Resource unavailable
Interface Name  : 1/1/2
VLAN Id         : 100
MAC Address     : 00:50:56:96:04:4d
IP Address      : 120.168.43.52

```

```
Time Remaining      : 115 seconds
Lockdown Status     : Yes
Hardware Status      : No
Hardware Error Reason : Resource unavailable
```

Showing the detailed binding record and related information for interface lag112 (identified in this example command by the IP address):

```
switch# show ipv4 source-lockdown binding ip 120.168.76.182
Interface Name      : lag112
VLAN Id             : 12
MAC Address         : 00:50:56:96:d8:3d
IP Address          : 120.168.76.182
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status      : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/1 (identified in this example command by the MAC address):

```
switch# show ipv4 source-lockdown binding mac 00:50:56:96:e4:cf
Interface Name      : 1/1/1
VLAN Id             : 2000
MAC Address         : 00:50:56:96:e4:cf
IP Address          : 192.168.142.113
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status      : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/2 (identified in this example command by the VLAN):

```
switch# show ipv4 source-lockdown binding vlan 100
Interface Name      : 1/1/2
VLAN Id             : 100
MAC Address         : 00:50:56:96:04:4d
IP Address          : 120.168.43.52
Time Remaining      : 115 seconds
Lockdown Status     : Yes
Hardware Status      : No
Hardware Error Reason : Resource unavailable
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

IPv6 source lockdown commands

Subtopics

IPv6 source lockdown commands

IPv6 source lockdown commands

Select a command from the list in the left navigation menu.

Subtopics

ipv6 source-binding

ipv6 source-lockdown

ipv6 source-lockdown hardware retry

show ipv6 source-binding

show ipv6 source-lockdown

ipv6 source-binding

Syntax

```
ipv6 source-binding <VLAN-ID>
    <IPV6-ADDR>
    <MAC-ADDR>
    <IFNAME>
no ipv6 source-binding <VLAN-ID>
    <IPV6-ADDR>
    <MAC-ADDR>
    <IFNAME>
```

Description

Adds static IPv6 client source binding information to the switch IPv6 binding database. Although DHCPv6 snooping is often used to dynamically populate the binding database, this command is available for manually adding entries to the switch IPv6 binding database.



NOTE

Statically configured IPv6 binding information supersedes any dynamically collected binding information for the same client.

The no form of this command removes the specified binding that was statically configured with the **ipv6 source-binding** command. The no form has no effect on bindings that were dynamically configured with DHCPv6 snooping.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPV6-ADDR>	Specifies the client IPv6 address.
<MAC-ADDR>	Specifies the client MAC address.

Parameter	Description
<IFNAME>	Specifies the interface on which the client is connected.

Examples

Adding a static IPv6 binding:

```
switch(config) # ipv6 source-binding 2 2000::2 00:12:11:44:55:12 1/1/28
```

Removing a IPv6 binding:

```
switch(config) # no ipv6 source-binding 2 2000::2 00:12:11:44:55:12 1/1/28
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.
6000		
6100		

ipv6 source-lockdown

Syntax

```
ipv6 source-lockdown
no ipv6 source-lockdown
```

Description

Enables IPv6 source lockdown for all VLANs on the selected interface (port).

The no form of this command disables IPv6 source lockdown for the selected interface (port).



CAUTION

This configuration will disable flow tracking statistics collection.

Examples

Enabling IPv6 source lockdown on interface 1/1/1:

```
switch(config) # interface 1/1/1
switch(config-if) # ipv6 source-lockdown
```

Enabling IPv6 source lockdown on interface lag112:

```
switch(config) # interface lag112
switch(config-if) # ipv6 source-lockdown
```

Disabling IPv6 source lockdown on interface 1/1/1:

```
switch(config) # interface 1/1/1
switch(config-if) # no ipv6 source-lockdown
```

Command History

Release	Modification
10.14	Added information related to role based IPFIX.
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

ipv6 source-lockdown hardware retry

Syntax

```
ipv6 source-lockdown hardware retry <VLAN-ID>  
                                <IPv6-ADDR>
```

Description

Retries the IPv6 source lockdown hardware programming for a client identified by VLAN and IPv6 address.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPv6-ADDR>	Specifies the client IPv6 address.

Example

Configure IPv6 source lockdown hardware retry for the client on VLAN 1.

```
switch(config)# ipv6 source-lockdown hardware retry 1 2000::2
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show ipv6 source-binding

Syntax

```
show ipv6 source-binding
```

Description

Shows all IPv6 static source binding information irrespective of source lockdown configuration.

Examples

Showing all IPv6 source binding information:

```
switch#
```

show ipv6 source-binding

```
PORT VLAN MAC-ADDRESS HW-STATUS FROM IPv6-ADDRESS -----  
----- 1/1/1 1234 00:50:56:96:e4:cf Yes/No static 3000::1 1/1/1 1 00:50:56:96:04:4d Yes/No  
static 3000::2 1/1/24 1 00:01:01:00:00:01 Yes static 1001::1
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 source-lockdown

Syntax

```
show ipv6 source-lockdown [binding [interface <IFNAME> | ip <IPV6-ADDR> |  
mac <MAC-ADDR> | vlan <VLAN-ID>] | interface <IFNAME>]
```

Description

Shows summary or detailed IPv6 source lockdown information. When entered without parameters, summary status information for all interfaces (ports) in the binding database is shown.

Parameter	Description
binding	Specifies that detailed lockdown binding record information is to be displayed. The binding database record can be identified by any one of interface (port), ip , mac , or vlan .
interface <IFNAME>	Specifies the client interface (port). When entered without the binding parameter, the summary status information is displayed for the specified interface.
ip <IPV6-ADDR>	Specifies the client IPv6 address.
mac <MAC-ADDR>	Specifies the client MAC address.
vlan <VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

Examples

Showing the summary status information for all interfaces in the binding database:

```
switch# show ipv6 source-lockdown
```

INTERFACE	LOCKDOWN	HW-STATUS
-----	-----	-----
1/1/1	Yes	Yes
1/1/2	Yes	Yes
lag112	Yes	Yes

Showing the summary status information for the specified interface in the binding database:

```
switch# show ipv6 source-lockdown interface 1/1/2
```

INTERFACE	LOCKDOWN	HW-STATUS
-----	-----	-----
1/1/2	Yes	No

Showing the detailed binding record and related information for all interfaces in the binding database:

```
switch# show ipv6 source-lockdown binding
```

Interface Name	: 1/1/1
VLAN Id	: 1234
MAC Address	: 00:50:56:96:e4:cf
IP Address	: aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining	: static
Lockdown Status	: Yes
Hardware Status	: Yes
Hardware Error Reason	: --
Interface Name	: 1/1/2
VLAN Id	: 1234
MAC Address	: 00:50:56:96:04:4d
IP Address	: 4000::1
Time Remaining	: 3290 seconds
Lockdown Status	: Yes
Hardware Status	: No
Hardware Error Reason	: Resource unavailable
Interface Name	: lag112
VLAN Id	: 151
MAC Address	: 00:50:56:96:d8:3d
IP Address	: 1001::5
Time Remaining	: 1200 seconds
Lockdown Status	: No
Hardware Status	: Yes
Hardware Error Reason	: --
Interface Name	: 1/1/1
VLAN Id	: 1234
MAC Address	: 00:50:56:96:e4:cf
IP Address	: aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining	: static
Lockdown Status	: Yes

```

Hardware Status      : Yes
Hardware Error Reason : --
Interface Name       : 1/1/2
VLAN Id              : 1234
MAC Address          : 00:50:56:96:04:4d
IP Address           : 4000::1
Time Remaining       : 3290 seconds
Lockdown Status      : Yes
Hardware Status      : No
Hardware Error Reason : Resource unavailable
Interface Name       : lag112
VLAN Id              : 151
MAC Address          : 00:50:56:96:d8:3d
IP Address           : 1001::5
Time Remaining       : 1200 seconds
Lockdown Status      : No
Hardware Status      : Yes
Hardware Error Reason : --

```

Showing the detailed binding record and related information for interface 1/1/2:

```
switch# show ipv6 source-lockdown binding interface 1/1/2
```

```

Interface Name      : 1/1/2
VLAN Id             : 1234
MAC Address         : 00:50:56:96:04:4d
IP Address          : 4000::1
Time Remaining      : 3290 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable
Interface Name      : 1/1/2
VLAN Id             : 1234
MAC Address         : 00:50:56:96:04:4d
IP Address          : 4000::1
Time Remaining      : 3290 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable

```

Showing the detailed binding record and related information for interface 1/1/2 (identified in this example command by the IP address):

```
switch# show ipv6 source-lockdown binding ip 4000::1
```

```

Interface Name      : 1/1/2
VLAN Id             : 1234
MAC Address         : 00:50:56:96:04:4d
IP Address          : 4000::1

```

```
Time Remaining      : 515 seconds
Lockdown Status     : No
Hardware Status      : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/1 (identified in this example command by the MAC address):

```
switch# show ipv6 source-lockdown binding mac 00:50:56:96:e4:cf
Interface Name      : 1/1/1
VLAN Id             : 1234
MAC Address         : 00:50:56:96:e4:cf
IP Address          : aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status      : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface lag112 (identified in this example command by the VLAN):

```
switch# show ipv6 source-lockdown binding vlan 151
Interface Name      : lag112
VLAN Id             : 151
MAC Address         : 00:50:56:96:d8:3d
IP Address          : 1001::5
Time Remaining      : 1200 seconds
Lockdown Status     : No
Hardware Status      : Yes
Hardware Error Reason : --
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000	Operator () or Manager ()	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Platforms	Command context	Authority
6100		

Internet Control Message Protocol (ICMP)

The Internet Control Message Protocol (ICMP) is a supporting protocol in the Internet protocol suite. The protocol is used by network devices, including routers, to send error messages and operational information. For example, an ICMP message might indicate that a requested service is not available. Another example of an ICMP message might be that a host or router could not be reached.

Subtopics

- [ICMP message types](#)
- [When ICMP messages are sent](#)
- [ICMP redirect messages](#)
- [When ICMP redirect messages are sent](#)
- [ICMP commands](#)

ICMP message types

The type field identifies the type of message sent by the host or gateway.

Type	ICMP messages
0	Echo Reply (Ping Reply, used with Type 8, Ping Request)
3	Destination Unreachable
4	Source Quench
5	Redirect
8	Echo Request (Ping Request, used with Type 0, Ping Reply)
9	Router Advertisement (Used with Type 9)
10	Router Solicitation (Used with Type 10)

Type	ICMP messages
11	Time Exceeded
12	Parameter Problem
13	Timestamp Request (Used with Type 14)
14	Timestamp Reply (Used with Type 13)
15	Information Request (obsolete) (Used with Type 16)
16	Information Reply (obsolete) (Used with Type 15)
17	Address Mask Request (Used with Type 17)
18	Address Mask Reply (Used with Type 18)

When ICMP messages are sent

ICMP messages are sent when one or more of the following scenarios occur:

- A datagram cannot reach its destination.
- The gateway does not have the buffering capacity to forward a datagram.
- The gateway can direct the host to send traffic on a shorter route.

ICMP redirect messages

ICMP redirect messages are used by routers to notify the hosts on the data link that a better route is available for a particular destination.

When ICMP redirect messages are sent

The switch is configured to send redirects by default. ICMP redirect messages are sent when one or more of the following scenarios occur:

- The interface on which the packet comes into the router is the same interface on which the packet gets routed out.

- The subnet or network of the source IP address is on the same subnet or network of the next-hop IP address of the routed packet.
- The datagram is not source-routed.
- The destination unicast address is unreachable. In this case, the router generates the ICMP destination unreachable message to inform the source host about the situation.

ICMP commands

Subtopics

ICMP commands

ICMP commands

Select a command from the list in the left navigation menu.

Subtopics

ip icmp redirect

ip icmp throttle

ip icmp unreachable

ip icmp redirect

Syntax

```
ip icmp redirect  
no ip icmp redirect
```

Description

Enables the sending of ICMPv4 and ICMPv6 redirect messages to the source host. Enabled by default.

The **no** form of this command disables ICMPv4 and ICMPv6 redirect messages to the source host.

Examples

Enabling ICMP redirect messages:


```
switch(config)# ip icmp redirect
```

Disabling ICMP redirect messages:

```
switch(config)# no ip icmp redirect
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip icmp throttle

Syntax

```
ip icmp throttle <PACKET-INTERVAL>  
no ip icmp throttle [<PACKET-INTERVAL>]
```

Description

Used to configure the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

The **no** form of this command disables the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

Parameter	Description
<PACKET-INTERVAL>	Specifies the ICMPv4/v6 packet interval in seconds. Default: 1 second. Range: 1 - 86400.

Examples

Enabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# ip icmp throttle 3000
```

Disabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# no ip icmp throttle
```

Command History

Release	Modification
10.8	Added the optional <code><PACKET - INTERVAL></code> parameter to the no form of the command.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip icmp unreachable

Syntax

```
ip icmp unreachable  
no ip icmp unreachable
```

Description

Enables the sending of ICMPv4 and ICMPv6 destination unreachable messages on the switch to a source host when a specific host is unreachable. The unreachable host address originates from the failed packet. Default setting.

The **no** form of this command disables the sending of ICMPv4 and ICMPv6 destination unreachable messages from the switch to a source host when a specific host is unreachable. This command does not prevent other hosts from sending an ICMP unreachable message.

Examples

Enabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# ip icmp unreachable
```

Disabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# no ip icmp unreachable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

DNS

The Domain Name System (DNS) is the Internet protocol for mapping a hostname to its IP address. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

Subtopics

DNS client

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Syntax

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```

switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns
VRF Name : default
Domain Name: switch.com
Name Server(s): 1.1.1.1
Host Name                                     Address
-----
-----
myhost1
3.3.3.3
switch#

```

Fully Qualified Domain Name Resolver

The FQDN (Fully Qualified Domain Name) resolver is a subsystem within the DNS client that enables dynamic resolution of domain names to their corresponding IP addresses (IPv4 and/or IPv6). The task of name resolutions can be offloaded to **FQDN Resolver** by the applications. **FQDN Resolver** performs the name resolution and updates the status back to the interested applications. Periodic refresh for name resolution will be carried by this service.

With the command line interface it's possible to configure refresh Interval and force refresh for FQDN resolver entries. The following show commands are also available:

Show FQDN Resolver

Field	Description
Total FQDN	Total count of FQDN resolver entries in the database.
Refresh Time.	Refresh interval for FQDN resolver entries in second s.
Source	Source of the resolution requests.
VRF	Virtual Routing and Forwarding instance.
FQDN	Fully Qualified Domain Name being resolved.
Status	Resolution status (resolved, unresolved or stale).

Field	Description
Addr - Family	Address family requested for resolution (IPv4, IPv6, or both).
ip - address	Resolved IP addresses.

Show FQDN Resolver Detail

Field	Description
Source	Source of the resolution requests.
VRF	Virtual Routing and Forwarding instance.
FQDN	Fully Qualified Domain Name being resolved.
Status	Resolution status (resolved, unresolved or stale).
Requested Address Family	Address family requested for resolution (IPv4, IPv6, or both).
IPv4 Addresses	Resolved IPv4 addresses.
IPv6 Addresses	Resolved IPv6 addresses.

Show FQDN Resolver Refresh Interval

Field	Description
Refresh Interval	Current refresh interval for FQDN resolvers in seconds.

DNS client commands

Subtopics

DNS client commands

DNS client commands

Select a command from the list in the left navigation menu.

Subtopics

- [ip dns domain-list](#)
- [ip dns domain-name](#)
- [ip dns host](#)
- [ip dns server address](#)
- [ip dns fqdn-resolver refresh-interval](#)
- [ip dns fqdn-resolver force-refresh](#)
- [show ip dns](#)
- [Show FQDN Resolver](#)
- [Show IP DNS FQDN Resolver Detail](#)
- [Show IP DNS FQDN Resolver Refresh Interval](#)

ip dns domain-list

Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf
                                default]
no ip dns domain-list <DOMAIN-NAME> [vrf
                                default]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Parameter	Description
<code>list <DOMAIN-NAME></code>	Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com

switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain5.com vrf mainvrf
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns domain-name

Syntax

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Parameter	Description
<code><DOMAIN-NAME></code>	Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ip dns host

Syntax

```
ip dns host <HOST-NAME>
           <IP-ADDR> [ vrf <VRF-NAME> ]
```

```
no ip dns host <HOST-NAME>
        <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Parameter	Description
host <HOST-NAME>	Specifies the name of a host. Length: 1 to 256 characters.
<IP-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ip dns server address

Syntax

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]  
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Parameter	Description
<code><IP-ADDR></code>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
-----------	-----------------	-----------

All platforms	config	Administrators or local user group members with execution rights for this command.
---------------	--------	--

ip dns fqdn-resolver refresh-interval

Syntax

```
[no] ip dns fqdn-resolver refresh-interval <3600-86400>
```

Description

Sets the refresh interval for FQDN resolvers. The interval specifies the time in seconds after which the FQDN resolver entries are refreshed. The valid range for the interval is between 3600 seconds (1 hour) to 86400 seconds (1 day). For each FQDN entry refresh timer is applicable from the time entry is installed.

The **no** form of this command removes the refresh interval and sets it back to default value which is 1 day.

Parameter	Description
-----------	-------------

<3600-86400>	Refresh interval in seconds (default: 86400 seconds). Required
--------------	---

Examples

This example sets a custom refresh interval.

```
switch(config)# ip dns fqdn-resolver refresh-interval 6000
```

```
switch(config)# ip dns fqdn-resolver refresh-interval 85400
```

Resetting the refresh interval to the default value.

```
switch(config)# no ip dns fqdn-resolver refresh-interval
```

Attempting to set an invalid interval:

```
switch(config)# ip dns fqdn-resolver refresh-interval 20
Invalid interval. Valid range: 3600-86400 seconds.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns fqdn-resolver force-refresh

Syntax

```
ip dns fqdn-resolver force-refresh { fqdn WORD | source port-access | vrf WORD | status (resolved | stale | unresolved) }
```

Description

This command initiates a force refresh for FQDN resolver entry/entries. It can refresh all FQDN entries, or entries filtered by FQDN, source, VRF or status. If no FQDN is specified, all entries are refreshed.

Parameter	Description
fqdn	Refresh entries associated with specific FQDN.

Parameter	Description
	Optional
source	Refresh entries based on the source of the resolution request. Optional
vrf	Refresh entries associated with a specific VRF. Optional
status	Refresh entries associated with a specific resolution status. Optional

Examples

Refreshing all FQDN entries:

```
switch(config)# ip dns fqdn-resolver force-refresh
```

Refreshing a specific FQDN:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.example.com
```

Refreshing all FQDNs from specific source:

```
switch(config)# ip dns fqdn-resolver force-refresh source port-access
```

Refreshing FQDN in a specific VRF:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.example.com vrf
red
```

Refreshing FQDN for a specific source and vrf:

```
switch(config)# ip dns fqdn-resolver force-refresh source classifier vrf red
```

When a FQDN entry is not found:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.nonexistent.com
No FQDN entries matching the given criteria.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ip dns

Syntax

```
show ip dns [vrf <VRF-NAME>]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Parameter	Description
vrf <VRF-NAME>	Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

Usage

AOS-CX gives the following priority levels to these three update methods.

- Priority 1 - standalone CLI configuration
- Priority 2 - static ip dns configuration
- Priority 3 - Dynamic config

Examples

These examples define DNS settings and then show how they are displayed with the **show ip dns** command.

```
switch(config) # ip dns domain-name domain.com
switch(config) # ip dns domain-list domain5.com
switch(config) # ip dns domain-list domain8.com
switch(config) # ip dns server-address 4.4.4.4
switch(config) # ip dns server-address 6.6.6.6
switch(config) # ip dns host host3 5.5.5.5
```

```

switch(config)# ip dns host host2 2.2.2.2
switch(config)# ip dns host host3 c::12
switch(config)# ip dns domain-name reddomain.com vrf default
switch(config)# ip dns domain-list reddomain5.com vrf default
switch(config)# ip dns domain-list reddomain8.com vrf default
switch(config)# ip dns server-address 4.4.4.5 vrf default
switch(config)# ip dns server-address 6.6.6.7 vrf default
switch(config)# ip dns host host3 5.5.5.6 vrf default
switch(config)# ip dns host host2 2.2.2.3 vrf default
switch(config)# ip dns host host3 c::13 vrf

```

default

```

switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name      Address
-----
host2          2.2.2.2
host3          5.5.5.5
host3          c::12

```

```

switch(config)# ip dns domain-name domain.com vrf red
switch(config)# ip dns domain-list domain5.com vrf red
switch(config)# ip dns domain-list domain8.com vrf red
switch(config)# ip dns server-address 4.4.4.4 vrf red
switch(config)# ip dns server-address 6.6.6.6 vrf red
switch(config)# ip dns host host3 5.5.5.5 vrf red
switch(config)# no ip dns host host2 2.2.2.2 vrf red
switch(config)# ip dns host host3 c::12 vrf red
switch# show ip dns vrf red

```

```

VRF Name : red
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name      Address
-----
host3          5.5.5.5
host3          c::12

```

```

switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com

```



```

switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host3 c::12
switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name      Address
-----
host3          5.5.5.5
host3          c::12

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Show FQDN Resolver

Syntax

```
show ip dns fqdn-resolver
```

Description

This command displays a summary table of all FQDN resolver entries, including the total count of FQDN entries, refresh time, source information, and a tabular display of all entries.

Examples

Show unresolved FQDN entries with ipv4 requested:

```
switch# show ip dns fqdn-resolver
Total FQDN: 1
Refresh Time: 86400 seconds
Source: port-access
VRF          FQDN          Status          Addr-Family     ip-
address
-----
default      host1.search.com        unresolved      ipv4            -
```

Show resolved FQDN entries:

```
switch# show ip dns fqdn-resolver
Total FQDN: 2
Refresh Time: 7200 seconds
Source: port-access
VRF          FQDN          Status          Addr-Family
ip-address
-----
default      www.example.com        resolved        ipv4
192.168.1.1, 192.168.1.2
default      api.test.com           resolved        ipv4
10.0.0.1
2001:db8::1, 2001:db8::2
long_vrf_na  a_very_long_fqdn_name  resolved        ipv4
10.0.0.2
```

When no FQDN entries exist:

```
switch(config)# show ip dns fqdn-resolver
No FQDN resolver entries found.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Show IP DNS FQDN Resolver Detail

Syntax

```
show ip dns fqdn-resolver detail {fqdn WORD | source (port-access) | vrf WORD | requested-address-family (ipv4|ipv6|both)}
```

Description

Displays detailed hierarchical information for FQDN resolver entries, including the FQDN, VRF, requested address family, resolved IPv4 and IPv6 addresses, resolution status, and remaining refresh timer. The command supports filtering by FQDN, source, VRF, and address family. If no filters are provided, all entries are displayed.

Examples

Show detailed view with unresolved entry:

```
switch# show ip dns fqdn-resolver detail
```

```
-----  
----  
Source: port-access  
-----  
----  
FQDN: host1.search.com  
VRF: default  
Requested Address Family: ipv4  
Status: unresolved  
-----  
----
```

Show detailed view with resolved entries:

```
switch# show ip dns fqdn-resolver detail
```

```
Source: port-access  
  FQDN: www.example.com  
  VRF: default
```

```
Status: resolved
Requested Address Family: both
  IPv4 Addresses:
    192.168.1.1
    192.168.1.2
  IPv6 Addresses:
    2001:db8::1
    2001:db8::2
```

```
-----
FQDN: www.hpe.com
VRF: default
Status: resolved
Requested Address Family: ipv4
  IPv4 Addresses:
    10.0.0.1
```

Show detailed view with filter:

```
switch# show ip dns fqdn-resolver detail fqdn www.example.com
Source: port-access
  FQDN: www.example.com
  VRF: default
  Status: resolved
  Requested Address Family: both
  IPv4 Addresses:
    192.168.1.1
    192.168.1.2
  IPv6 Addresses:
    2001:db8::1
    2001:db8::2
```

Show detailed view when no entries exist:

```
switch# show ip dns fqdn-resolver detail
```

```
No FQDN resolver entries found
```

**NOTE**

The **source** parameter is not supported in 8320, 8400 or 9300 switch series.

**NOTE**

For more information on features that use this command, refer to the Security Guide for your switch model.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Show IP DNS FQDN Resolver Refresh Interval

Syntax

```
show ip dns fqdn-resolver refresh-interval
```

Description

Displays the current refresh interval for FQDN resolvers.

Examples

Show current refresh interval (default):

```
switch# show ip dns fqdn-resolver refresh-interval
```

```
Refresh Interval: 86400 seconds
```

Show refresh interval after configuration change:

```
switch(config)# ip dns fqdn-resolver refresh-interval 7200
switch(config)# exit
switch# show ip dns fqdn-resolver refresh-interval
Refresh Interval: 7200 seconds
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

ARP

ARP (Address Resolution Protocol) is used to map the network address assigned to a device to its physical address. For example, on an Ethernet network, ARP maps layer 3 IPv4 network addresses to layer 2 MAC addresses. (ARP does not work with IPv6 addresses. Instead, the Neighbor discovery protocol is used.)

ARP operates at layer 2. ARP requests are broadcast to all devices on the local network segment and are not forwarded by routers. ARP is enabled by default and cannot be disabled.

Proxy ARP

Proxy ARP allows a routing switch to answer ARP requests from devices on one network on behalf of devices on another network. The ARP proxy is aware of the location of the traffic destination, and offers its own MAC address as the final destination.

For example, if Proxy ARP is enabled on a routing switch connected to two subnets (10.10.10.0/24 and 20.20.20.0/24), the routing switch can respond to an ARP request from 10.10.10.69 for the MAC address of the device with IP address 20.20.20.69.

Typically, the host that sent the ARP request then sends its packets to the switch that has the ARP proxy. This switch then forwards the packets to the intended host through a mechanism such as a tunnel.

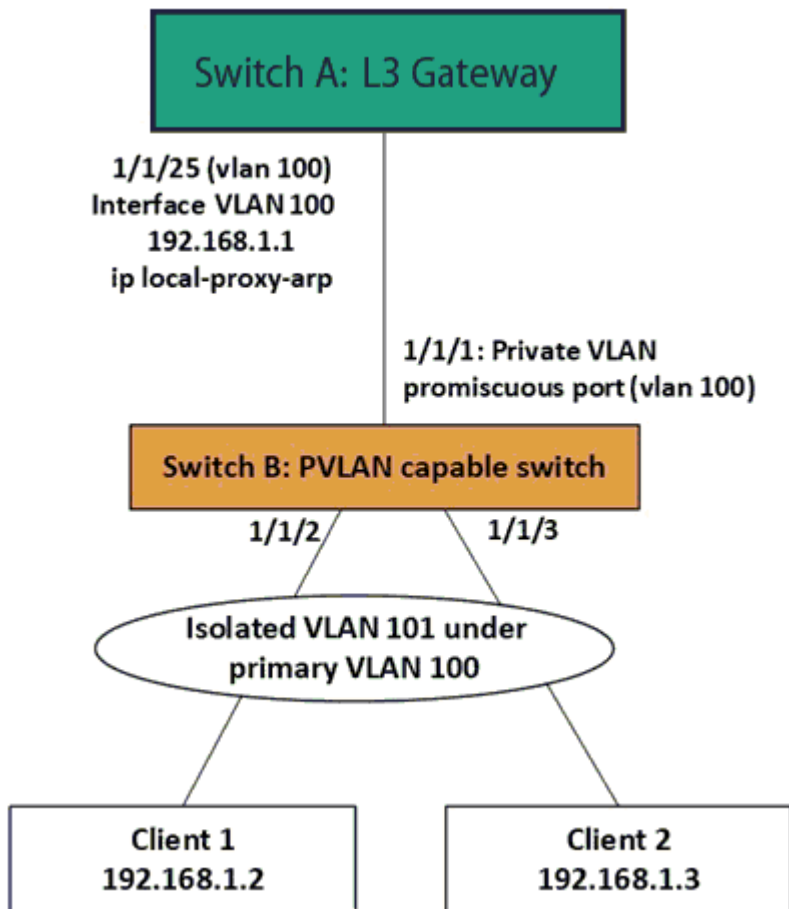
Proxy ARP is supported on L3 physical and VLAN interfaces. It is disabled by default. To enable proxy ARP, routing must be enabled on the interface.

Local proxy ARP

Local proxy ARP is a technique by which a device on a given network answers the ARP queries for a host address that is on the same network. It is primarily used to enable layer 3 communication between hosts within a common subnet that are separated by layer 2 boundaries (Example: PVLAN). Local proxy ARP is supported on L3 physical and VLAN interfaces and is disabled by default. Routing must be enabled on the interface to enable local proxy ARP.

Local proxy ARP can be used along with private VLAN deployments to allow layer 3 communication where clients are isolated at layer 2.

Figure 1. Local Proxy ARP Use Case Example



NOTE

The local proxy ARP feature is to be configured only if there is a need for the clients to communicate while still maintaining layer 2 isolation.

The following steps configure the local ARP proxy deployment shown in [Figure 1, Local Proxy ARP Use Case Example](#).

Step 1: Configuring private VLAN in switch B

1. Configure the primary and isolated VLAN first

```
PVLAN-Switch(config)# vlan 100  
PVLAN-Switch(config-vlan-100)# private-vlan primary  
PVLAN-Switch(config-vlan-100)# vlan 101  
PVLAN-Switch(config-vlan-101)# private-vlan isolated primary-vlan 100
```

2. Configure the PVLAN secondary ports to which clients are connected

```
PVLAN-Switch(config)# interface 1/1/2,1/1/3  
PVLAN-Switch(config-if-<1/1/1,1/1/2>)# vlan trunk allowed 101  
PVLAN-Switch(config-if-<1/1/1,1/1/2>)# private-vlan port-type secondary
```

3. Configure the PVLAN promiscuous port on the uplink which connects to the L3 gateway

```
PVLAN-Switch(config)# interface 1/1/1  
PVLAN-Switch(config-if)# vlan trunk allowed 100  
PVLAN-Switch(config-if)# private-vlan port-type promiscuous  
PVLAN-Switch(config-if)# no shutdown
```

For details about private VLAN configuration, refer to the AOS-CX Layer-2 Bridging guide.

Step 2: Configuring L3 gateway in switch A

```
L3-GW(config)# vlan 100  
L3-GW(config-vlan-100)# exit  
L3-GW(config)# interface 1/1/25  
L3-GW(config-if)# vlan trunk allowed 100  
L3-GW(config-if)# no shutdown  
L3-GW(config-if)# exit  
L3-GW(config)# interface vlan 100  
L3-GW(config-if-vlan)# ip address 192.168.1.1/24
```

Now the clients will be able to ping the gateway IP; but will not be able to communicate with each other due to the isolation enforced with private VLAN. As expected, the ping from client 1 to client 2 will fail in the above example, although the clients are in the same subnet.

Step 3: Configuring local proxy ARP in switch A

If local proxy ARP is enabled in the L3 gateway, the L3 gateway responds to ARP requests for addresses in the same subnet. In other words, the L3 gateway will be doing proxy for ARP resolution.

```
L3-GW(config)# no ip icmp redirect  
L3-GW(config)# interface vlan 100  
L3-GW(config-if-vlan)# ip local-proxy-arp
```


With this, the L3 communication will be enabled between client 1 and client 2. Now the ping from client 1 to client 2 will be successful in the above example. The ARP table in client 1 will be populated with the MAC of L3 gateway; not the MAC of client 2. Similar ARP table entry for client 2 as well. In other words, the L3 gateway is doing proxy for ARP resolution for the local network where clients are isolated at layer 2.

Dynamic ARP inspection

Dynamic arp inspection is provided as a mechanism for making ARP more secure.

ARP is used for resolving IP against MAC addresses on a broadcast network segment like the Ethernet and was originally defined by Internet Standard RFC 826. ARP does not support any inherent security mechanism and as such depends on simple datagram exchanges for the resolution, with many of these being broadcast.

Because it is an unreliable and non-secure protocol, ARP is vulnerable to attacks. Some attacks may be targeted toward the networks whereas other attacks may be targeted toward the switch itself. The attacks primarily intend to create denial of service (DoS) for the other entities present in the network.

Most of the attacks are carried out in one of the following three forms:

- Overwhelming the switch control plane with too many ARP packets.
- Overwhelming the switch control plane with too many unresolved data packets.
- Masquerading as a trusted gateway/server by wrongly advertising ARPs.

Several defense mechanisms can be put in place on a switch to protect against attacks:

- Limit the amount of ARP activity allowed from a host or on a port.
- Ensure that all ARP packets are consistent with one or more binding databases, which can be created through various means.
- Enforce integrity checks on the ARP packets to check against different MAC or IP addresses in the Ethernet or IP header and ARP header.

The following is supported:

- Enabling and disabling of Dynamic ARP Inspection on a VLAN level (it does not have to be SVI).
- Defining the member ports of a VLAN as either trusted or untrusted.
- Only ARP traffic on untrusted ports subjected to checks.
- Routed ports (RoPs) always treated as trusted.
- Listening to the DHCP Bindings table and check every ARP packet to match against the binding.

ARP ACLs are not supported and the DHCP snooping table is the only source of binding.

Subtopics

ARP commands

ARP commands

Select a command from the list in the left navigation menu.

Subtopics

- arp inspection
- arp inspection trust
- arp ip
- arp process-grat-arp
- clear arp
- debug arp-security
- ip proxy-arp
- ipv6 local-proxy-nd
- ipv6 neighbor mac
- show arp
- show arp inspection interface
- show arp inspection statistics
- show arp inspection vlan
- show arp state
- show arp summary
- show arp timeout
- show arp vrf
- show ipv6 neighbors
- show ipv6 neighbors state
- show tech arp-security

arp inspection

Syntax

```
arp inspection
```

Description

Enables Dynamic ARP inspection on the current VLAN, which means that ARP packets received from untrusted interfaces are discarded if they have an Invalid IP-to-MAC address binding.

The **no** form of this command disables Dynamic ARP Inspection on the VLAN.

Examples

Enabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# arp inspection
```

Disabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# no arp inspection
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-vlan- <VLAN-ID>	Administrators or local user group members with execution rights for this command.

arp inspection trust

Syntax

```
arp inspection trust
no arp inspection trust
```

Description

Configures the interface as a trusted. All interfaces are untrusted by default.

The **no** form of this command returns the interface to the default state (untrusted).

Example

Setting an interface as trusted:

```
switch(config-if) # arp inspection trust
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

arp ip

Syntax

```
arp ip <IP_ADDR> mac <MAC_ADDR>  
no arp ip <IP_ADDR> mac <MAC_ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv4 neighbors).

The no form of this command deletes a permanent static neighbor entry from the ARP table.

Parameter	Description
<code>ip <IP-ADDR></code>	Specifies the IP address of the neighbor or the virtual IP addresses of the cluster in IP format (x.x.x.x), where x is a decimal number from 0 to 255. . Range: 4096 to 131072. Default: 131072.
<code>mac <MAC-ADDR></code>	Specifies the MAC address of the neighbor or the multicast MAC address in IANA format (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Example

Configuring a static ARP entry on a interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# arp ip 2.2.2.2 mac 01:00:5e:00:00:01
```

Removing a static ARP entry on interface VLAN**10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no arp ip 2.2.2.2 mac 01:00:5e:00:00:01
```

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if config-if-vlan	Administrators or local user group members with execution rights for this command.

arp process-grat-arp

Syntax

```
arp process-grat-arp
no arp process-grat-arp
```

Description

Enables the processing of gratuitous ARP packets on the individual port or group of L3 ports together.

By default, the gratuitous ARP processing is enabled. When gratuitous ARP (GARP) processing is enabled, a switch that is advertising any changes in its MAC through the GARP will reflect in the neighbor table of the switch. However, the switch will not be able to learn the neighbor through the GARP. This configuration is applicable only on L3 interfaces such as ROPs, subinterfaces, and SVIs.

The **no** form of this command disables the processing of gratuitous ARP packets.

Example

Enabling the processing of gratuitous ARP packets on the interface **1/1/1**:

```
switch(config) # interface 1/1/1
switch(config-if) # no shutdown
switch(config-if) # arp process-grat-arp
```

Enabling the processing of gratuitous ARP packets on interfaces **1/1/1** to **1/1/5**:

```
switch(config) # interface 1/1/1-1/1/5
switch(config-if<1/1/1-1/1/5>) # no shutdown
switch(config-if<1/1/1-1/1/5>) # arp process-grat-arp
```

Enabling the processing of gratuitous ARP packets on sub-interface **1/1/1.10**:



NOTE

Applies only to the HPE Aruba Networking 6300, 6400, 8100, and 8360 Switch Series.

```
switch(config) # interface 1/1/1.10
switch(config-subif) # no shutdown
switch(config-subif) # arp process-grat-arp
```

Disabling the processing of gratuitous ARP packets on VLANs **2** to **100**:

```
switch(config) # interface vlan 2-100
switch(config-if-vlan<2-100>) # no shutdown
switch(config-if-vlan<2-100>) # no arp process-grat-arp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if config-if-vlan config-subif	Administrators or local user group members with execution rights for this command.

clear arp

Syntax

```
clear arp
  port <PORT-ID> [ip <A.B.C.D>|all] [[ipv6 <X:X::X:X>|all]vla
  vrf [all-vrfs|{<VRF-NAME> [ip <A.B.C.D>] [[ipv6 <X:X::X:X>]]]
```

Description

Clears IPv4 and IPv6 neighbor entries from the ARP table. If you do not specify any VRF or port parameters, ARP table entries are cleared for the **default** VRF.

Parameter	Description
port <PORT-ID>	Specifies a port on the switch. For example: vlan 2.
ip <A.B.C.D> all]	(Optional) Include an IP address to clear neighbor entries for that specific address, or use the all parameter to clear entries for all IP addresses.
ipv6 <X:X::X:X> all	(Optional) Include an IPv6 address to clear neighbor entries for that specific address, or use the all parameter to clear entries for all IPv6 addresses.
vrf	Clears IPv4 and IPv6 neighbor entries for the specified VRF or for all VRFs. If no VRF is specified the default VRF is cleared.

Parameter	Description
all-vrfs	Clear neighbor entries for all VRFs
<VRF-NAME>	Clear neighbor entries for the specified VRF.
ip <A.B.C.D>	(Optional) Include an IP address to clear just the neighbor entries for the specified IP address.
ipv6 <X:X::X:	(Optional) Include an IPv6 address to clear the neighbor entries for the specified IPv6 address.

Examples

Clearing all IPv4 and IPv6 neighbor ARP entries for the default VRF:

```
switch# clear arp
```

Clearing all ARP neighbor entries for a port:

```
switch#
```

```
clear arp vlan 2
```

```
switch# clear arp vrf all-vrfs
```

```
switch# clear arp vrf RED
```

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

debug arp-security

Syntax

```
debug arp-security <LOG-CATEGORY> [severity <LEVEL>]  
no debug arp-security [<LOG-CATEGORY>] [severity <LEVEL>]
```

Description

Enables ARP security debug logs. If <SEVERITY> is omitted, all severities are logged.

The **no** form of this command disables ARP security debug logs.

Parameter	Description
<code><LOG-CATEGORY></code>	Selects the ARP security debug log category. Available categories are: • all: Selects all ARP security debug log categories. • config: Selects the ARP security config debug log category. • inspection: Selects the ARP security inspection debug log category. • packet: Selects the ARP security packet debug log category.
<code>severity <LEVEL></code>	Specifies how to filter the ARP security debug logging by setting the minimum severity level for which debug logging will be performed. The selected severity level and all severities above (more severe) will be included in the logging. • emerg: Sets ARP security debug log filtering to Emergency only. • alert: Sets ARP security debug log filtering to Alert and above. • critical: Sets ARP security debug log filtering to Critical and above.

Parameter	Description
	• error: Sets ARP security debug log filtering to Error and above. • warning: Sets ARP security debug log filtering to Warning and above. • notice: Sets ARP security debug log filtering to Notice and above. • info: Sets ARP security debug log filtering to Info and above. • debug: Sets ARP security debug log filtering to all severities.

Examples

Enable ARP security debug logging for all categories and all severities:

```
switch# debug arp-security all
```

Enable ARP security config debug log for severity level Error and above:

```
switch# debug arp-security config severity error
```

Enable ARP security inspection debug log for severity level Notice and above:

```
switch# debug arp-security inspection severity notice
```

Enable ARP security debug packet for severity level Critical and above:

```
switch# debug arp-security packet severity critical
```

Enable ARP security debug logging for all categories and severity level Alert and above:

```
switch# debug arp-security all severity alert
```

Disable ARP security debug logging:

```
switch# no debug arp-security
```

Command History

Release	Modification
10.11	Command introduced on 4100i Switch Series
10.09	Command introduced on 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ip proxy-arp

Syntax

```
ip proxy-arp
no ip proxy-arp
```

Description

Enables proxy ARP for the specified Layer 3 interface. Proxy ARP is supported on Layer 3 physical interfaces, LAG interfaces, and VLAN interfaces. It is disabled by default. To enable proxy ARP on an interface, routing must be enabled on that interface.

The **no** form of this command disables proxy ARP for the specified interface.

Examples

Enabling proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1
switch(config-if)# ip proxy-arp
```

Enabling proxy ARP on VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip proxy-arp
```

Enabling proxy ARP on a LAG **11**:

```
switch(config)# int lag 11
switch(config-lag-if)# ip proxy-arp
```

Disabling proxy ARP on interface 1/1/1:

```
switch# interface 1/1/1
switch(config-if)# no ip proxy-arp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if config-if-vlan config-lag-vlan	Administrators or local user group members with execution rights for this command.

ipv6 local-proxy-nd

Syntax

```
ipv6 local-proxy-nd  
no ipv6 local-proxy-nd
```

Description

Enables local proxy ND for the specified interface.

The **noipv6 local-proxy-nd** command disables local proxy ND for the specified interface.

Usage

Before enabling local proxy ND, disable Internet Control Message Protocol (ICMP) redirect messages. Disabling the ICMP redirect ensures that hosts don't communicate through other gateways which could possibly have better routes and thereby skip the proxy.

Example

Enable local proxy ND on a L3 physical interface.

```
switch(config)# interface 1/1/1  
switch(config-if)# ipv6 local-proxy-nd
```

Enable local proxy ND on a VLAN interface.

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 local-proxy-nd
```

```
switch(config)# interface 1/1/1.10
switch(config-subif)# ipv6 local-proxy-nd
```

Disable local proxy ND for the specified interface.

```
switch(config-if)# no ipv6 local-proxy-nd
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

ipv6 neighbor mac

Syntax

```
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
no ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv6 neighbors).

The **no** form of this command deletes a permanent static neighbor entry from the ARP table.

Parameter	Description
<IPV6-ADDR>>	Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Parameter	Description
mac <MAC-ADDR>>	Specifies the MAC address of the neighbor (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Example

Creates a static ARP entry on interface **vlan 2**.

```
switch(config)#
    interface vlan 2
switch(config-if-vlan)# arp ipv6 neighbor 2001:0db8:85a3::8a2e:0370:7334
mac 00:50:56:96:df:c8
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

show arp

Syntax

```
show arp
```

Description

Shows the entries in the ARP (Address Resolution Protocol) table.

Usage

This command displays information about ARP entries, including the IP address, MAC address, port, and state.

When no parameters are specified, the `show arp` command shows all ARP entries for the default VRF (Virtual Router Forwarding) instance.

Examples

```
switch# show arp
IPv4 Address      MAC                Port      Physical Port
State
-----
---
192.168.1.2       00:50:56:96:7b:e0  vlan10    1/1/29      stale
192.168.1.3       00:50:56:96:7b:ac  vlan10    1/1/1
reachable
Total Number Of ARP Entries Listed- 2.
-----
---
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show arp inspection interface

Syntax

```
show arp inspection interface [<IFNAME>] [vlan <VLAN-ID>]
```

Description

Shows the current configuration of dynamic ARP inspection on an interface.

Parameter	Description
<code><IFNAME></code>	Specifies the interface.
<code><VLAN-ID></code>	Specifies the VLAN ID. Range: 1 to 4094.

Examples

Showing current configuration of dynamic ARP inspection on all interfaces:

```
switch# show arp inspection interface
```

```
-----  
Interface          Trust-State  
-----  
1/1/1              Untrusted  
-----
```

switch# show arp inspection interface vsx-peer

```
-----  
Interface          Trust-State  
-----  
1/1/1              Untrusted  
lag100             Trusted  
-----
```

Showing current configuration of dynamic ARP inspection on a particular interface:

```
switch# show arp inspection interface 1/1/1
```

```
-----  
Interface          Trust-State  
-----  
1/1/1              Untrusted  
-----
```

Showing current configuration of dynamic ARP inspection on interface VLAN 2:

```
switch# show arp inspection interface vlan 2
```


Interface	Trust-State
-----	-----
vlan2	Trusted
-----	-----

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6000		
6100		

show arp inspection statistics

Syntax

```
show arp inspection statistics vlan [<VLAN-ID>]
```

Description

Shows statistics about forwarded and dropped ARP packets. When <VLAN-ID> is not specified, information is shown for all configured VLANs.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID or range of IDs separated by a dash "-". Range: 1 to 4094.

Examples

Showing ARP packet statistics for a range of VLANs:

```
switch# show arp inspection statistics vlan 1-100
```

VLAN	Name	Forwarded	Dropped
1	DEFAULT_VLAN_1	0	0

```
switch# show arp inspection statistics vlan vsx-peer
```

VLAN	Name	Forwarded	Dropped
1	DEFAULT_VLAN_1	0	0
200	VLAN200	0	0

Command History

Release

Modification

10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms

Command context

Authority

4100i	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6000		
6100		

show arp inspection vlan

Syntax

```
show arp inspection vlan [<VLAN-ID>]
```

Description

Shows the current configuration of dynamic ARP inspection on a VLAN. When <VLAN-ID> is not specified, information is shown for all configured VLANs.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID or range of IDs separated by a dash "-". Range: 1 to 4094.

Examples

Showing dynamic ARP configuration for all VLANs:

```
switch# show arp inspection vlan
-----
VLAN   Name                ARP Inspection
-----
1      DEFAULT_VLAN_1     -
100    VLAN100             -
200    VLAN200             Enabled
-----
```

Showing dynamic ARP configuration for a particular VLAN:

```
switch# show arp inspection vlan 1
-----
VLAN   Name                ARP Inspection
-----
1      DEFAULT_VLAN_1     -
-----

switch# show arp inspection vlan vsx-peer
-----
VLAN   Name                ARP Inspection
-----
1      DEFAULT_VLAN_1     -
-----
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series

Release	Modification
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

show arp state

Syntax

```
show arp state {all | failed | incomplete | permanent | reachable | stale}
```

Description

Shows ARP (Address Resolution Protocol) cache entries that are in the specified state.

Parameter	Description
all	Shows the ARP cache entries for all VRF (Virtual Router Forwarding) instances.
failed	Shows the ARP cache entries that are in <code>failed</code> state. The neighbor might have been deleted.
incomplete	Shows the ARP cache entries that are in <code>incomplete</code> state. An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. A solicitation request was sent, and the switch is waiting for a solicitation reply or a timeout.
permanent	Shows the ARP cache entries that are in <code>permanent</code> state. ARP entries that are in a permanent state can be removed by an administrative action only.

Parameter	Description
reachable	Shows the ARP cache entries that are in <code>reachable</code> state, meaning that the neighbor is known to have been reachable recently.
stale	Shows ARP cache entries that are in <code>stale</code> state. ARP cache entries are in the <code>stale</code> state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Examples

```
switch# show arp state failed
```

IPv4 Address	MAC	Port	Physical Port	State
192.168.1.4		vlan10		failed

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

show arp summary

Syntax

```
show arp summary [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows a summary of the IPv4 and IPv6 neighbor entries on the switch for all VRFs or a specific VRF.

Parameter	Description
<code>all-vrfs</code>	Selects all VRFs.
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF.

Examples

Showing summary ARP information for all VRFs:

```
switch# show arp summary all-vrfs
ARP Entry's State           : IPv4      IPv6
-----
Number of Reachable ARP entries : 2          0
Number of Stale ARP entries    : 0          0
Number of Failed ARP entries   : 2          2
Number of Incomplete ARP entries : 0          0
Number of Permanent ARP entries : 0          0
-----
Total ARP Entries: 6           : 4          2
-----
```

Showing a summary of all IPv4 and IPv6 neighbor entries on the primary and secondary (peer) switches:

```
vsx-primary# show arp summary
ARP Entry's State           IPv4      IPv6
-----
Number of Reachable ARP entries 25858    32231
Number of Stale ARP entries      0         1
Number of Failed ARP entries     0        257
Number of Incomplete ARP entries 0         0
Number of Permanent ARP entries  0         0
-----
Total ARP Entries- 58347         25858    32489
vsx-primary# show arp summary vsx-peer
ARP Entry's State           IPv4      IPv6
-----
Number of Reachable ARP entries 25858    32168
Number of Stale ARP entries      0         3
Number of Failed ARP entries     0        317
Number of Incomplete ARP entries 0         0
Number of Permanent ARP entries  0         0
```

Total ARP Entries-	58346	25858	32488
--------------------	-------	-------	-------

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show arp timeout

Syntax

```
show arp timeout [<INTERFACE>]
```

Description

Shows the age-out period for each ARP (Address Resolution Protocol) entry for a port, LAG, or VLAN interface.

Parameter	Description
<INTERFACE>	Specifies a physical port, VLAN, or LAG on the switch. For physical ports, use the format (for example, 1/3/1).

Examples

Showing ARP timeout information for a VLAN:

```
switch# show arp timeout vlan2
Port          VRF          Timeout
-----
vlan2         default      1800
switch# show arp timeout 1/1/1
ARP Timeout:
-----
Port          VRF          Timeout
1/1/1         default      600
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show arp vrf

Syntax

```
show arp {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows the ARP table for all VRF instances, or for the named VRF.

Parameter	Description
all-vrfs	Specifies all VRFs.

Parameter	Description
vrf <VRF-NAME>	Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

Examples

Showing ARP entries for VRF **vrf1**.

```
switch# show arp vrf vrf1
IPv4 Address      MAC                Port      Physical Port
State            VRF
-----
-----
100.1.250.50      00:50:56:8d:44:13  vlan1001   1/1/2
reachable vrf1
100.2.250.60      00:50:56:8d:45:63  vlan1002   vxlan1(1920:1680:1:1::2)
permanent vrf1
Total Number Of ARP Entries Listed: 2.
-----
-----
```

This example from a different network shows ARP entries for all VRFs.

```
switch# show arp all-vrfs
ARP IPv4 Entries:
-----
IPv4 Address      MAC                Port      Physical Port  State      VRF
192.168.120.10    00:50:56:bd:10:be  1/1/32    1/1/32         reachable  red
10.20.30.40       00:50:56:bd:6a:c5  1/1/29    1/1/29         reachable  test
-----
Total Number Of ARP Entries Listed: 2.
-----
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 neighbors

Syntax

```
show ipv6 neighbors {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows entries in the ARP table for all IPv6 neighbors for all VRFs or for a specific VRF.

When no parameters are specified, this command shows all ARP entries for the default VRF, and state information for `reachable` and `stale` entries only.

Parameter	Description
<code>all-vrfs</code>	Specifies all VRFs.
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

Examples

```
switch# show ipv6 neighbors
IPv6 Entries:
-----
IPv6 Address          MAC                Port      Physical Port
State
fe80::a21d:48ff:fe8f:2700  a0:1d:48:8f:27:00  vlan2300  1/1/31
reachable
fe80::f603:43ff:fe80:a600  f4:03:43:80:a6:00  vlan2300  1/1/30
reachable
-----
Total Number Of IPv6 Neighbors Entries Listed: 2.
```

```

switch# show ipv6 neighbors vrf vrf1
IPv6 Address                               MAC
Port           Physical Port      State      VRF
-----
1000:2:1:1::250:60    00:50:56:8d:45:63  vlan1002
vxlan1(1920:1680:1:1::2)  permanent  vrf1
1000:1:1:1::250:50    00:50:56:8d:44:13  vlan1001
1/1/2                      reachable  vrf1
Total Number Of IPv6 Neighbors Entries Listed: 2.

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ipv6 neighbors state

Syntax

```
show ipv6 neighbors state {all | failed | incomplete | permanent |
reachable | stale}
```

Description

Shows all IPv6 neighbor ARP (Address Resolution Protocol) cache entries, or those cache entries that are in the specified state.

Parameter	Description
all	Shows all ARP cache entries.
failed	Shows ARP cache entries that are in <code>failed</code> state. The neighbor might have been deleted. Set the neighbor to be unreachable.
incomplete	Shows ARP cache entries that are in <code>incomplete</code> state. An incomplete state means that address resolution is in progress and the link - layer address of the neighbor has not yet been determined. This means that a solicitation request was sent, and you are waiting for a solicitation reply or a timeout.
permanent	Shows ARP cache entries that are in <code>permanent</code> state.
reachable	Shows ARP cache entries that are in <code>reachable</code> state, meaning that the neighbor is known to have been reachable recently.
stale	Shows ARP cache entries that are in <code>stale</code> state. ARP cache entries are in the <code>stale</code> state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Example

```
switch# show ipv6 neighbors state all
IPv6 Address          MAC                Port      Physical Port
State
-----
----
100::2                48:0f:cf:af:f1:cc  lag1      lag1
reachable
300::3                48:0f:cf:af:33:be  vlan3     1/4/20
reachable
fe80::4a0f:cfff:feaf:f1cc 48:0f:cf:af:f1:cc  lag1      lag1
reachable
200::3                48:0f:cf:af:33:be  1/4/11    1/4/11
reachable
fe80::4a0f:cfff:feaf:33be 48:0f:cf:af:33:be  vlan3     1/4/20
reachable
Total Number Of IPv6 Neighbors Entries Listed- 5.
```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show tech arp-security

Syntax

```
show tech arp-security
```

Description

Shows the output of these three commands:

- `show arp inspection statistics vlan`
- `show arp inspection vlan`
- `show arp inspection interface`

Examples

Showing the output of the three ARP security show commands:

```
switch(config-if) # show tech arp-security
=====
Show Tech executed on Mon Nov 28 09:53:54 2019
=====
[Begin] Feature arp-security
```

```

=====
*****
Command : show arp inspection statistics vlan
*****

-----
VLAN      Name                Forwarded      Dropped
-----
1          DEFAULT_VLAN_1      0              0
200        VLAN200              0              0
-----

*****
Command : show arp inspection vlan
*****

-----
VLAN      Name                ARP-Inspection
-----
1          DEFAULT_VLAN_1      -
200        VLAN200              Enabled
-----

*****
Command : show arp inspection interface
*****

-----
Interface      Trust-State
-----
1/1/1          Untrusted
lag100         Trusted
-----

=====
[End] Feature arp-security
=====
=====
Show Tech commands executed successfully
=====

```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

[Accessing HPE Aruba Networking Support](#)

[Accessing Updates](#)

[Warranty Information](#)

[Regulatory Information](#)

[Documentation Feedback](#)

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	<u>https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html</u>
AOS - CX Switch Software Documentation Portal	<u>https://arubanetworking.hpe.com/techdocs/Aruba DocPortal/content/new-portal/aoscx.html</u>
HPE Aruba Networking Support Portal	<u>https://networkingsupport.hpe.com/home</u>
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	<u>https://www.hpe.com/psnow/doc/a50011948enw</u>

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End - of - Life information	https://networkingsupport.hpe.com/end - of - life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.